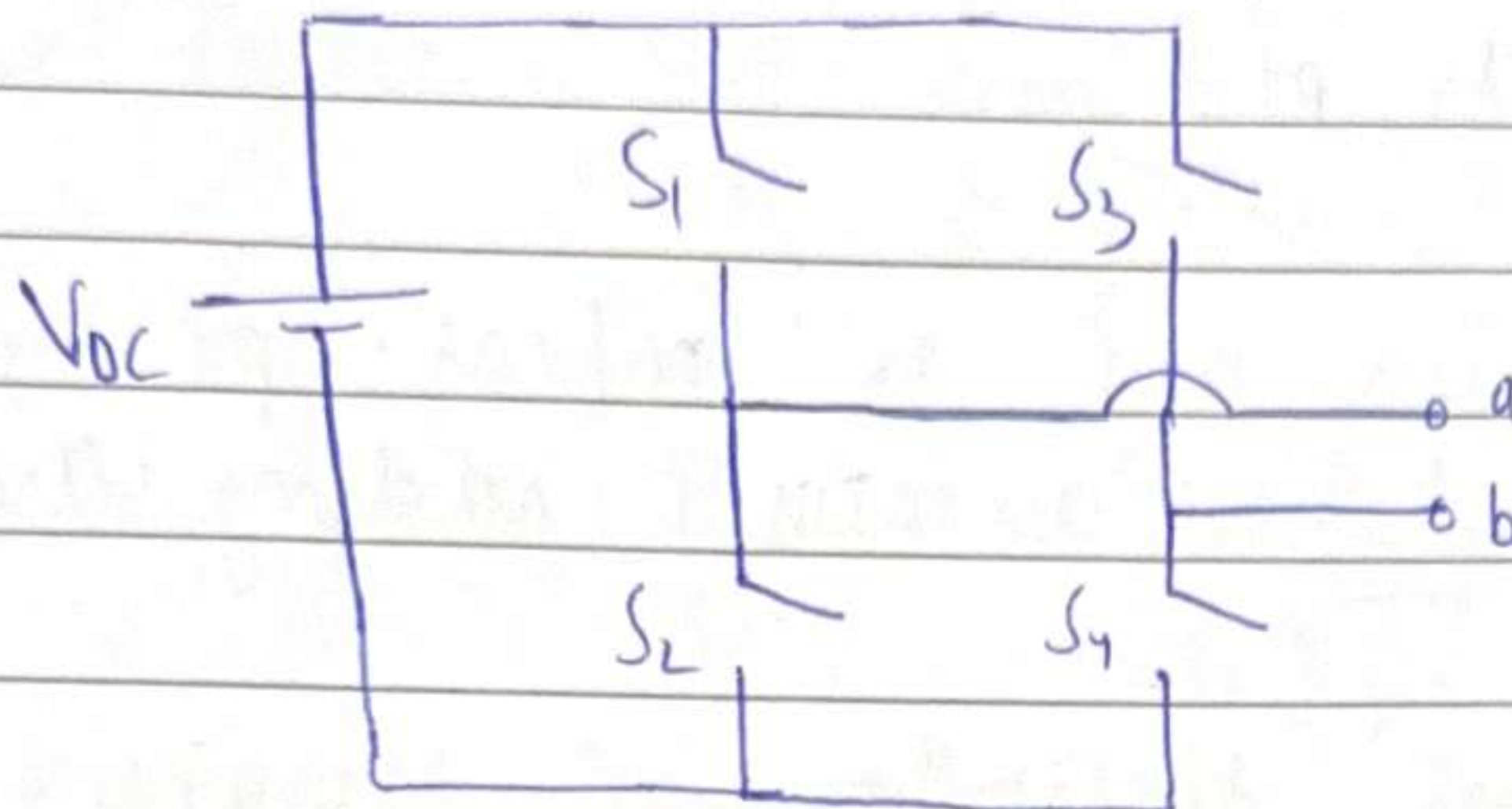


# Inverters



- H-bridge inverter (looks like an H)

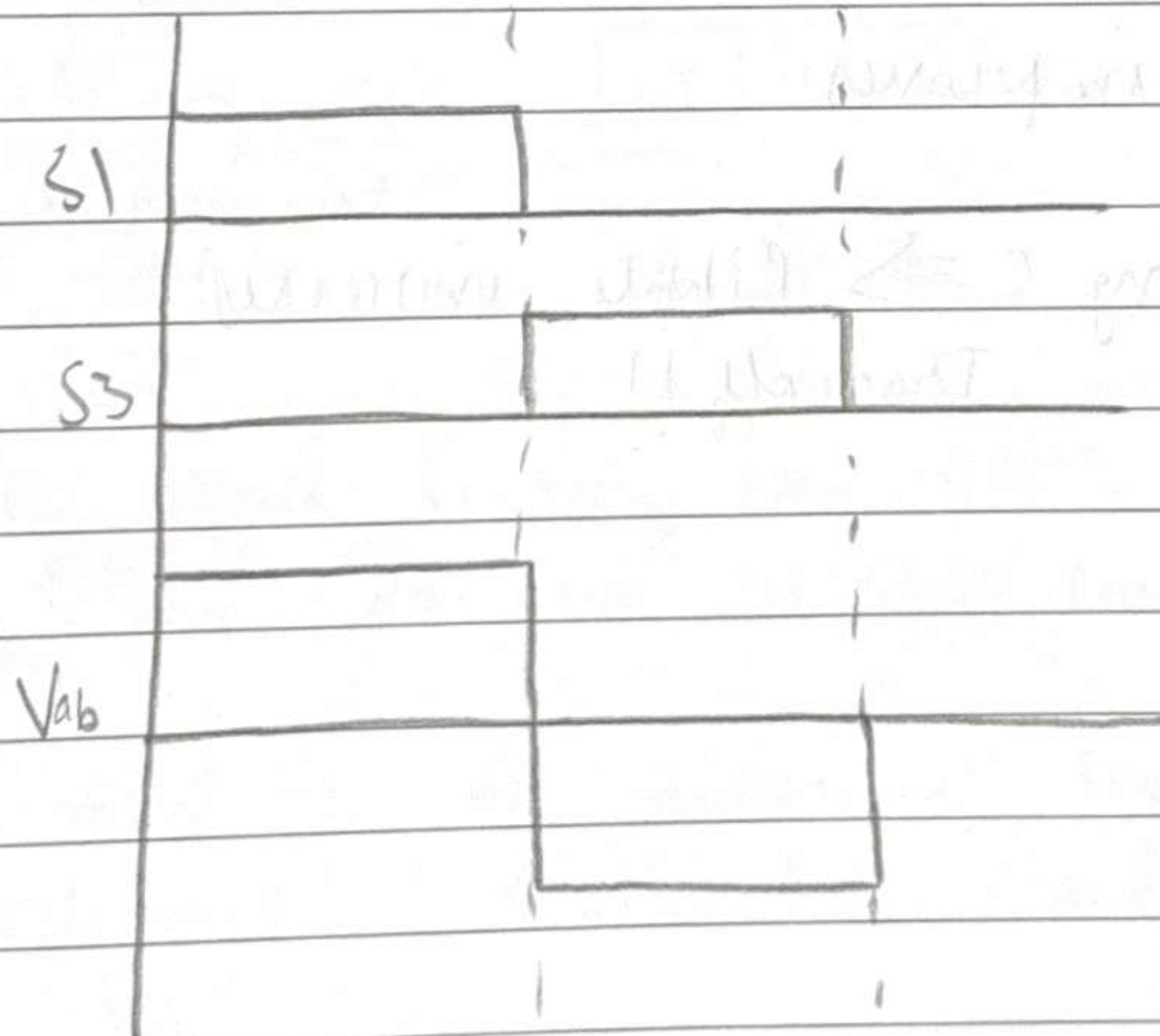
also called single-phase full bridge inverter

S1 and S2 can't be on together  
 S3 and S4 can't be on together

$$S1 = \overline{S2}$$

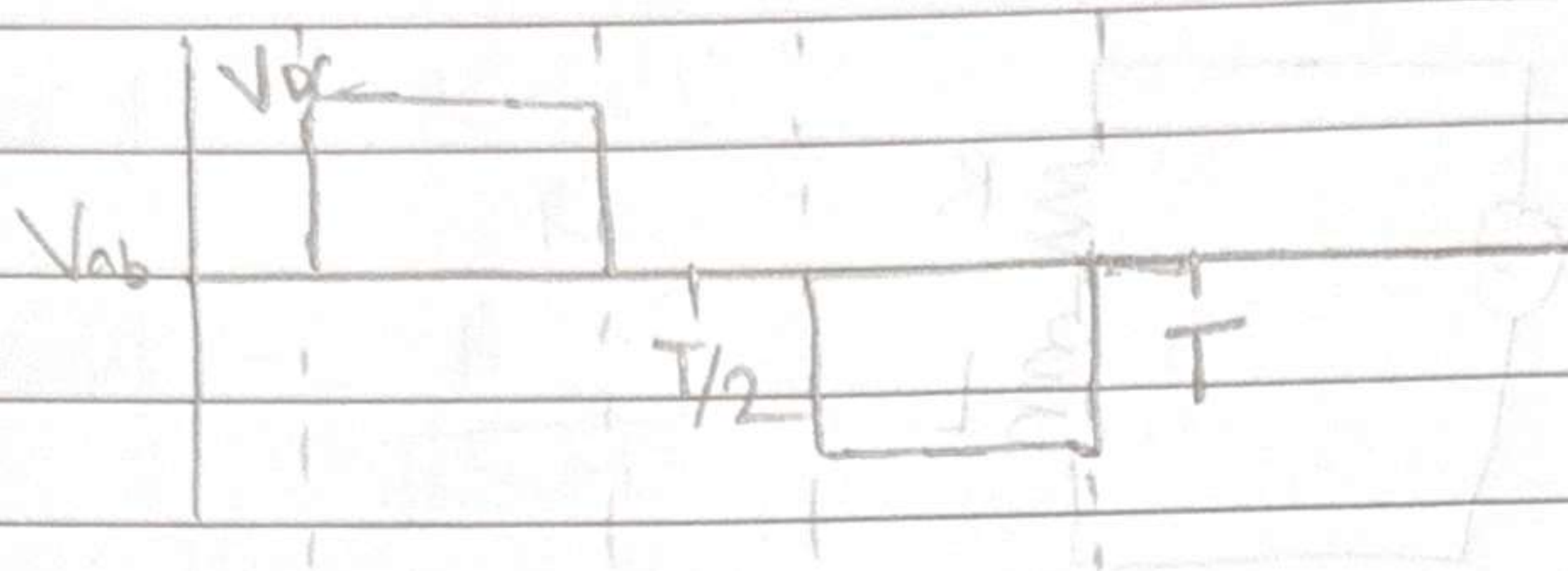
$$S3 = \overline{S4}$$

There are 4 allowed states of (S1, S2, S3, S4)





We can also do,



4 states

| $S_1$ | $S_2$ | $V_{ab}$  |               |
|-------|-------|-----------|---------------|
| 1     | 0     | $+V_{dc}$ | Active states |
| 0     | 1     | $-V_{dc}$ |               |
| 1     | 1     | 0         | Zero states   |
| 0     | 0     | 0         |               |

Advantages-

- Hence we can control  $V_{RMS}$  (and  $V_{avg}$ ) using zero states duration
- Can have variable frequency by adjusting switching frequency
- Can have adjustable phase

Disadvantages-

- High  $V_{THD}$

Solution-

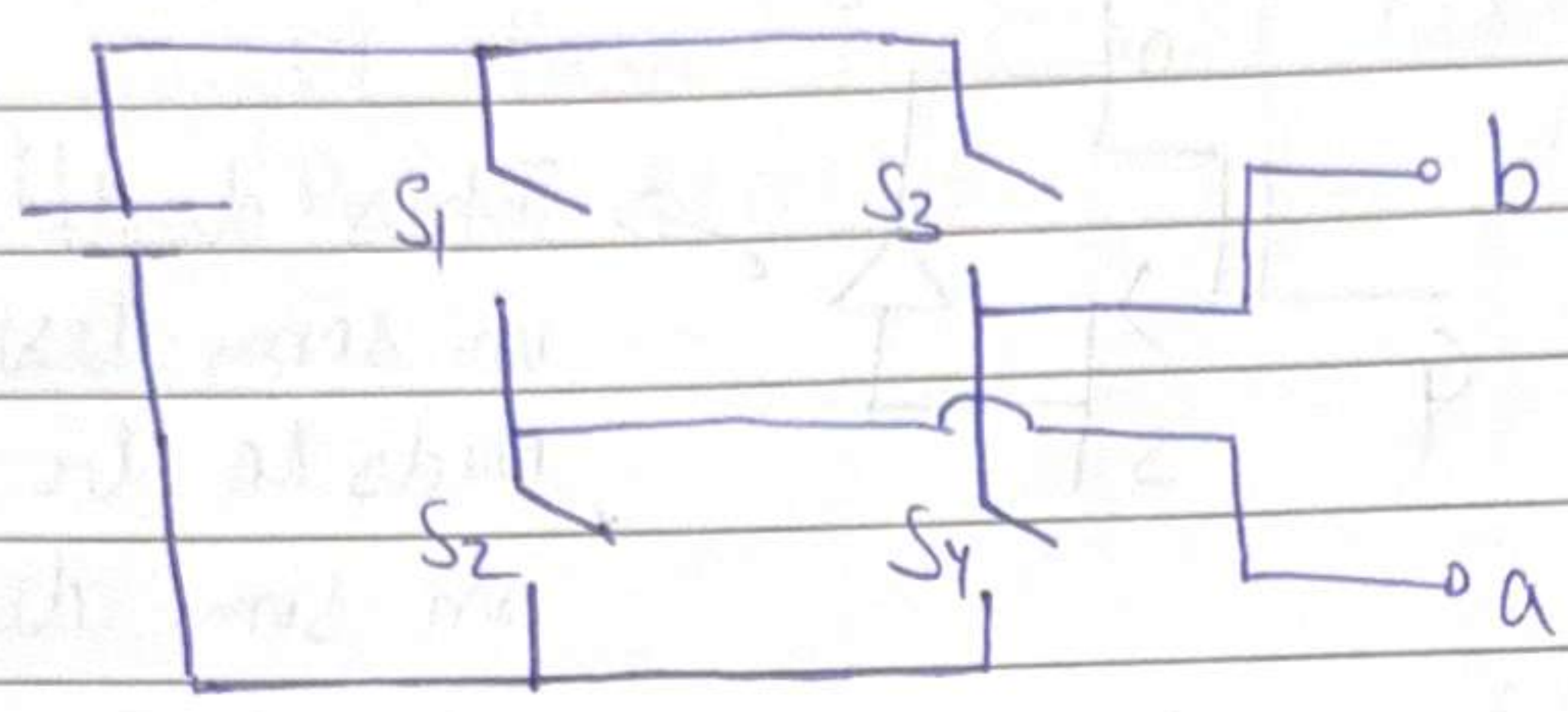
(i) LPF

(ii)



# Post Midsem

## Inverters



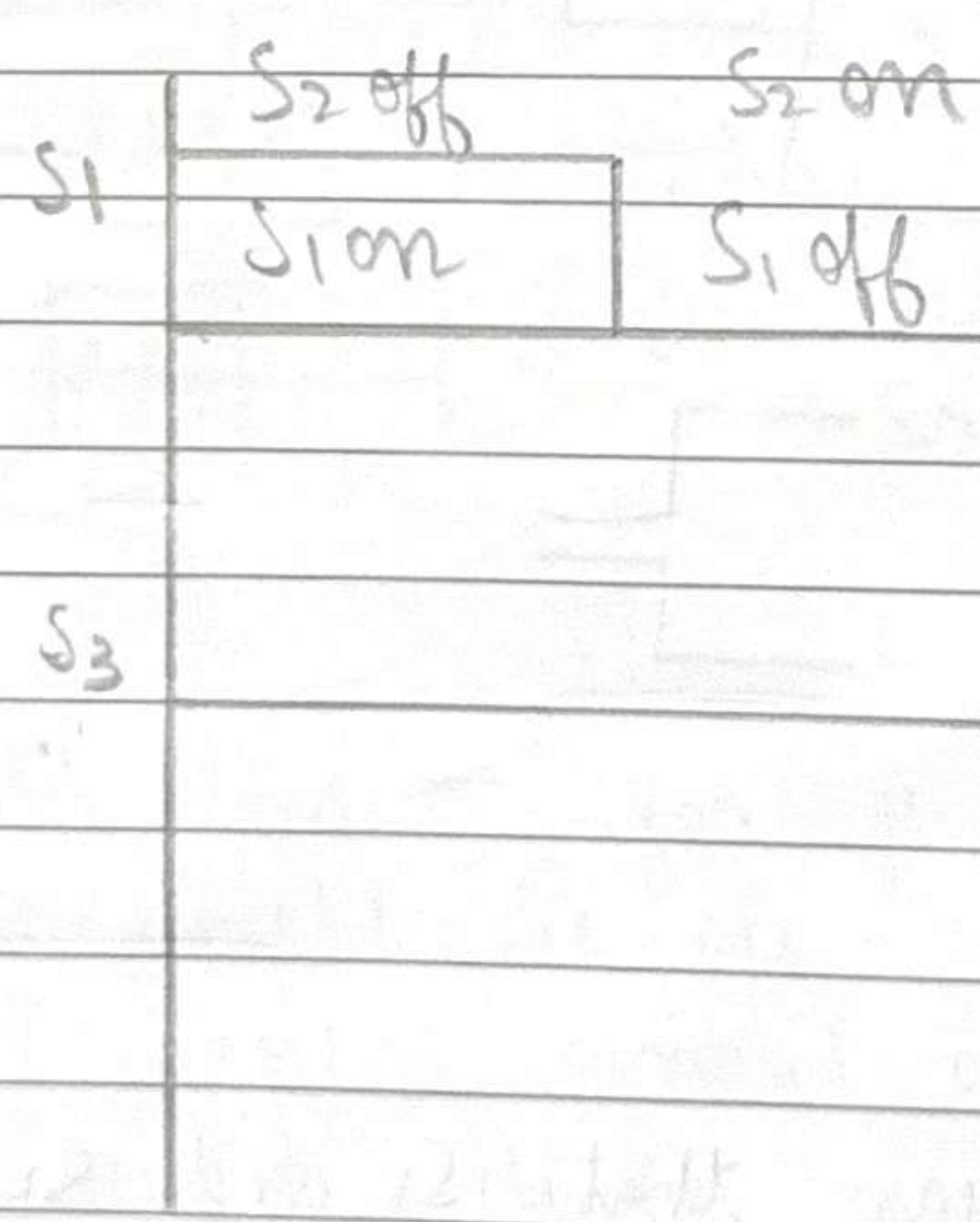
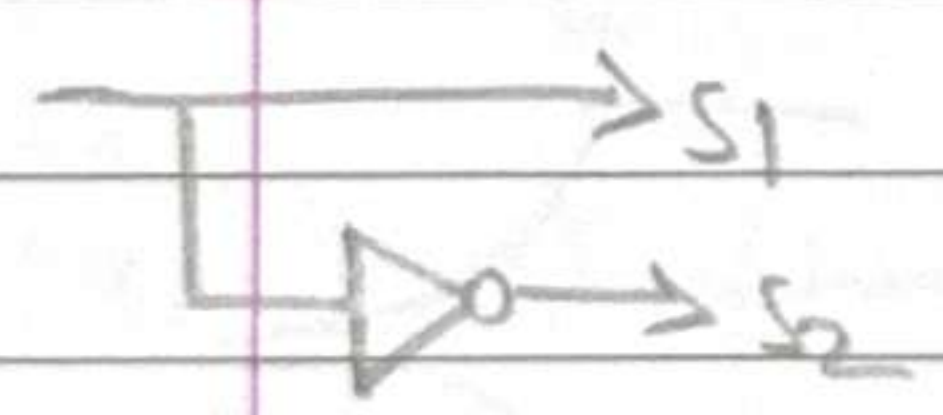
One application of inverter  
 — solar inverter

In past, majority of power generation  
 — thermal (generator)

Now shifting towards solar and wind  
 — power electronic circuits (inverter)

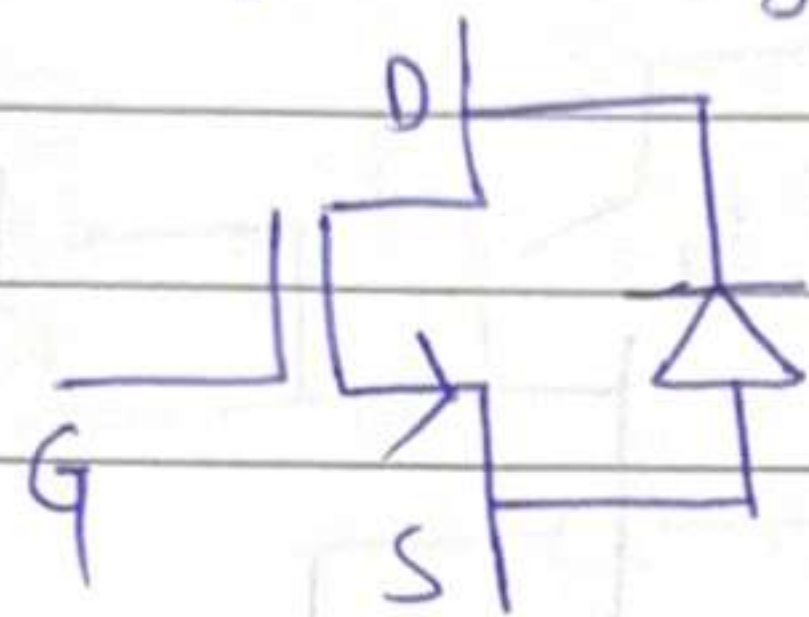
$$S_1 = \overline{S_2}$$

$$S_3 = \overline{S_4}$$

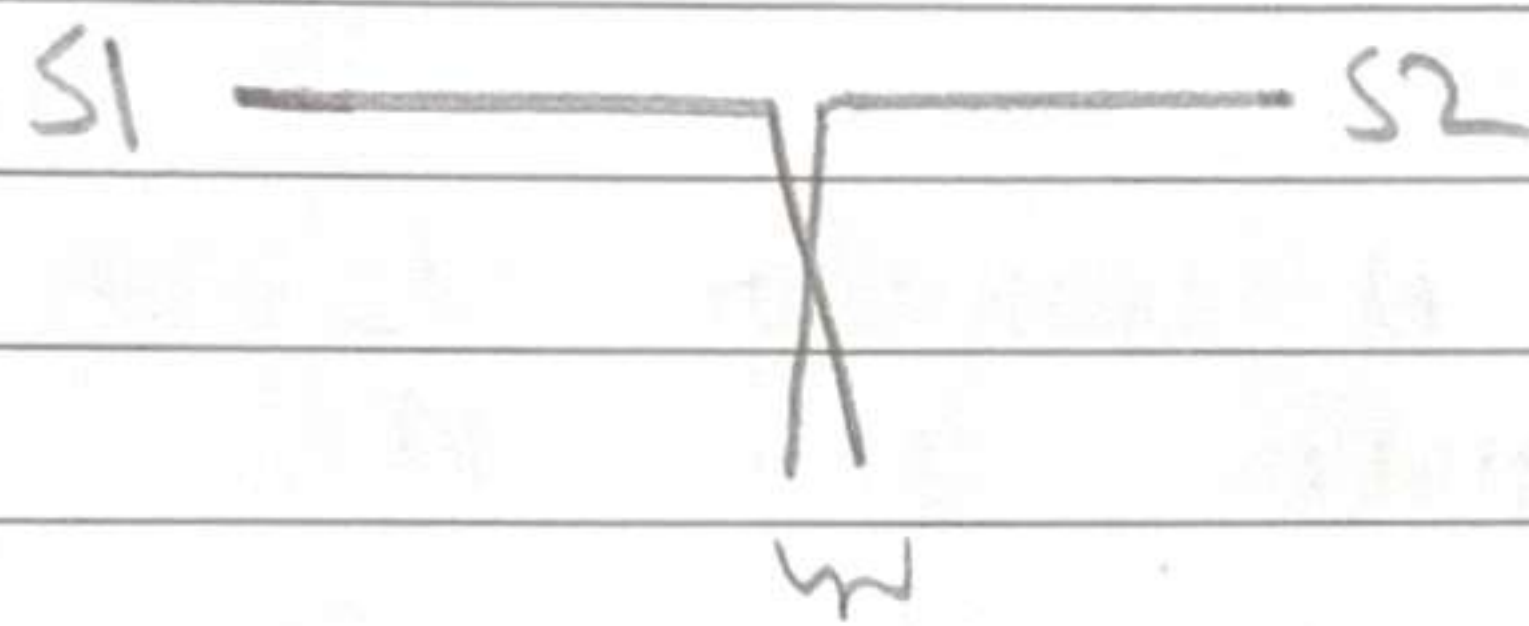




Switches are realized using MOSFETs, IGBTs etc.

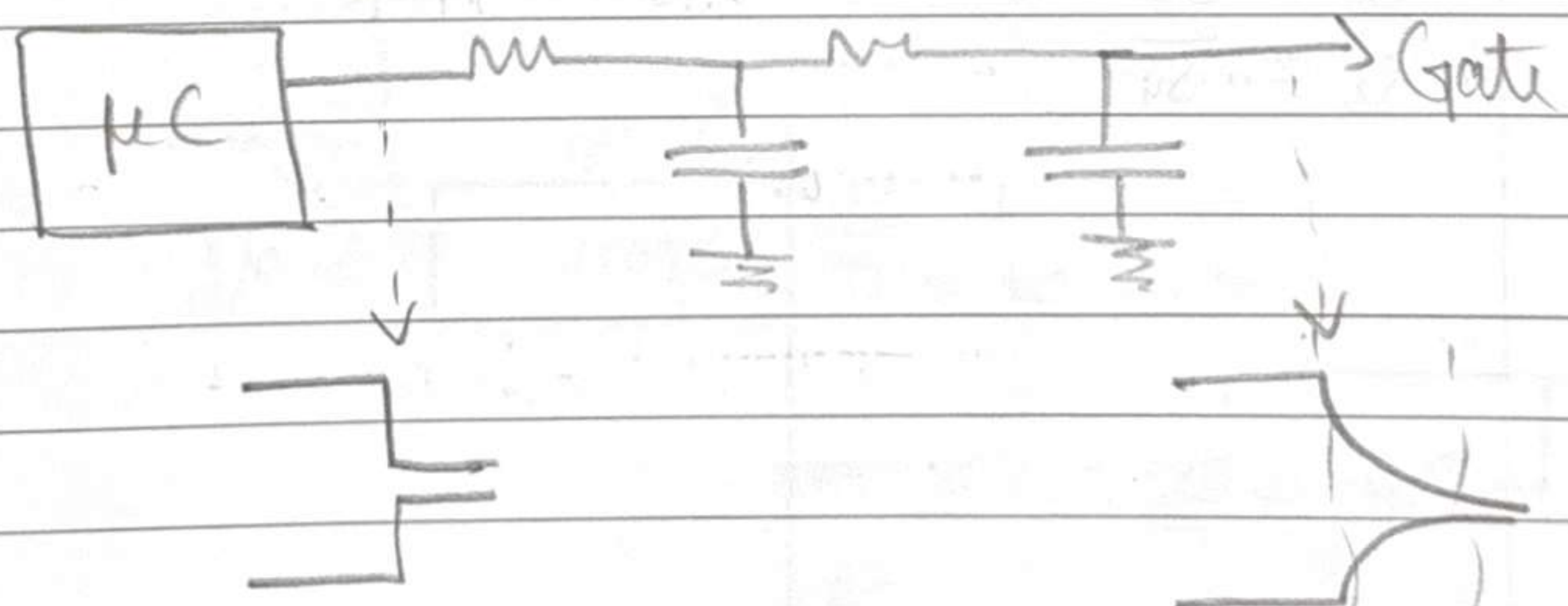


This diode is inherent in some devices. Needs to be added externally in some devices



We want the transitions of  $S_1$  and  $S_2$  to be ~~very~~ very sharp.

Practically, due to presence of  $C$  and  $L$  elements in circuit, the control signals for  $S_1$  and  $S_2$  may be distorted



possible overlap!

It might happen that  $S_1$  and  $S_2$  are both partially on at the same time  $\Rightarrow$  leading to short circuit  $\Rightarrow$  large current flow! called shoot-through problem

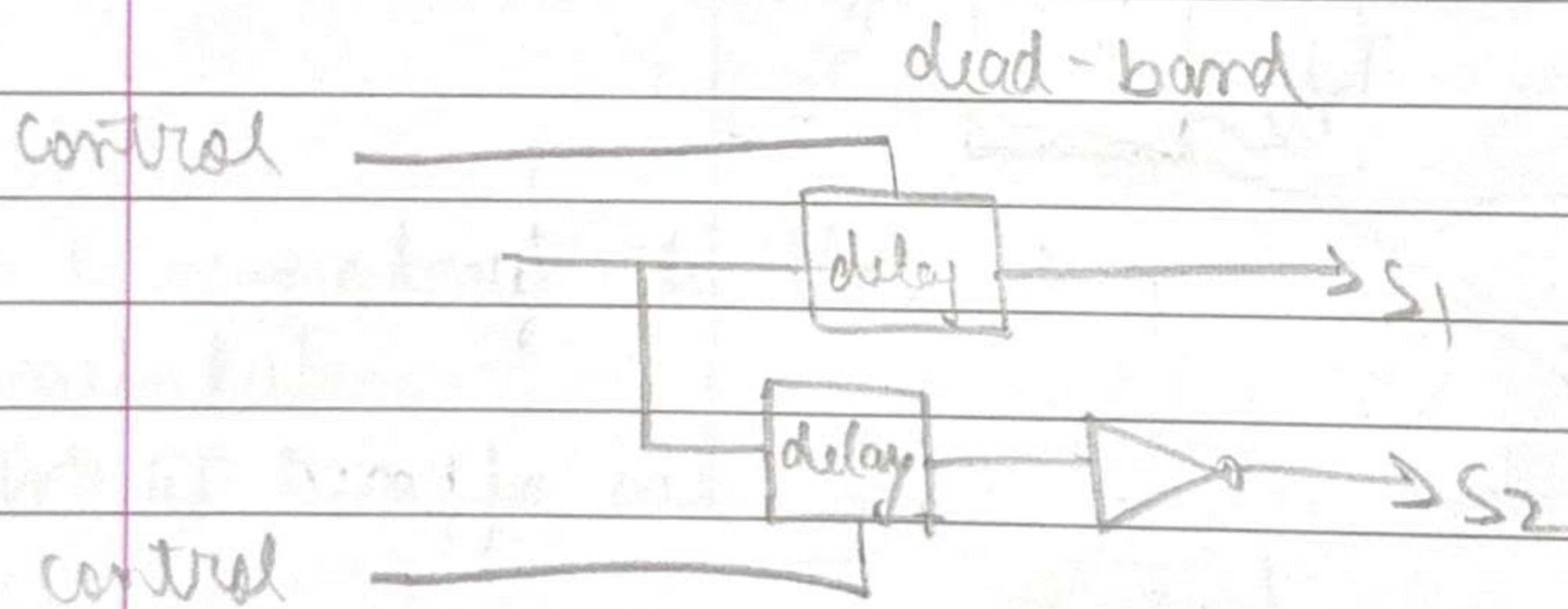


This might happen because the device (MOSFET) itself might have internal capacitances. Especially Power MOSFET has larger internal capacitances.

Solution?

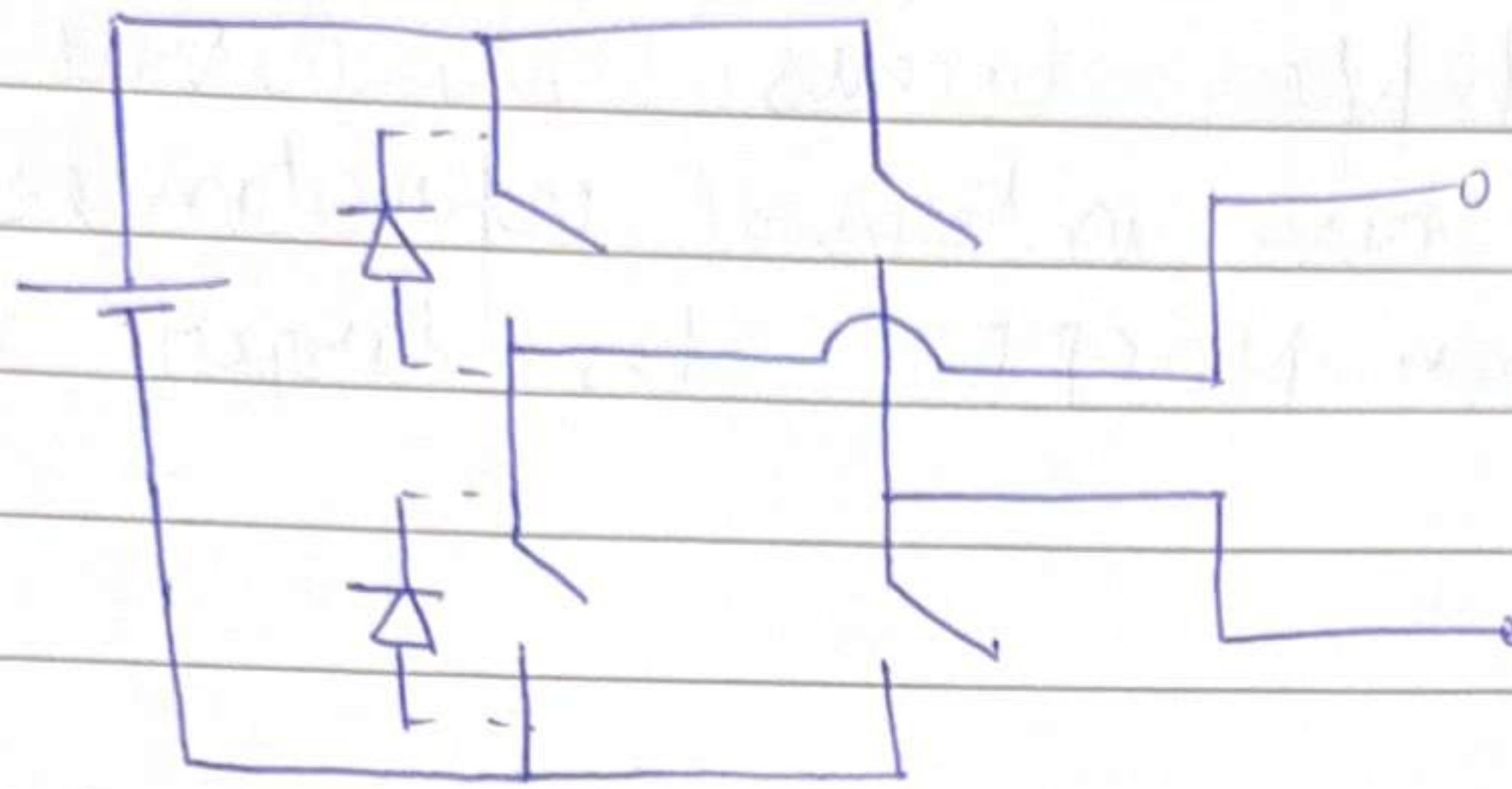
Delay the (S2) control signal a little

(whichever is turning ~~on~~ off) on)

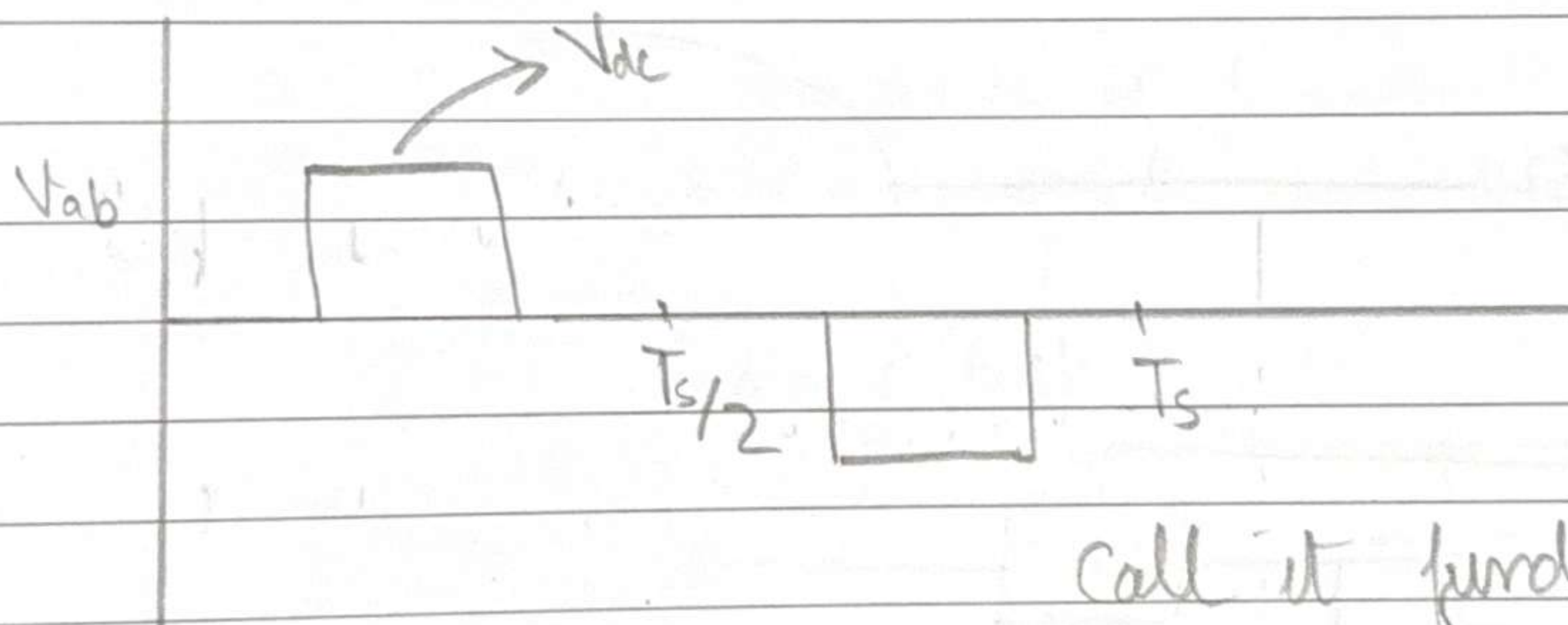


An issue with this — for just an instant, both switches might be off — inductor current cannot abruptly stop! Here the ~~di~~ switch-diode comes to the rescue! ~~the~~ Diode allows flow of current for that small duration of time, for which both switches are off.

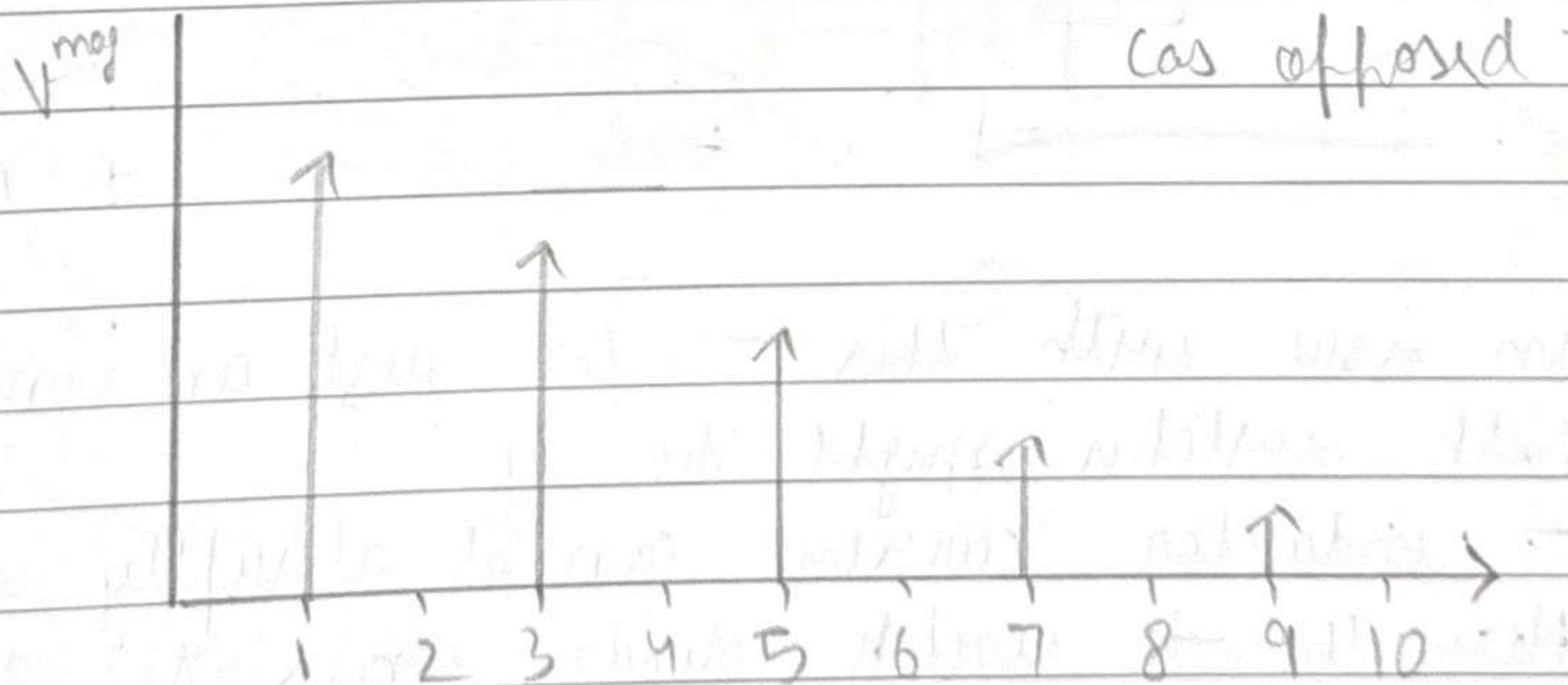




Note from the polarity of the diodes —  
 when ~~either~~ one of the switches is on,  
 none of the diodes conduct! → 'check once!'

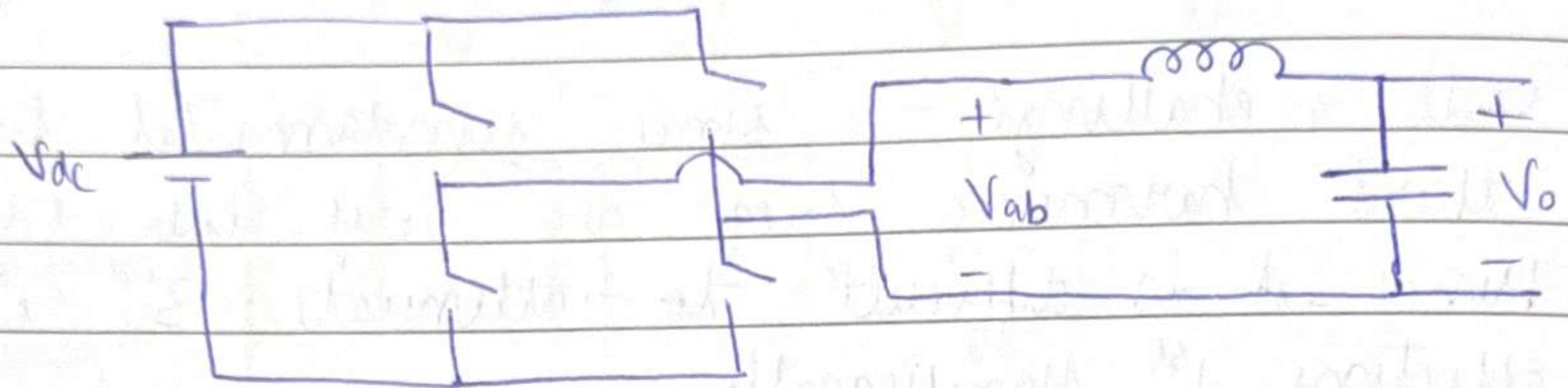


Call it fundamental  
 modulation  
 (as opposed to PWM)





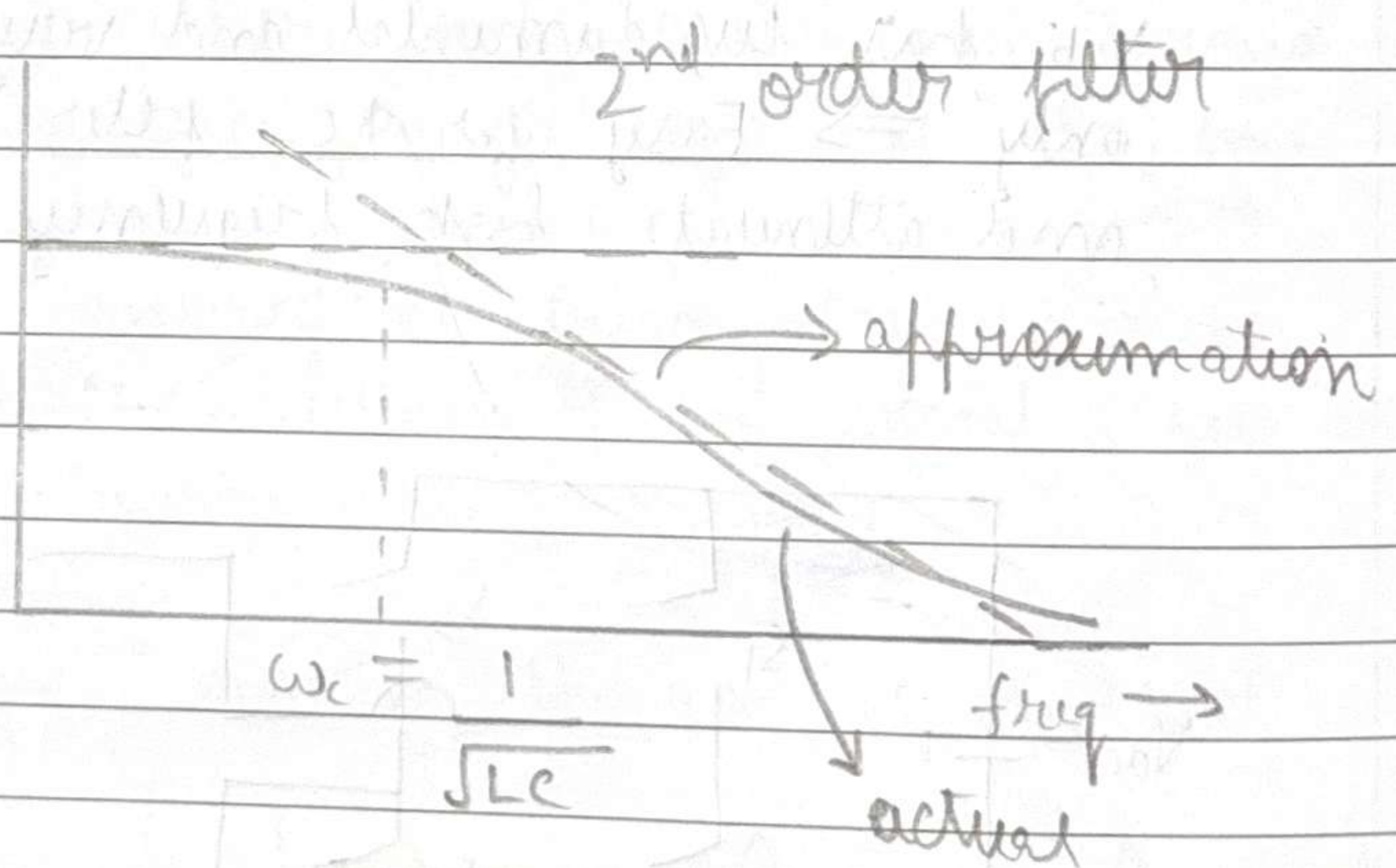
In order to ~~to~~ remove higher harmonics  
 → LPF!



$$\frac{V_o}{V_{ab}} = \frac{1}{1 + \Delta^2 LC}$$

Proof:

$$\frac{V_o}{V_{ab}} = \frac{\frac{1}{Cs}}{Ls + \frac{1}{Cs}} = \frac{1}{1 + LCs^2}$$



This will cause less attenuation to fundamental, and significant attenuation to higher harmonics



Why don't we use 1<sup>st</sup> order filter (RC)?

- Want to avoid R  $\rightarrow$  causes dissipation
- 2<sup>nd</sup> order has larger -ve roll-off slope

Still a challenge - since fundamental freq and third harmonic freq are very close by.

Hence it is difficult to attenuate 3<sup>rd</sup> without affecting 1<sup>st</sup> significantly.

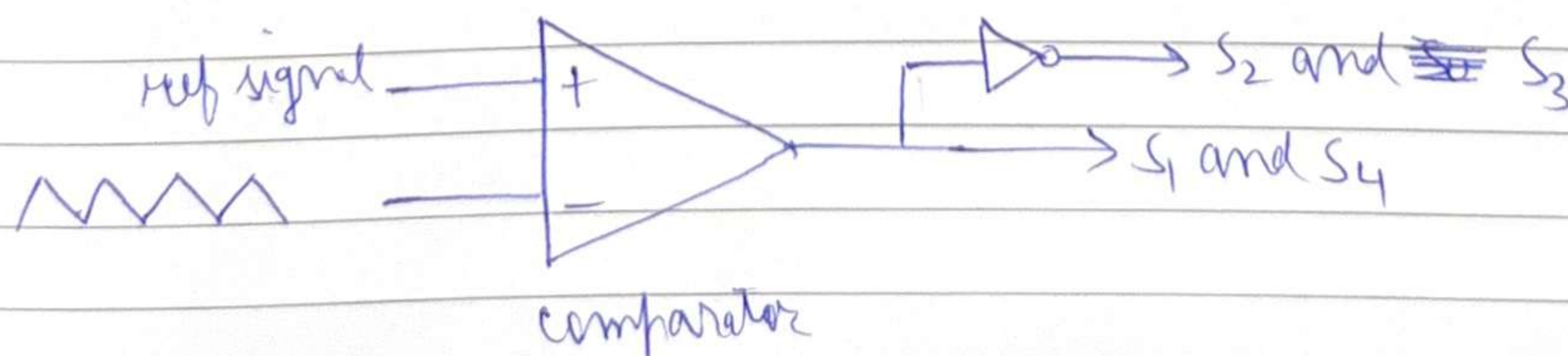
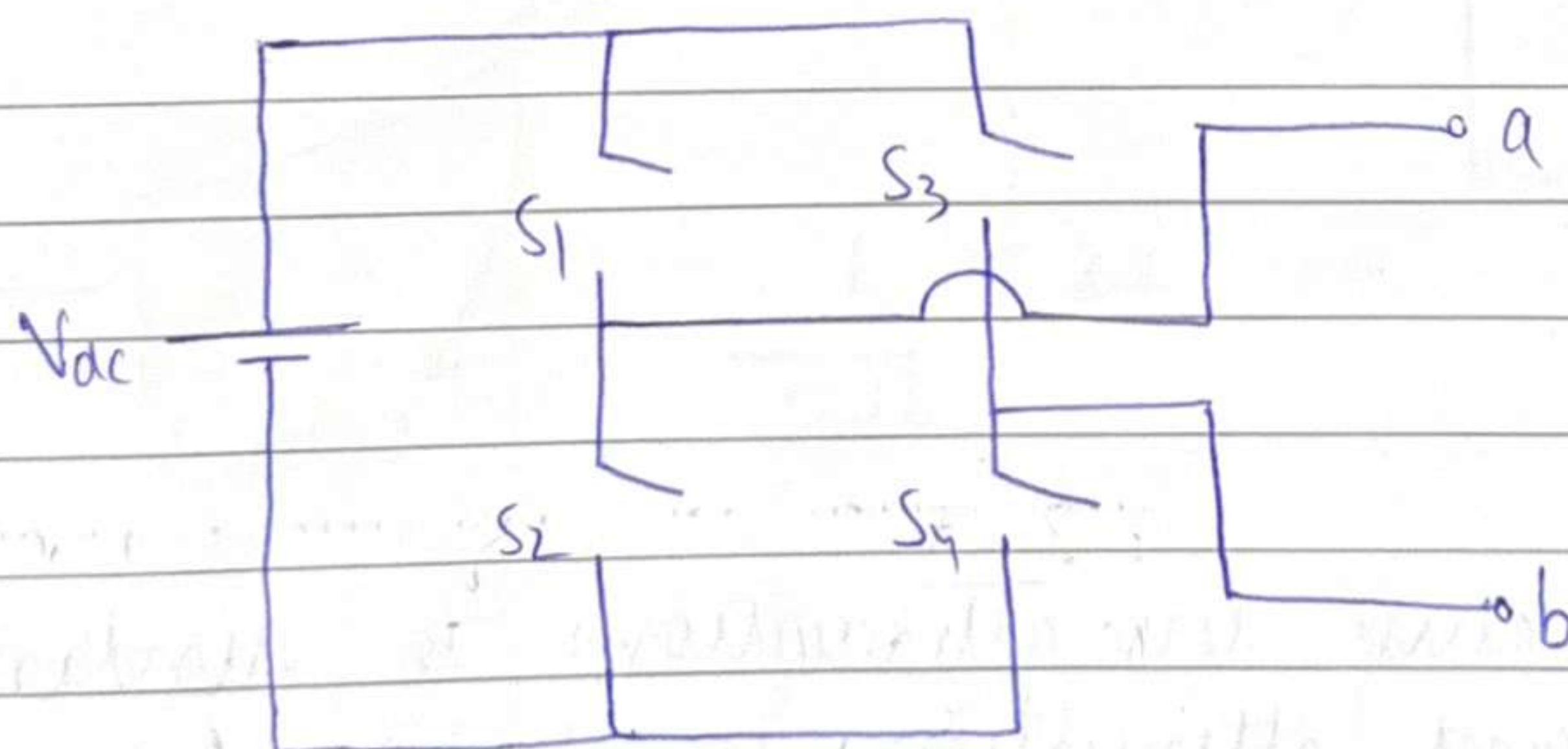
How to do it then?

PWM !!

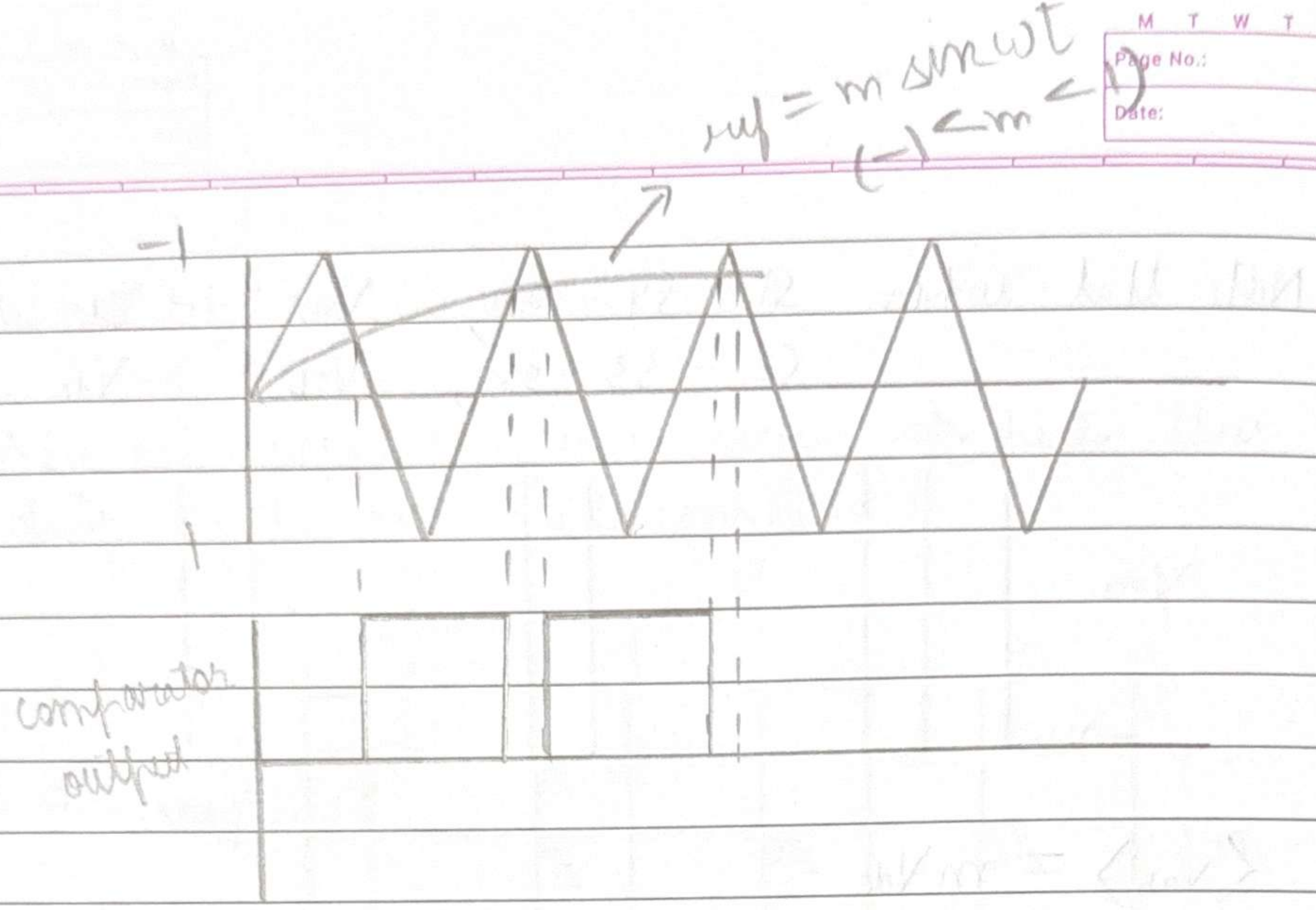
Pulse width modulation

Principle -

$V_{ab}$  has fundamental and very high frequency only  $\Rightarrow$  Easy for LC filter to separate and attenuate high frequency  $\Rightarrow$  sine output!





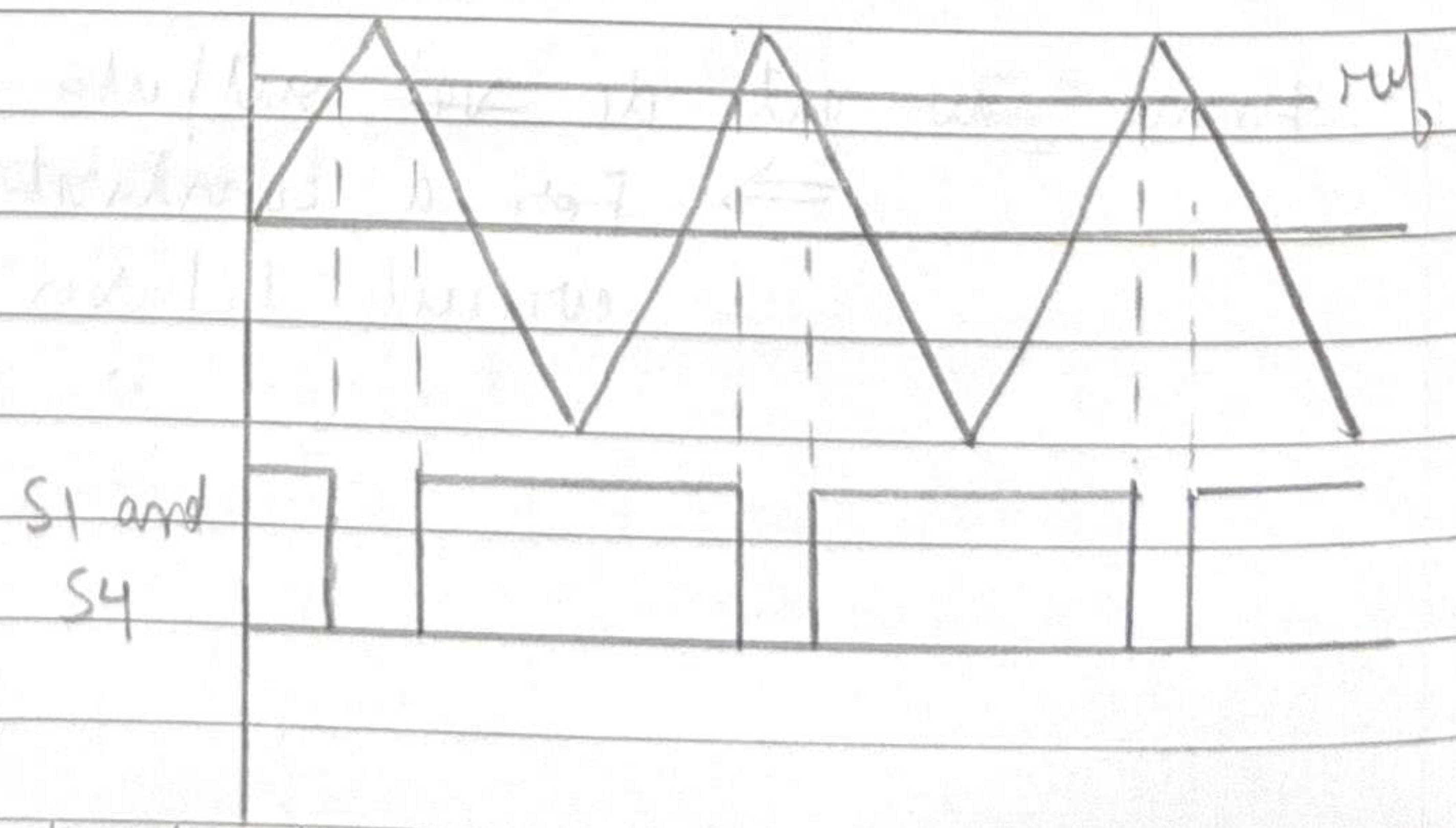


Note that each switch is turned on/off multiple times in one fundamental frequency of the system

→ Unlike earlier case (called fundamental modulation) where each switch is turned on/off only once in each cycle of the sine desired

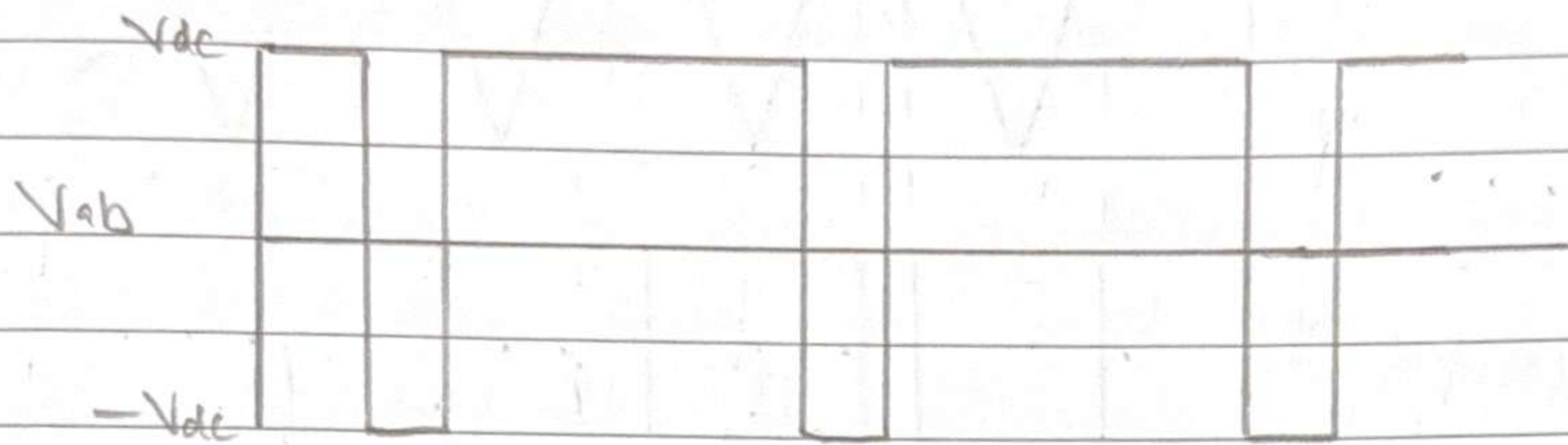
Note that frequency of triangular wave  $\gg$  frequency of ref signal (here sine)  
 (e.g.,  $f_{sine} = 50 \text{ Hz}$ ,  $f_{tri} = 10 \text{ kHz}$ )

Let us, for now, consider the case of constant ref signal  
 $r_{ref} = m$





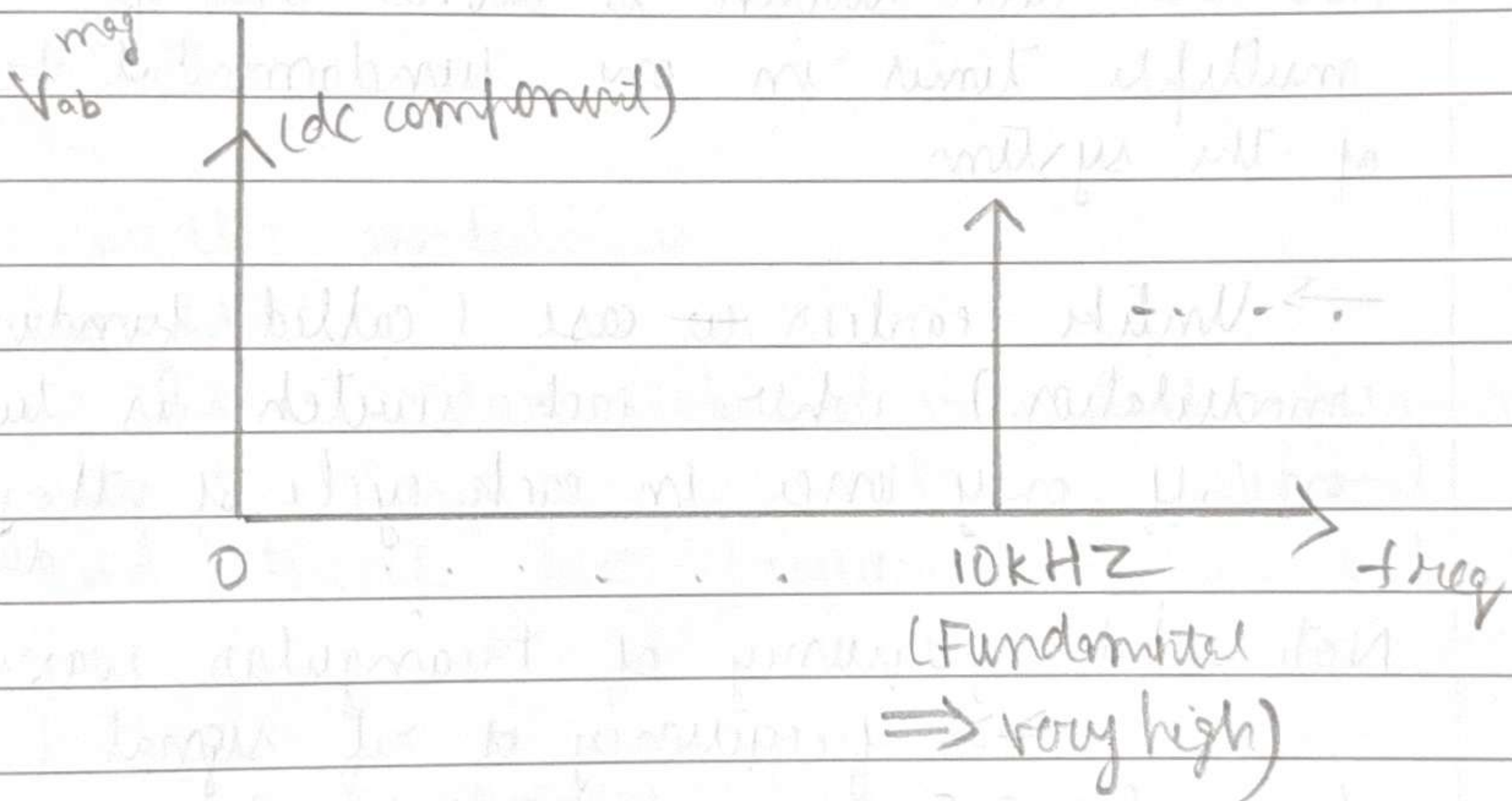
Note that when  $S1 = S4 = \text{on}$ ,  $V_{ab} = +V_{dc}$   
 $S2 = S3 = \text{on}$ ,  $V_{ab} = -V_{dc}$



Time period =  $\frac{1}{10\text{kHz}}$

$\langle V_{ab} \rangle = m V_{dc}$

$\therefore \text{Fundamental} = 10\text{kHz}$



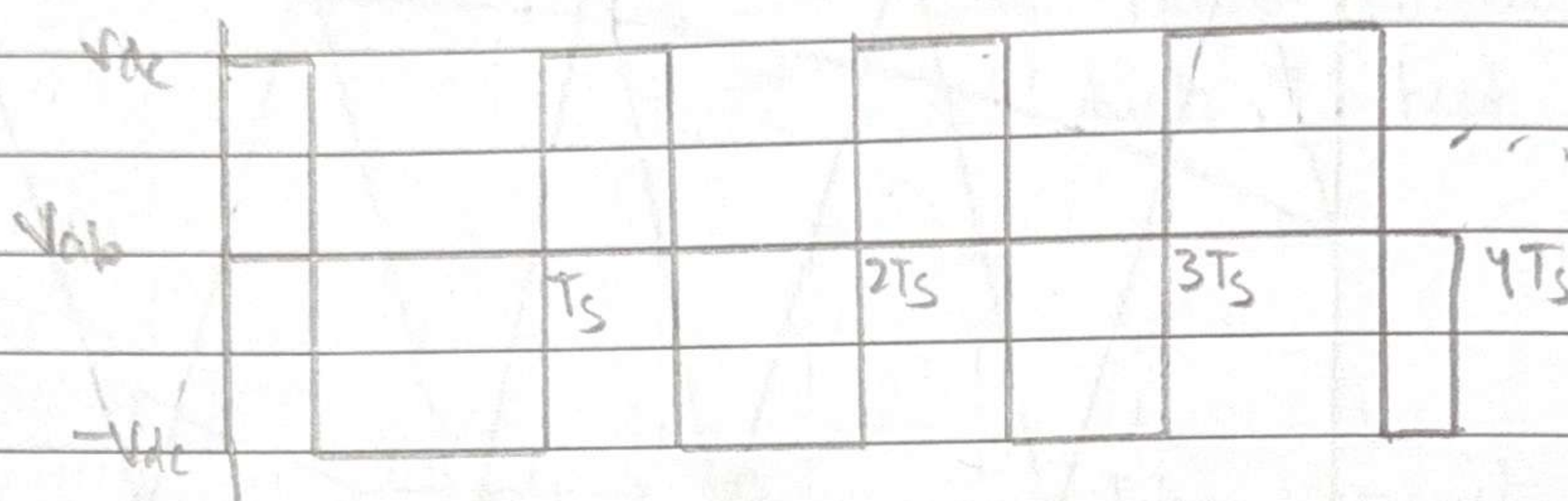
Easy for LPF to segregate dc component from 10kHz, 20kHz and so on!

Hence, we get dc ~~set~~ output  $m V_{dc}$   
 $\Rightarrow$  For a constant  $m$ ,  
 circuit behaves like a buck converter!



What if  $v_{ref} = m \sin(\omega t)$  ??

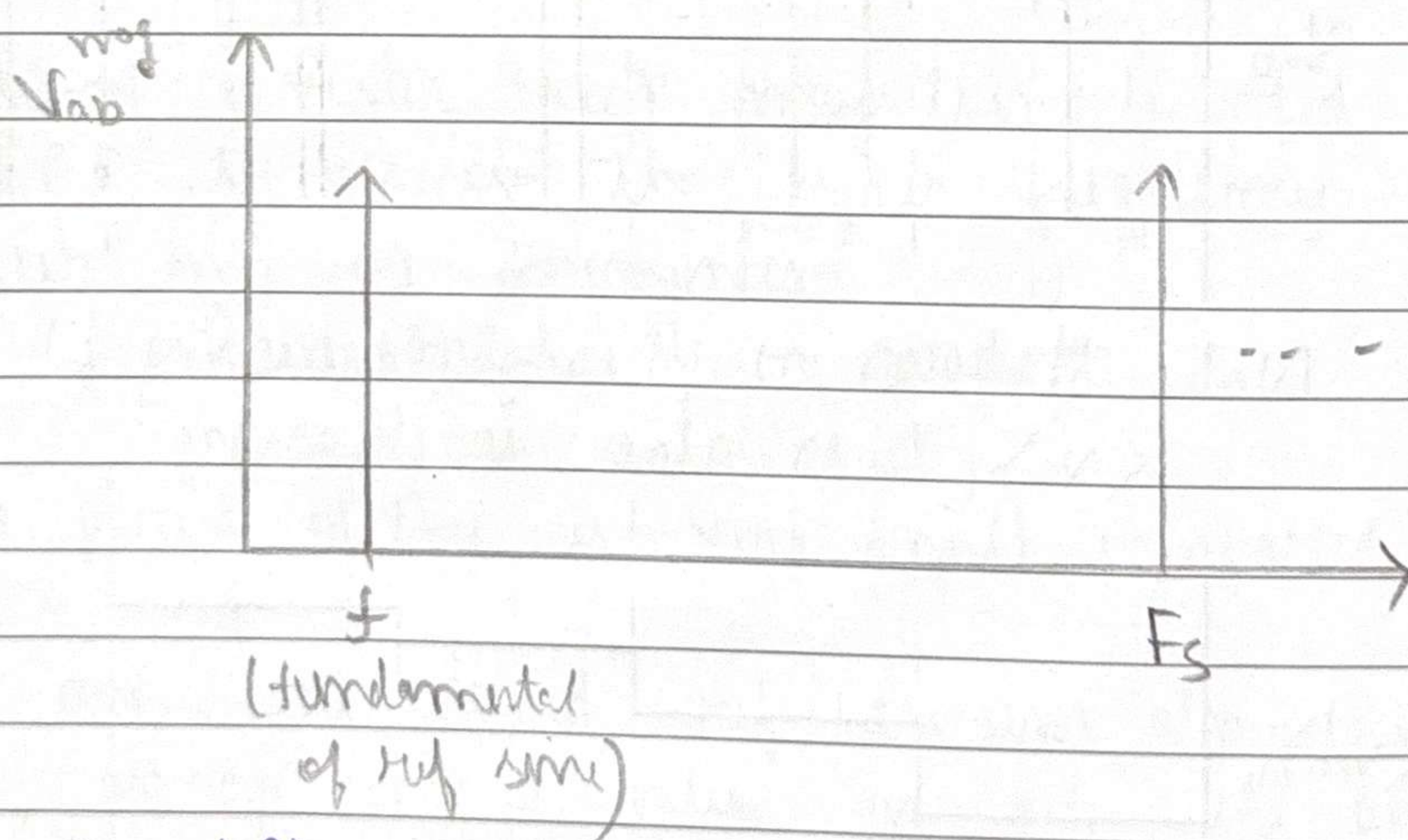
You get repeated square waves  $\Rightarrow$  but their duty cycles are not constant!



$T_s$  is the period of the triangular wave.

At some time  $t$ , the average value of  $v_{ab}$  in the  $T_s$  period containing that time is

$$\langle v_{ab} \rangle_{T_s} = m V_{ac} \sin(\omega_0 t)$$



Easy for LPF to segregate

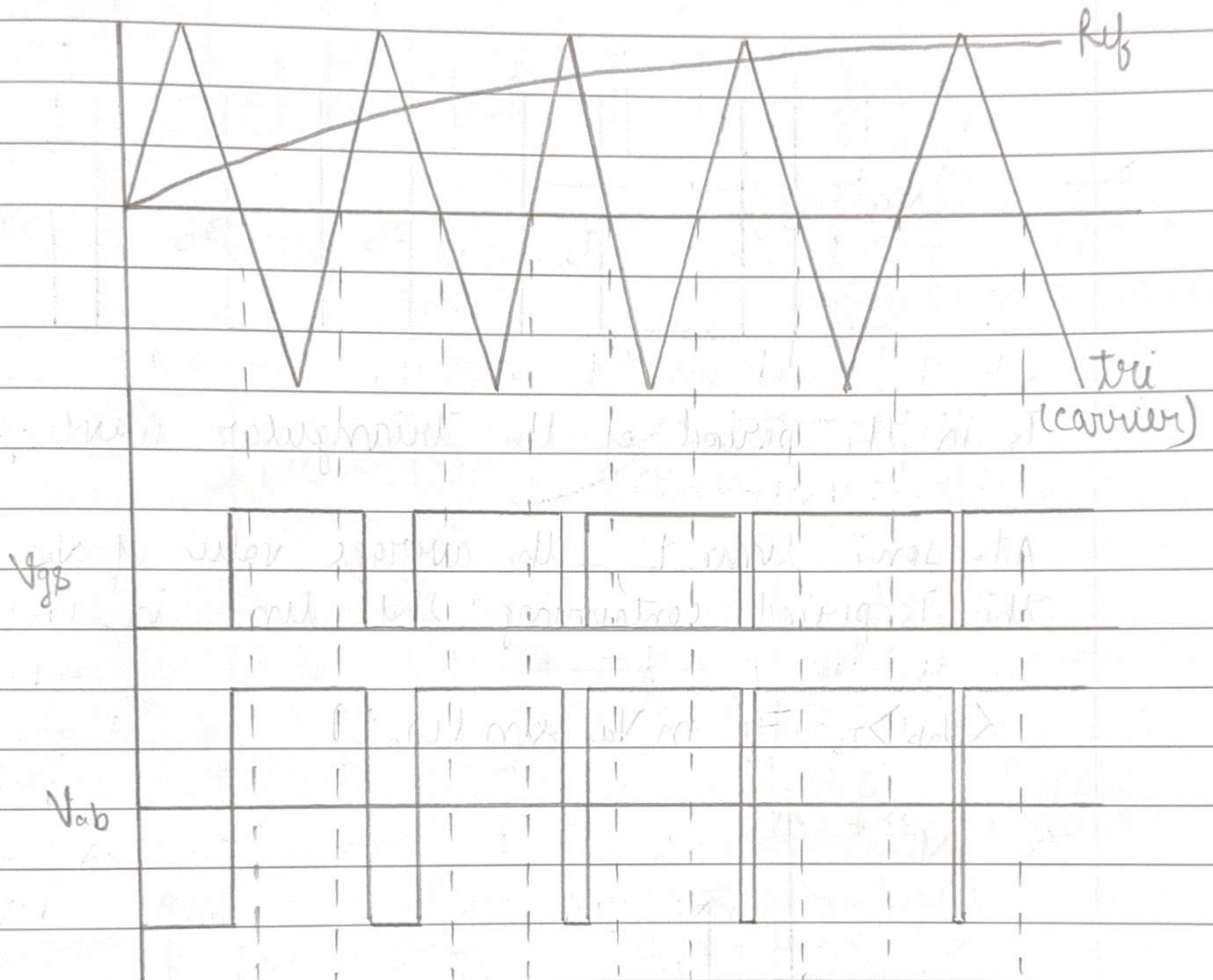
Note that harmonics of  $f$  don't appear

$\rightarrow$  difficult to prove!

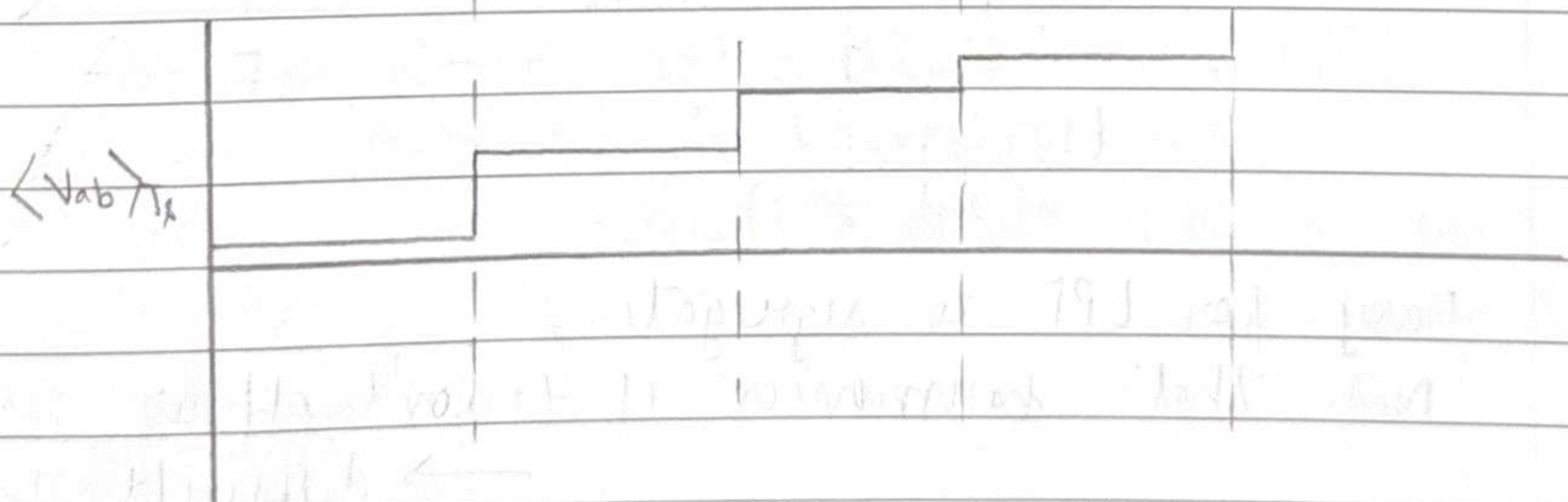
Harmonics of  $F_s$  appear



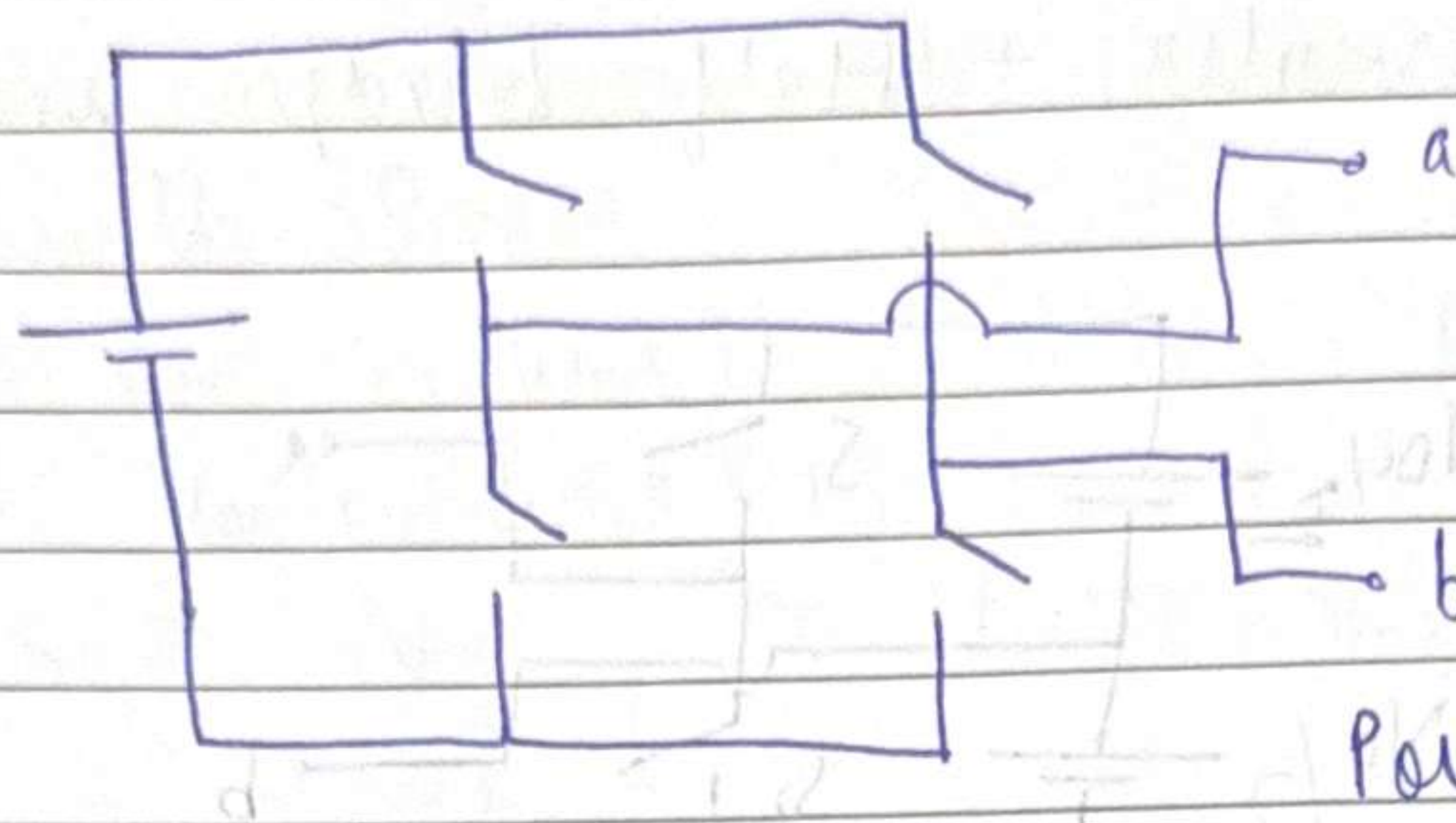
Note that  $v_{ref}$  is also a sinusoid, but output is a power output (high amplitude) but input  $v_{ref}$  is a small signal



Note that as  $v_{ref}$  is increasing,  $\langle V_{ab} \rangle_{T_s}$  is also increasing



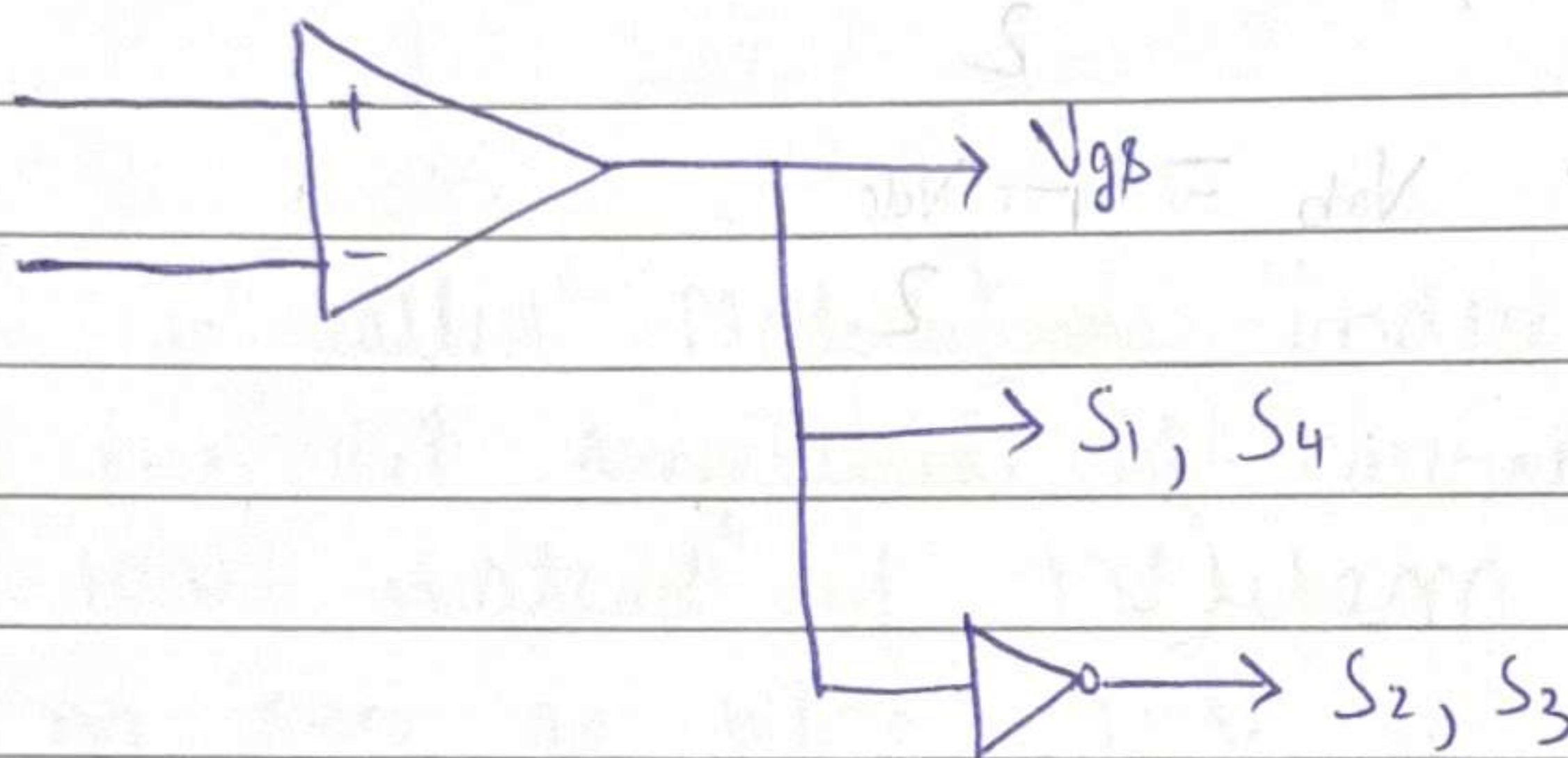




Inverter  
Power circuit  
(high V, high I)

(low freq)  
 $f_0 = 50\text{Hz}$

$\sim$  Ref  
 $\Delta$  Tri  
(Fs)  
(high freq)



This circuit is  
at signal level  
(low V, low I),  
can be  
implemented  
in analog or  
digital domain

Output of inverter ( $V_{ab}$ ) may then be fed into  
a LPF to remove the higher frequency  
carrier and its harmonics

(i.e. 'demodulate' the  $\pm$  wave)

Since period of Tri is very small compared to  
that of ref,  
in one Tri period, ref remains almost constant  
Hence, average output value in one Tri period;

$$\langle V_{ab} \rangle_{T_s} = m \cos(\omega_0 t) \times V_{oc}$$

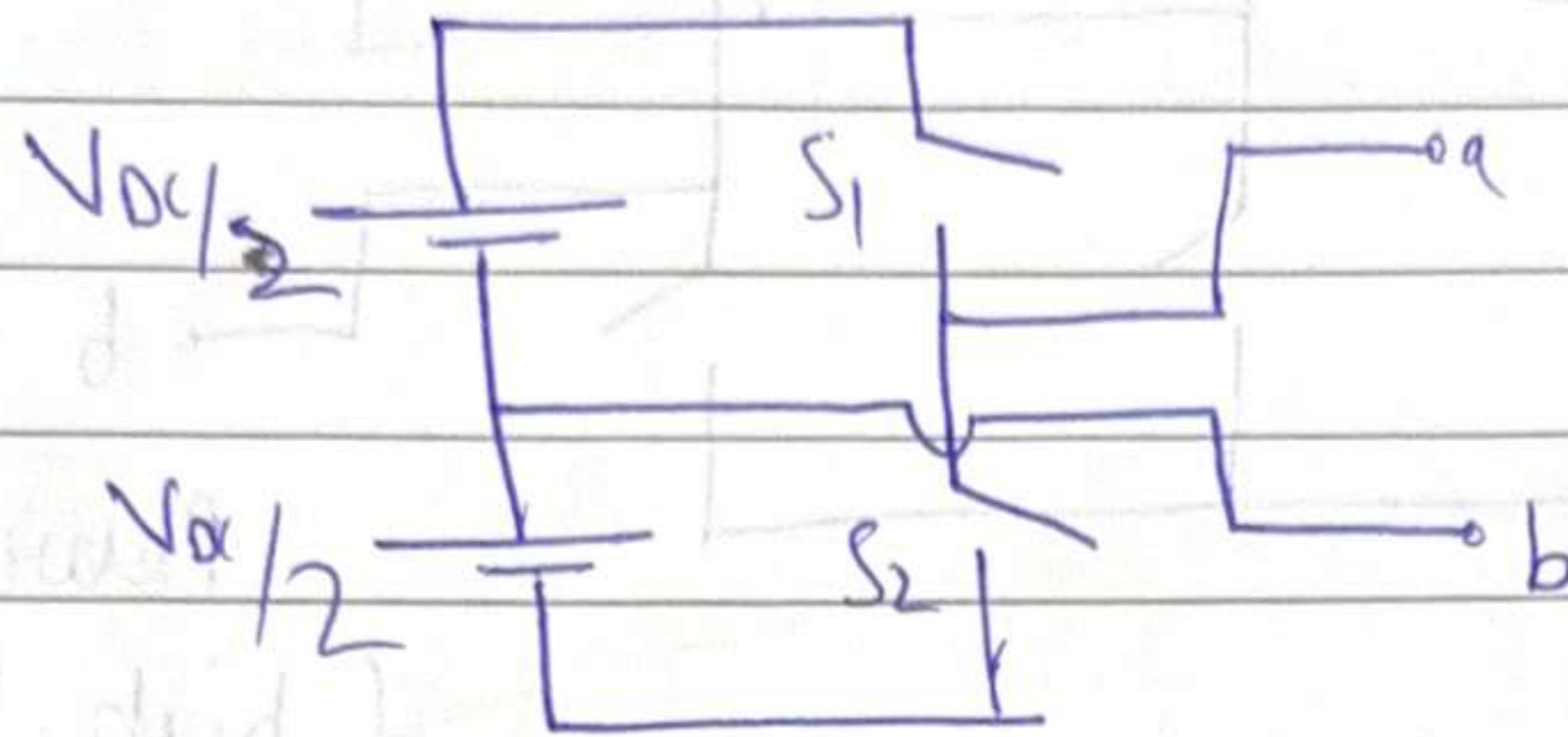
where  $\text{ref} = m \cos(\omega_0 t)$ , ( $m < 1$ )

This is called sin-triangle modulation  
or sin-pulse width modulation (SPWM)



# Try in Simulink

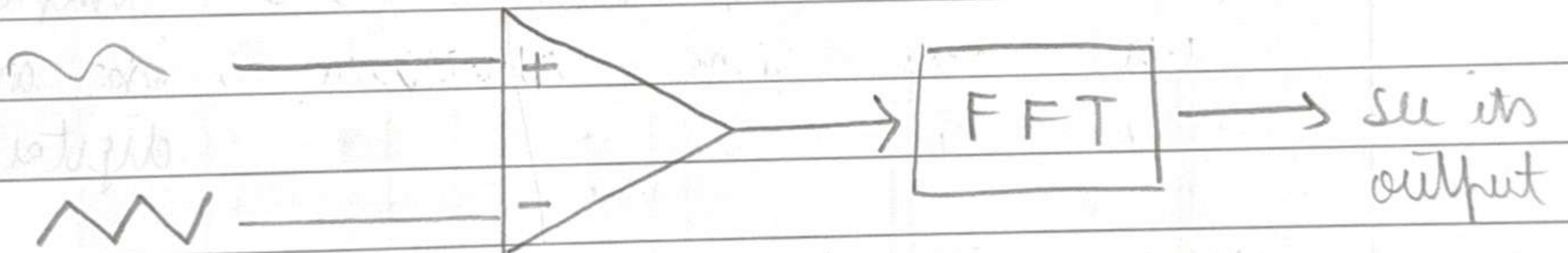
Another inverter - Half-bridge inverter



$$S_1 \text{ on} \Rightarrow V_{ab} = \frac{V_{dc}}{2}$$

$$S_2 \text{ on} \Rightarrow V_{ab} = -\frac{V_{dc}}{2}$$

Simulink model



Carrier is chosen during design stage.

The filter (LPF) is designed according to carrier (it must attenuate carrier frequency)

After design is complete, in order to fine-tune output, ref. signal is adjusted, (not carrier signal!)



Q. What is the maximum RMS value of  $V_{out}$  for

(a) H bridge with SPWM

$$V_{out} = m V_{oc} \cos(\omega t) \quad (m \leq 1)$$

$$(V_{out})_{max} = V_{oc} \cos(\omega t)$$

$$\therefore \text{Max RMS} = \frac{V_{oc}}{\sqrt{2}} = 0.707 V_{oc}$$

(b) Half-bridge with SPWM

$$\text{Max RMS} = \frac{V_{oc}}{2\sqrt{2}}$$

$m$  is called Modulation index

If we let amplitude of carrier ( $V_c$ ) be

10V instead of 1V,

$m$  can go upto 10V

But ~~that~~ after proportionally scaling up  $m$ ,

$V_{out}$  won't change

(c) H-bridge with fundamental modulation

$$\text{Max RMS} = \text{Fundamental RMS}$$

$$= \frac{4V_{oc}}{\pi\sqrt{2}}$$

$$= \frac{2\sqrt{2}}{\pi} V_{oc} = 0.9 V_{oc}$$

We get higher max output for fundamental modulation than SPWM!! But<sup>in</sup> fund. mod., difficult to remove harmonics!!

Which is better? — depends on application

— For low cost application, where it is not required to get very perfect sine wave (low  $V_{thd}$ ) fundamental modulation is used

⇒ E.g., home inverters ⇒ Humming noise



- For applications that require very precise sine output (low  $V_{THD}$ ), SPWM is used (costlier)  
 $\Rightarrow$  E.g., UPS in offices etc.

Note

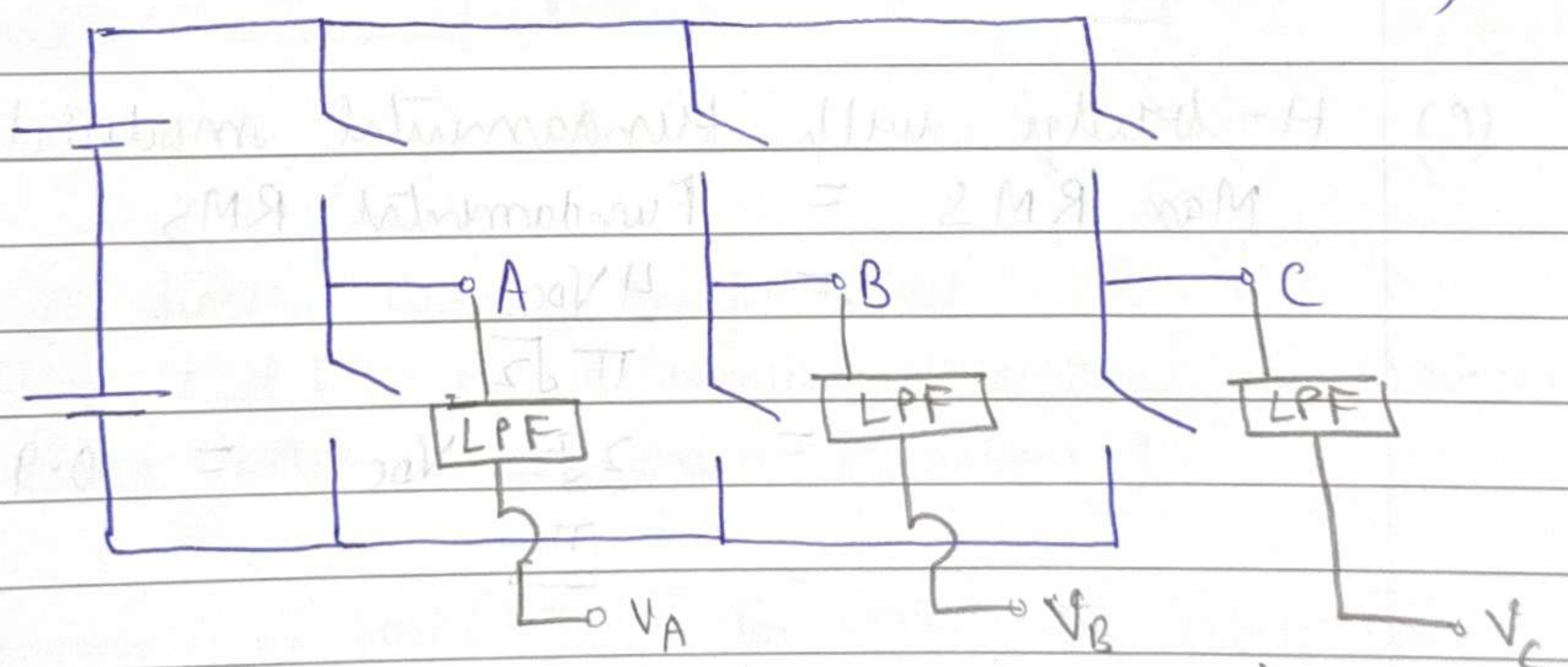
Can we apply fundamental modulation to Half-bridge inverter?

No, because we can't get a zero state in half-inverter

So, we can't adjust magnitude of fundamental component

$\Rightarrow$  can't adjust output magnitude

Three-phase inverter (Half-bridge inverter with SPWM)



We want  $V_A$ ,  $V_B$ ,  $V_C$  to have same magnitude, but  $120^\circ$  phase shifts

$$R_{refA} = m \cos(\omega t)$$

$$R_{refB} = m \cos(\omega t - 120^\circ)$$

$$R_{refC} = m \cos(\omega t + 120^\circ)$$



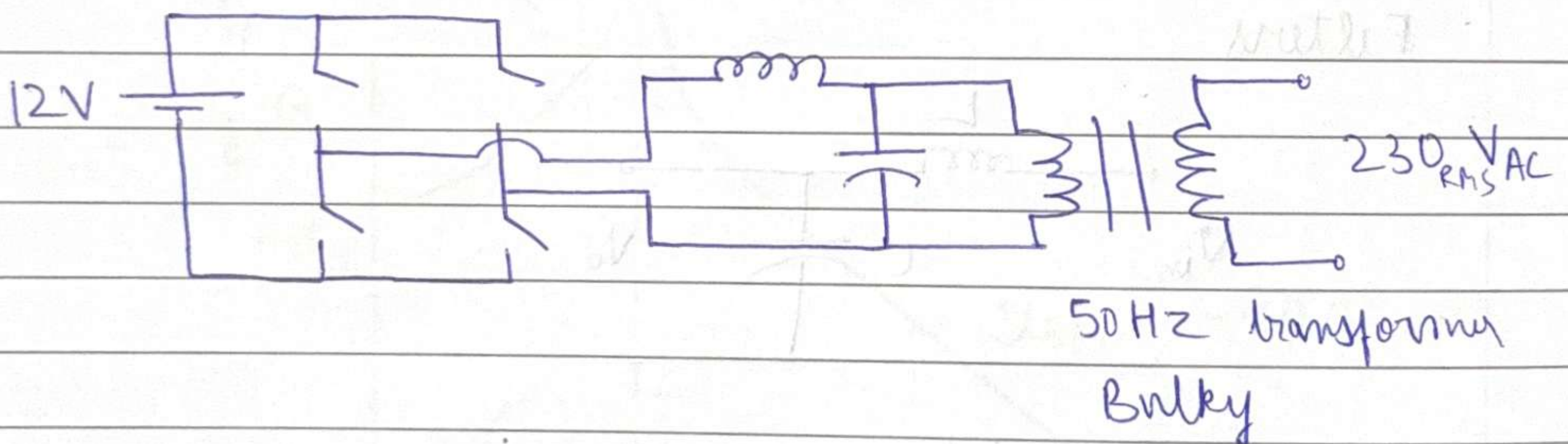
## Applications of Inverters

Few years back, when microcontroller's were expensive, and not very fast, it was difficult to realize the comparators circuit.

∴ In older times, ~~an~~ home inverters were mostly using fundamental modulation, which gave square wave output before LPF.

Nowadays, inverters with both options - fund. mod. and SPWM.

### (a) Home inverters

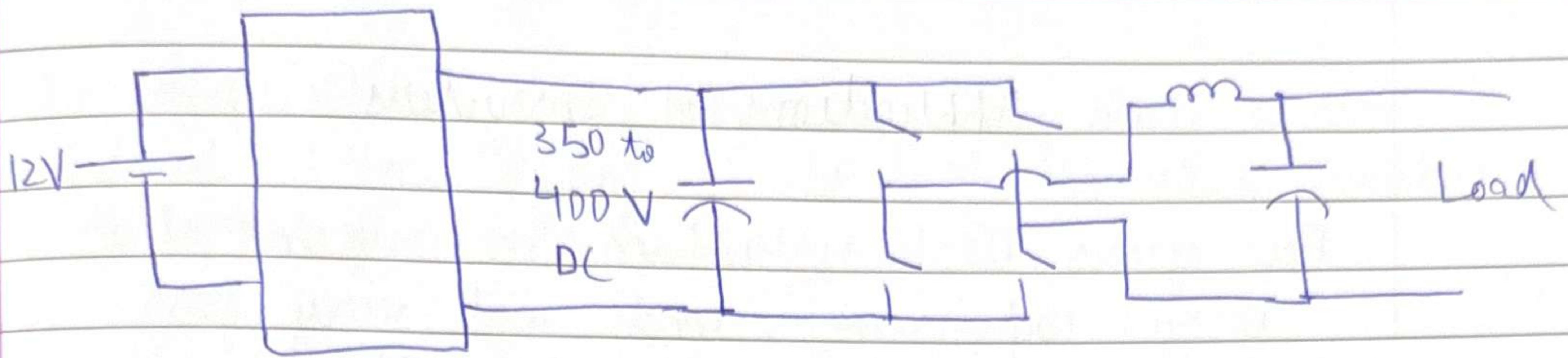


In older home inverter, low freq (50 Hz) transformer used  $\Rightarrow$  very bulky  
 Transformer took most of the volume of the inverter home

Better solution,

Boost on DC side rather than amplify on AC side



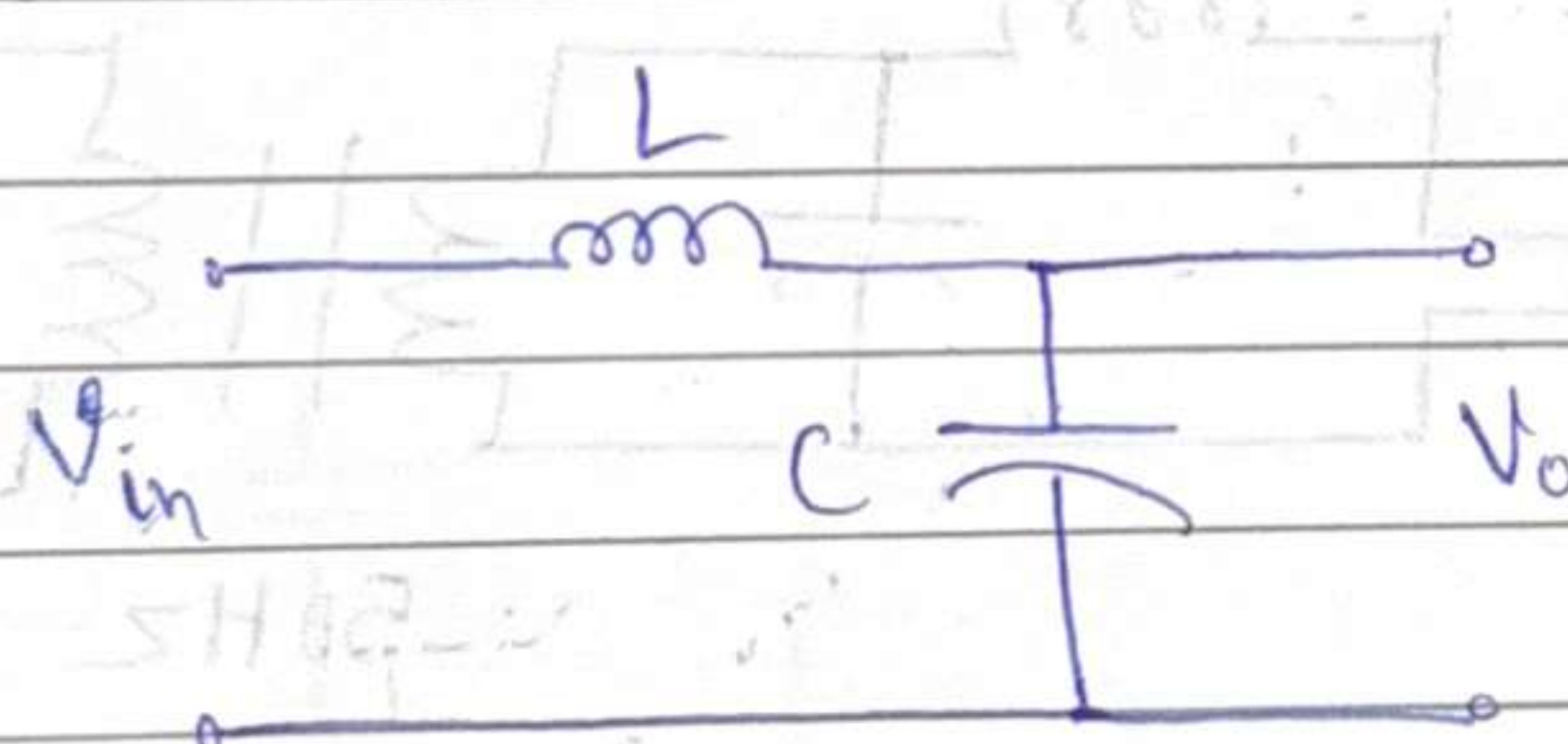


↓  
 High frequency  
 isolated dc/dc  
 converter  
 like PSFB

High  $f$  Transformer less bulky than  
 low  $f$

-3-26

## Filters



$$V_o = V_{in} \frac{1/j\omega C}{j\omega L + 1/j\omega C}$$

$$V_{in}$$

$$\therefore \text{Gain, } G = \frac{1}{1 - \omega^2 LC}$$

When plotting the gain magnitude, we typically  
 plot gain in dB,  $v/s$ ,  $\log \omega$

$$\text{Gain in dB} = 20 \log_{10} G$$

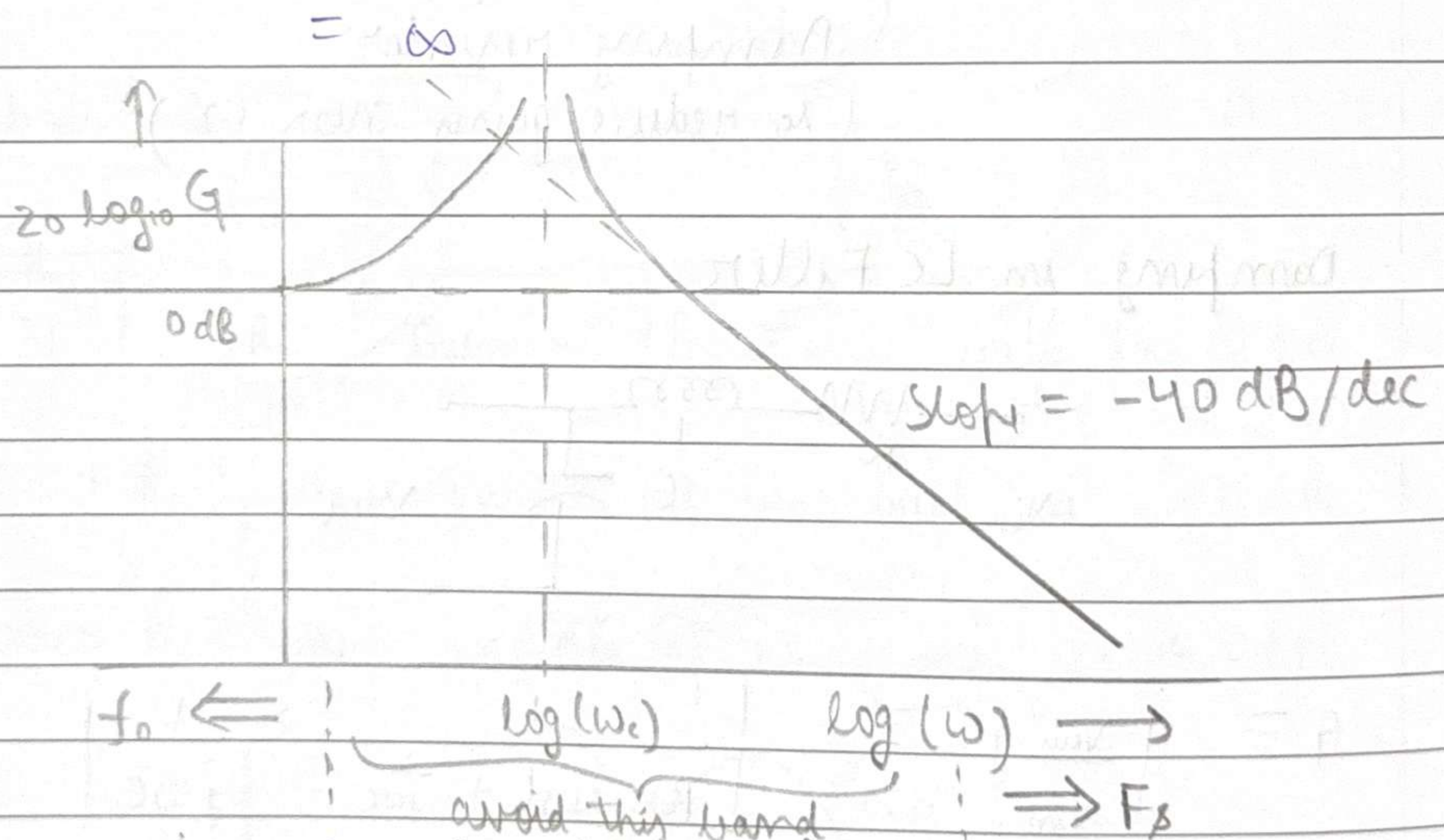


Here, Gain in dB =  $20 \log \left| \frac{1}{1 - \omega^2 LC} \right|$   
 $= -20 \log |1 - \omega^2 LC|$

When  $\omega$  is very small,  
 Gain in dB  $\approx -20 \log(1) = 0$

When  $\omega$  is very large,  
 Gain in dB  $\approx -20 \log(\omega^2 LC)$   
 $= -20 \log(LC) - 40 \log(\omega)$

When  $\omega = \frac{1}{\sqrt{LC}}$   
 Gain in dB =  $-20 \log \left( 1 - \frac{LC}{LC} \right) = -20 \log(0)$   
 $= \infty$



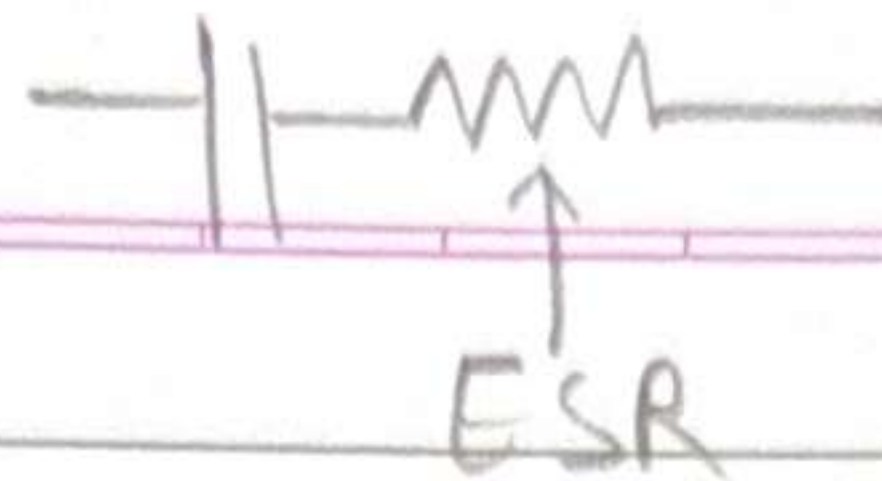
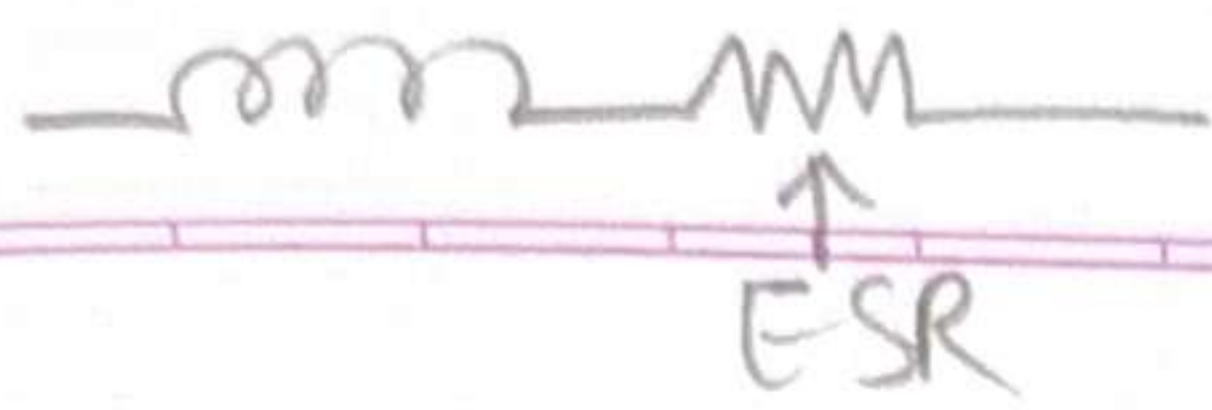
When using this circuit to filter PWM output, we keep  $f_0$  in lower end, and  $F_s$  on higher end.

We want to avoid any component of signal in middle band with very high gain.

Why? Gain very sensitive on  $\omega$  value and on  $L, C$ . Will show large variation with small changes in  $\omega, L, C$ .



## ESR - equivalent series resistance



Also, if any stray frequency component (maybe very small) comes near  $\omega_c$ , it will get largely amplified, giving undesired output.

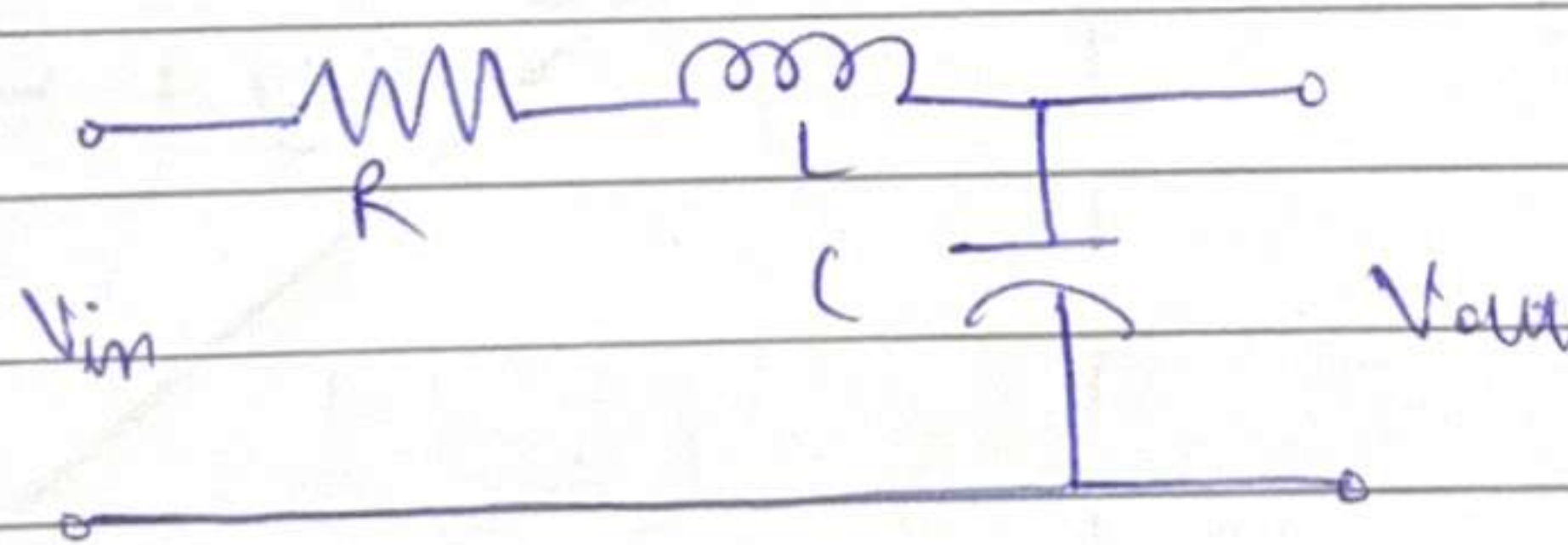
Hence,

- $F_s$  usually  $\gg f_c (= \frac{\omega_c}{2\pi}) \Rightarrow$  high attenuation to  $F_s$
- $F_o$  usually  $\ll f_c \Rightarrow$  To get unit-gain and predictable gain (not hyper-sensitive to experiment conditions)
- Avoid any frequency near  $f_c$  (resonant frequency)

+

Damping resistor  
(to reduce gains near  $\omega_c$ )

## Damping in LC Filter



$$\begin{aligned}
 G &= \left| \frac{V_{out}}{V_{in}} \right| = \left| \frac{1}{R + j\omega L + \frac{1}{j\omega C}} \times \frac{1}{j\omega C} \right| \\
 &= \left| \frac{1}{(1 - \omega^2 LC) + j\omega RC} \right| \\
 &= \frac{1}{\sqrt{(1 - \omega^2 LC)^2 + (\omega RC)^2}}
 \end{aligned}$$

$$\text{Gain in dB} = 20 \log G$$



$$\text{Gain in dB} = \frac{-20}{2} \log |(wRC)^2 + (1 - w^2LC)^2|$$

When  $w$  is very low,  
 Gain in dB  $\approx -10 \log(1) = 0$

When  $w$  is very high,  
 Gain in dB  $\approx -10 \log(w^4 L^2 C^2)$   
 $= -20 \log(LC) - 40 \log(w)$

When  $w = \frac{1}{\sqrt{LC}}$ ,  
 Gain in dB  $= -10 \log \left| \frac{R^2 C}{L} + 0 \right|$   
 $= -20 \log \left( R \sqrt{\frac{C}{L}} \right)$

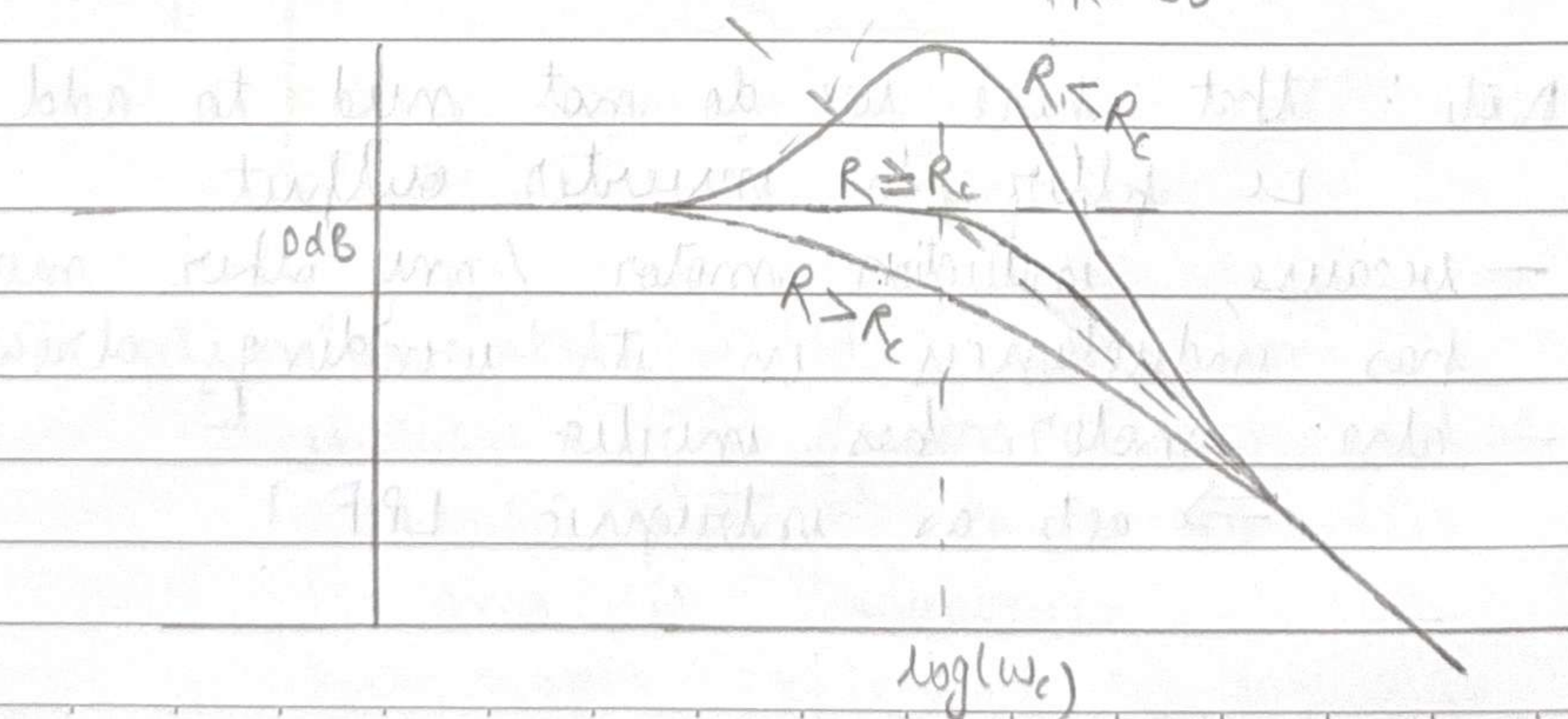
Define  $R_c := \sqrt{\frac{L}{C}}$

~~If  $R_c$~~

If  $R = R_c$ , Gain in dB at  $w_c$  will be 0 dB

If  $R < R_c$ , Gain in dB at  $w_c$  will be  $> 0$  dB

If  $R > R_c$ , Gain in dB at  $w_c$  will be  $< 0$  dB





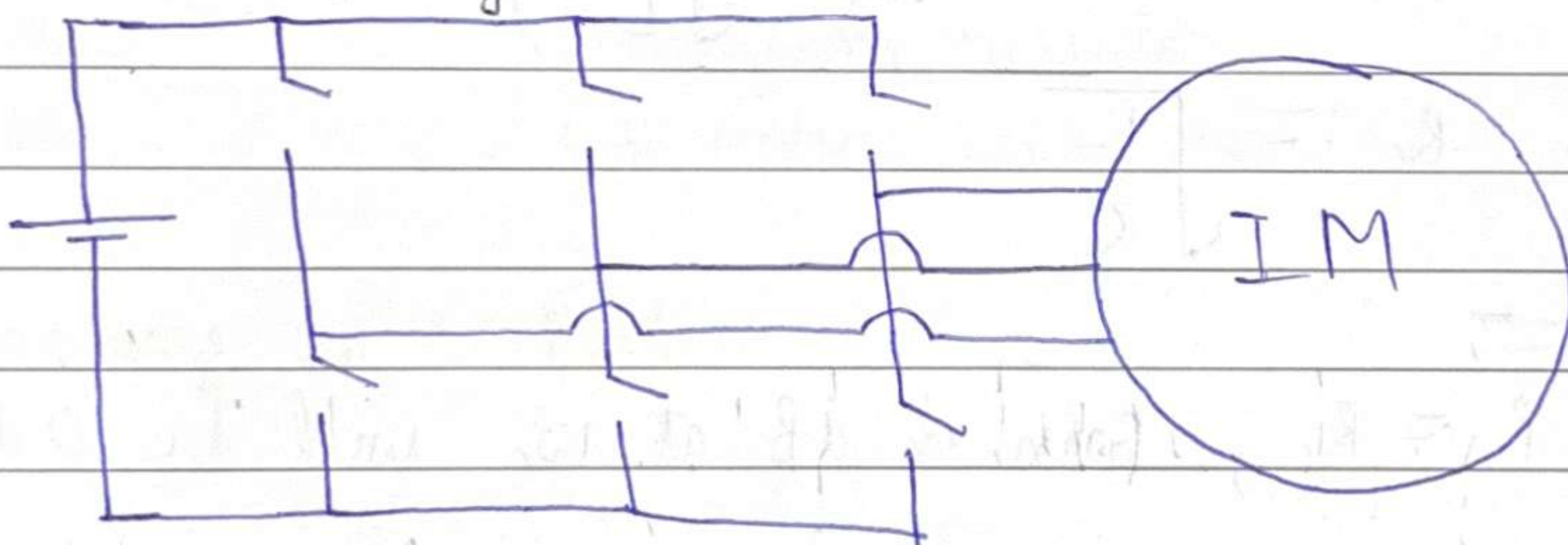
We don't want  $R = 0$ , but we also want small  $R$  to minimize  $I^2R$  loss. ( $R$  in series)

So we want some proper choice of  $R < R_c$ , but not too small.

In practice, the coil resistance <sup>(ESR)</sup> of inductor itself may provide sufficient damping, or we may have to add an additional physical resistor in series.

## Applications of Inverter circuits (continued -)

(b) Induction / any other motor drive



We can set the magnitude and frequency, both, of the inverter's output  $(V/f \text{ control})$  and phase

Note: that here we do not need to add an LC filter to inverter output

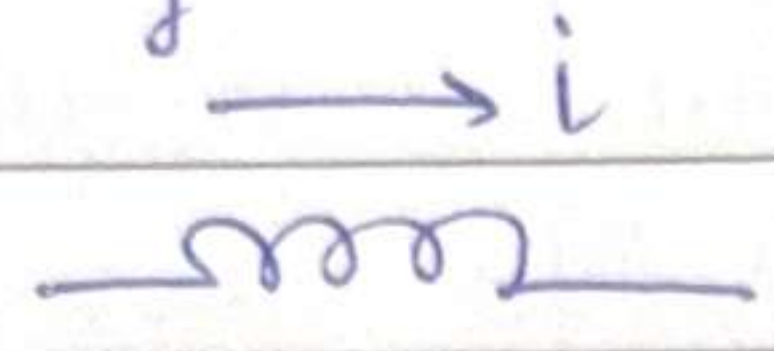
— because, induction motor / any other motor has inductances in its windings already

— also, motor has inertia

$\Rightarrow$  act as intrinsic LPF!



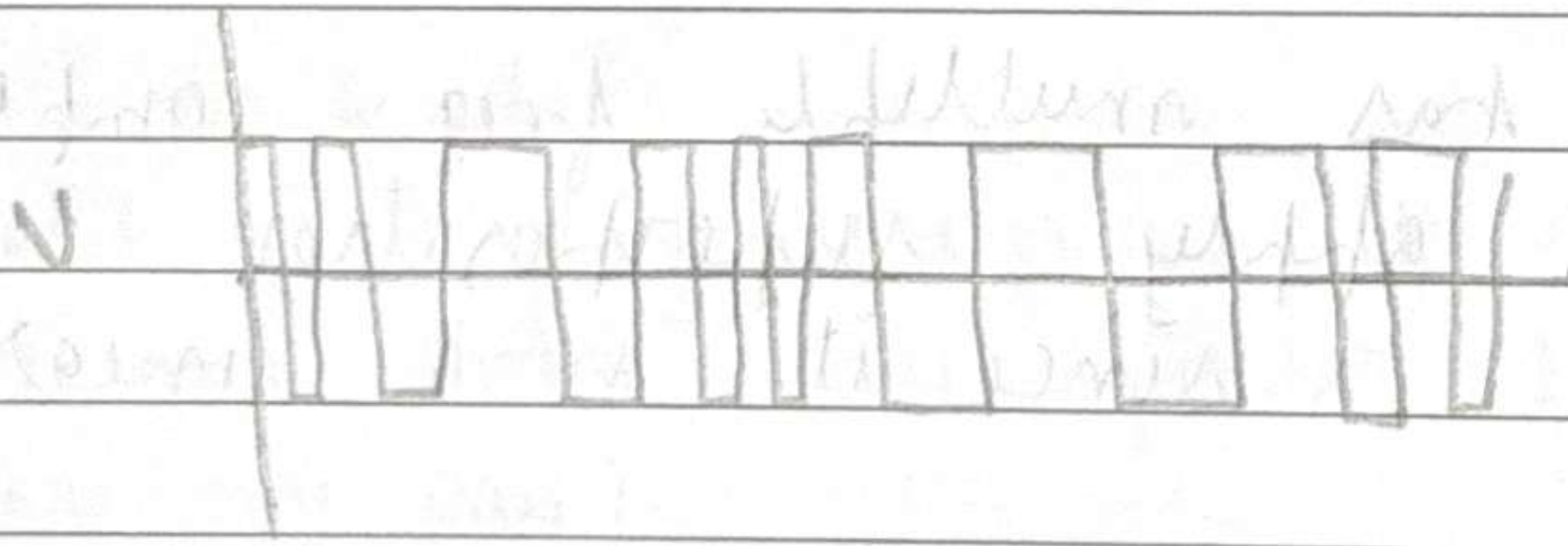
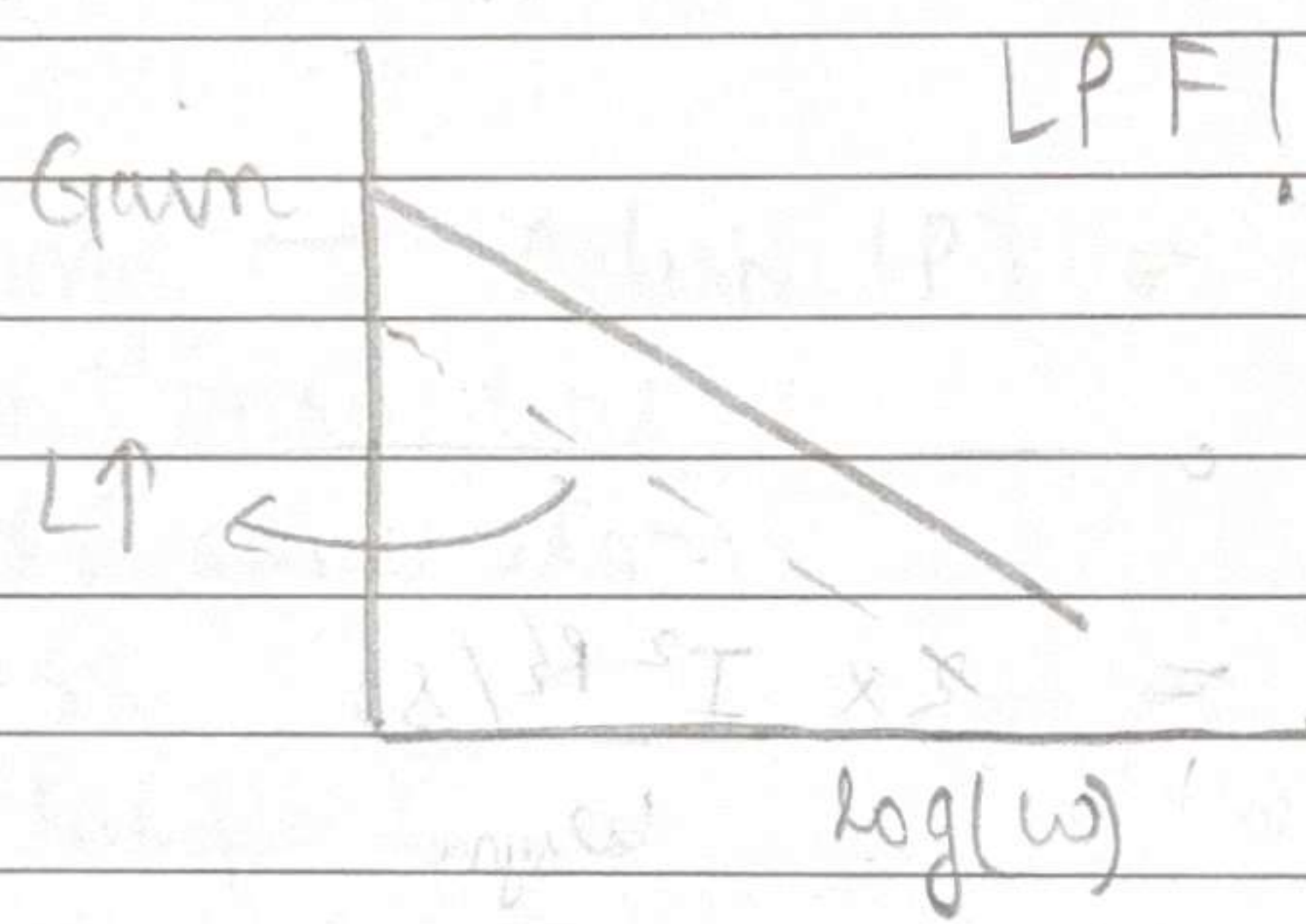
Why is inductor an LPF?



$$v = j\omega L i$$

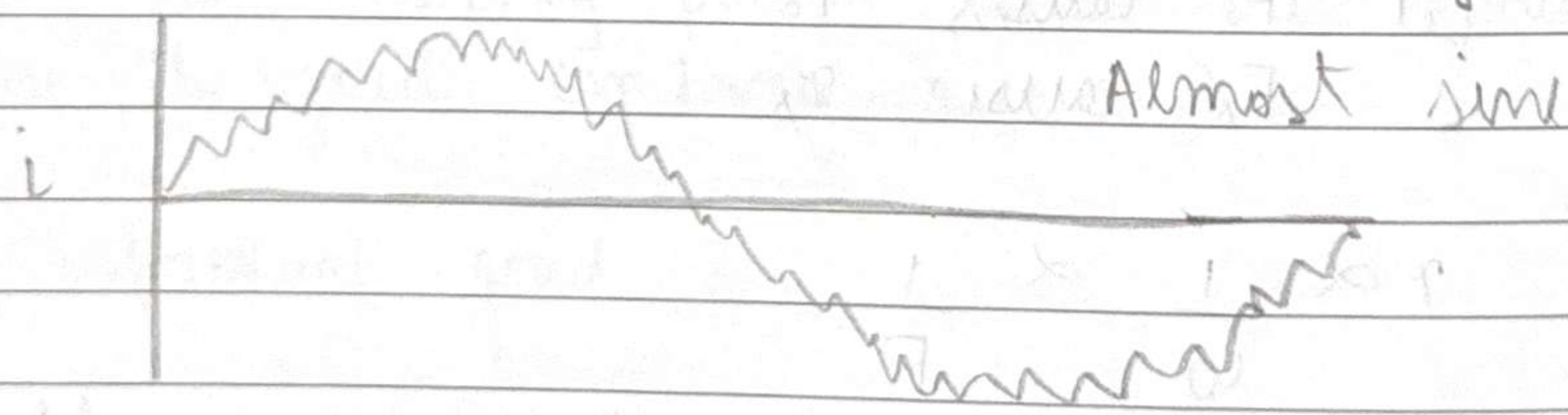
For a motor, output (mechanical motion) is because of the current.

$$\therefore \text{Gain (in dB)} = 20 \log \left| \frac{i}{v} \right| = -20 \log \omega + 20 \log L$$



Inverter output

50Hz,  
 Very high  $F_s$  and its harmonics



Almost sine

To get better shape of  $i(t)$ , two options -

- increase  $L \Rightarrow$  lowers the magnitude response plot  $\Rightarrow$  more attenuation of  $F_s$  and its harmonics

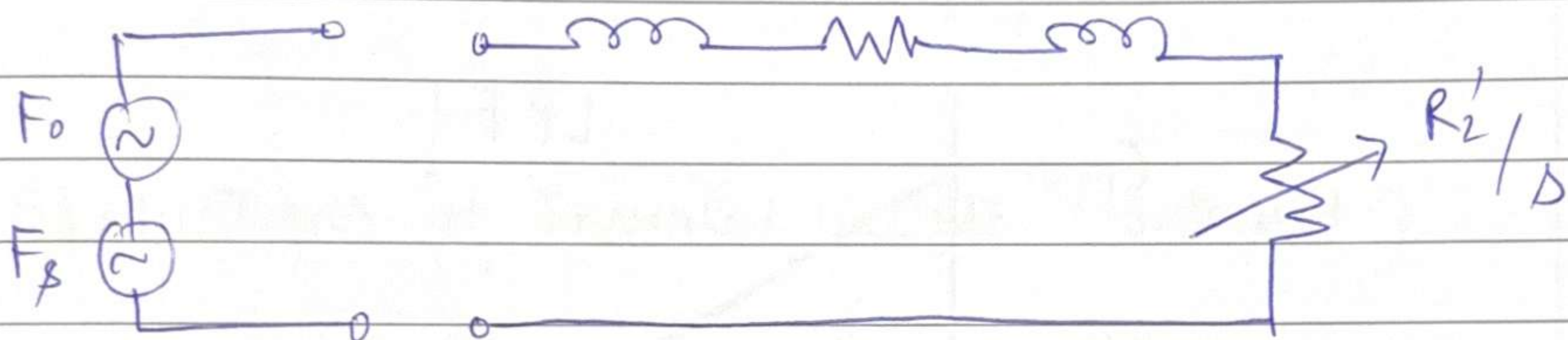
But maybe expensive to increase  $L \Rightarrow$  not practically done



— Increase  $F_s \Rightarrow$  more attenuation of  $F_s$   
 $\Rightarrow$  better  $i(t)$

High freq. components of  $i(t)$  cause humming noise while motor runs.

IM eq. ckt —



$$\text{Torque, } T = \frac{3 \times I^2 R_2'/s}{\omega_{\text{sync}}}$$

When  $I$  has multiple freq. components,  $F_0$  and  $F_s$ , apply superposition to get net torque (since it is a linear ckt).

Let  $F_0$  causes  $T_0$   
 $F_s$  causes  $T_s$

$$T \propto \frac{1}{\omega} \propto \frac{1}{F}$$

$T_0$  is large and  $T_s$  is very small

$T_s$  component causes only vibrations / dissipation



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Date:  
YOUVA

We may be satisfied with  $i(t)$  at intermediate value of  $F_s$ .

Here increasing  $F_s$  increasing losses.  
Hence, we may want to keep  $F_s$  at some optimally chosen value.

In application (a), we had an additional LC filter. So increasing  $F_s \uparrow$ ,  $\Rightarrow$  less  $L$  and  $C$  required  $\Rightarrow$  lower cost.

In that case, we may want very high  $F_s$ .

- (c) Grid Feeding — Active (P) or Reactive (Q) Power
- Solar inverters
  - Wind generators
  - BESS
  - Q Support

What is grid?

It is a big reservoir of electrical power. We can ~~take~~ draw power from it (in which case we pay).

We can supply power to it (in which case the grid company pays us).

We represent grid as

$\sim V_1$  with phase  $\angle 0^\circ$

assumed i.e.  $V_1 \sqrt{2} \cos(\omega t)$

The grid is 'stiff', which means that even if we supply some power to it, its  $V_0$ ,  $\phi_0$ ,  $\omega$  doesn't change.

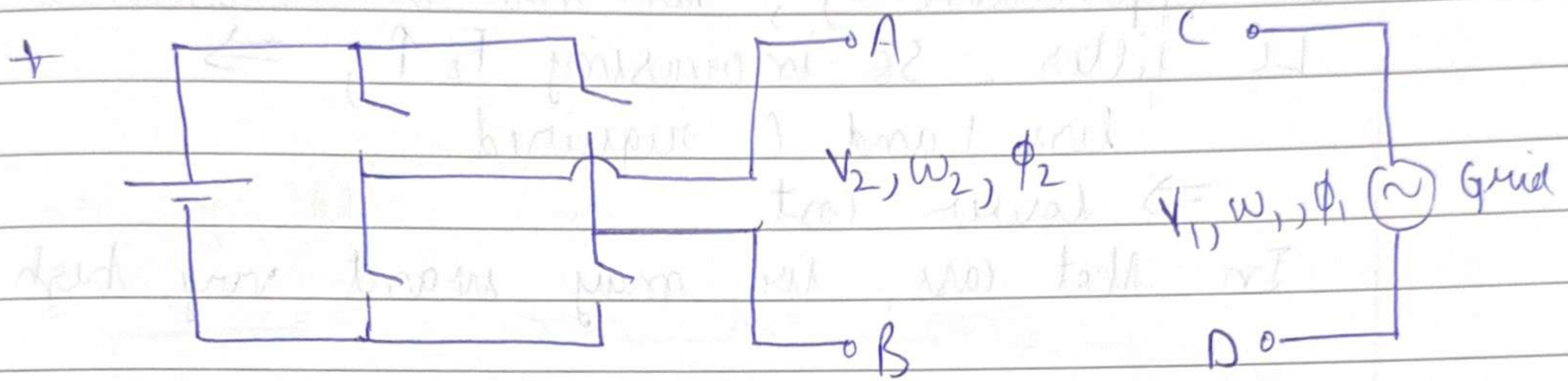
$\Rightarrow$  'Infinite bus'



# Phase Lock Loop (PLL)

This is assumed for  $P_{injected} \ll P_{grid}$

How do we use H-bridge inverter to feed power into grid?



We can control  $V_2, w_2, \phi_2$

Requirements —

$w_2 = w_1$ , very precisely

done by an algorithm called

Phase Locked Loop (PLL)

If  $w_1 \neq w_2$ , then integrating power over time gives average power supplied = 0

★ ★ How precise should  $w_2 = w_1$  be?

If  $f_{w1} = 50 \text{ Hz}$ ,  $f_{w2} = 30 \text{ Hz}$

$T_1 \sim 20 \text{ ms}$ ,  $T_2 \sim 33 \text{ ms}$

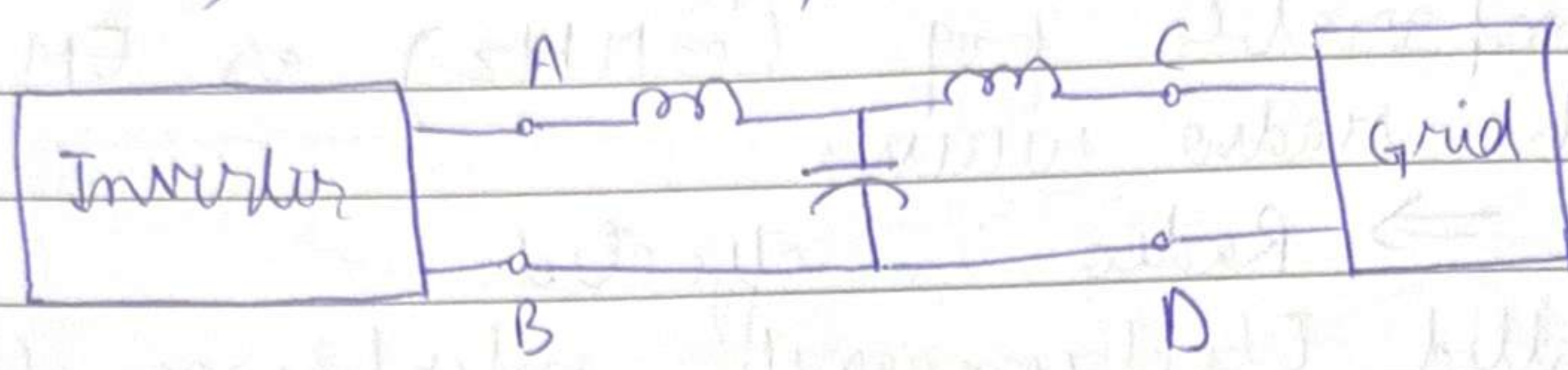
∴ Over a period of  $\text{lcm}(T_1, T_2) = 100 \text{ ms}$ ,  $\langle P \rangle = 0$

∴ Now, in case  $f_1 \approx f_2$  to very closely,  $T = \text{lcm}(T_1, T_2)$  will be very large

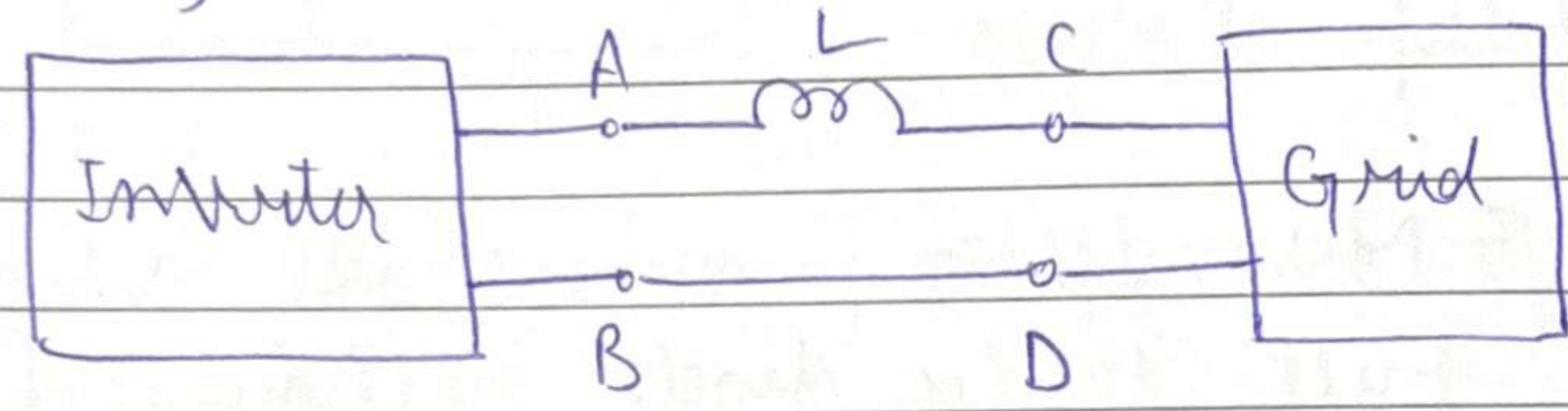
If  $T$  is of the order of a few days, then that's good enough. In theory, after a few days  $\langle P \rangle = 0$ . But by that time, we will reset system.



— Option 1) LCL Filter,



— Option 2) L



(Simpler to just use an L)

LC filter has resonant frequency which needs to be taken care of.

L ensures that if we don't add an L, due to shorting of grid ( $\sim$ ) with inverter output ( $\square\square\square$ ), there will be large magnitude large frequency current ( $F_x$  component of current)  $\Rightarrow$  need to avoid that.

L takes + removes  $F_x$  component of current.

Why avoid  $F_x$  component of current? to flow in the grid?

—  $I^2R$  losses

— other components (like IM motor) may get affected (IM makes noise)

(i) Motors in the grid

(ii) Components not in the grid

(like battery powered radios etc.)

are affected because high freq I in the PTO

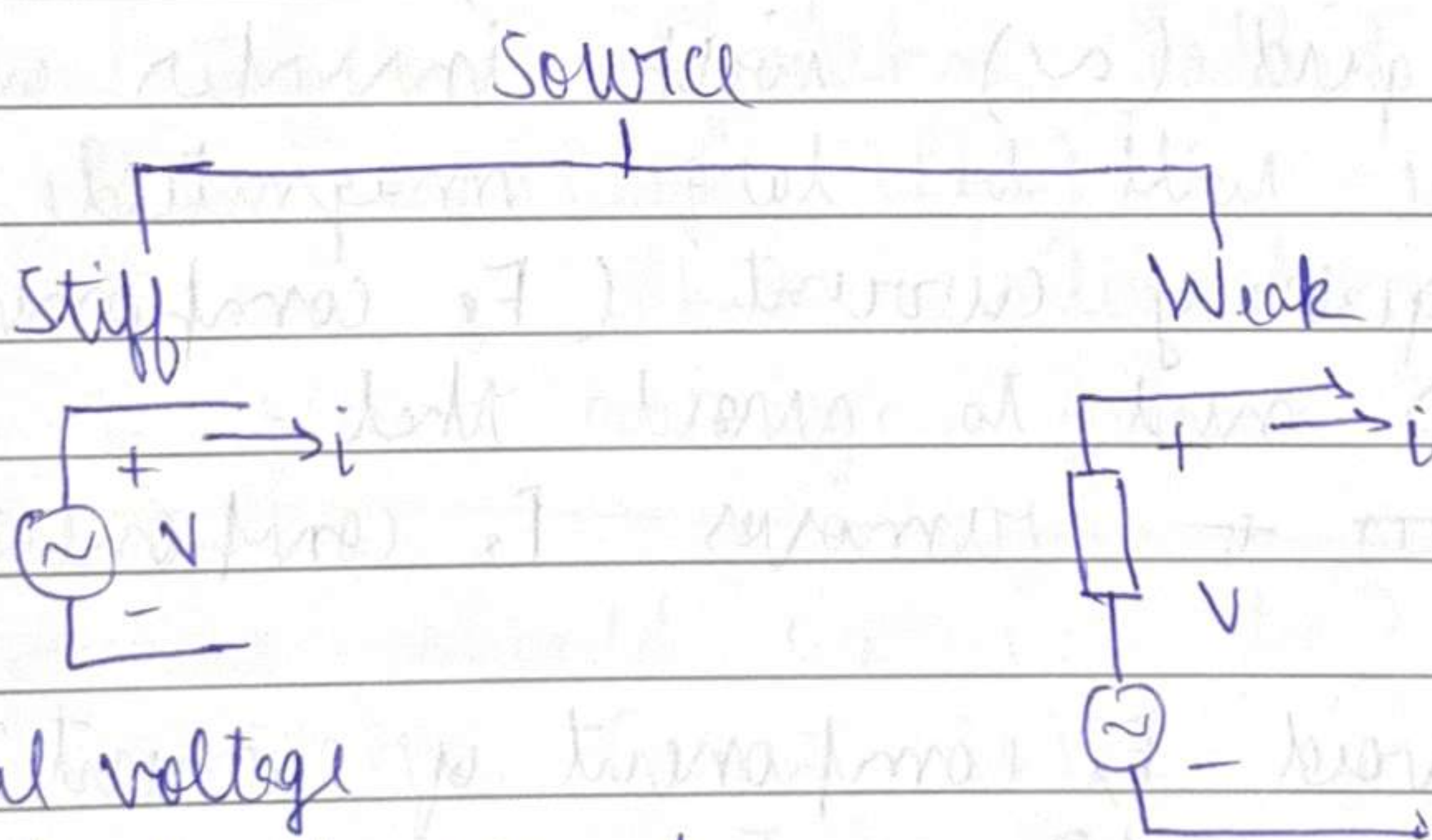


nearly grid with produces EM waves of comparable freq ( $\sim$  MHz) as EM signals the radio receives.

$\Rightarrow$  Radio is affected called Electromagnetic interference (EMI)

In laptop charger,

- i) EMI filter
  - ii) Full-bridge diode rectifier with large C at output
  - iii) Flyback converter
- ↓  
Laptop



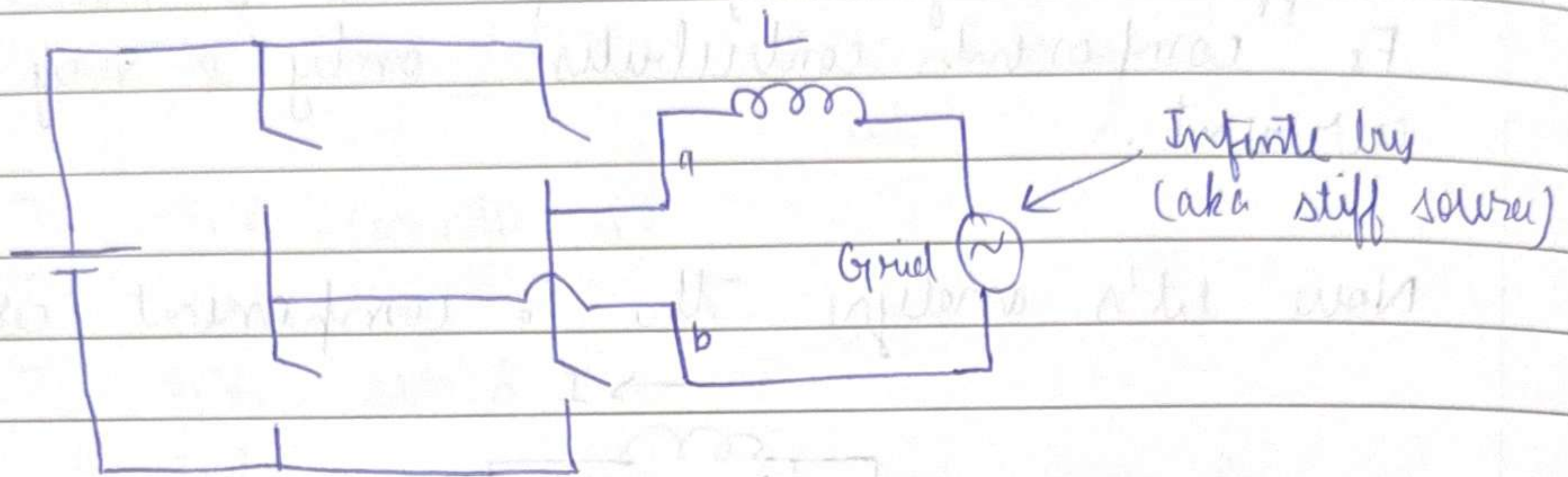
Terminal voltage independent of current

No/very low impedance

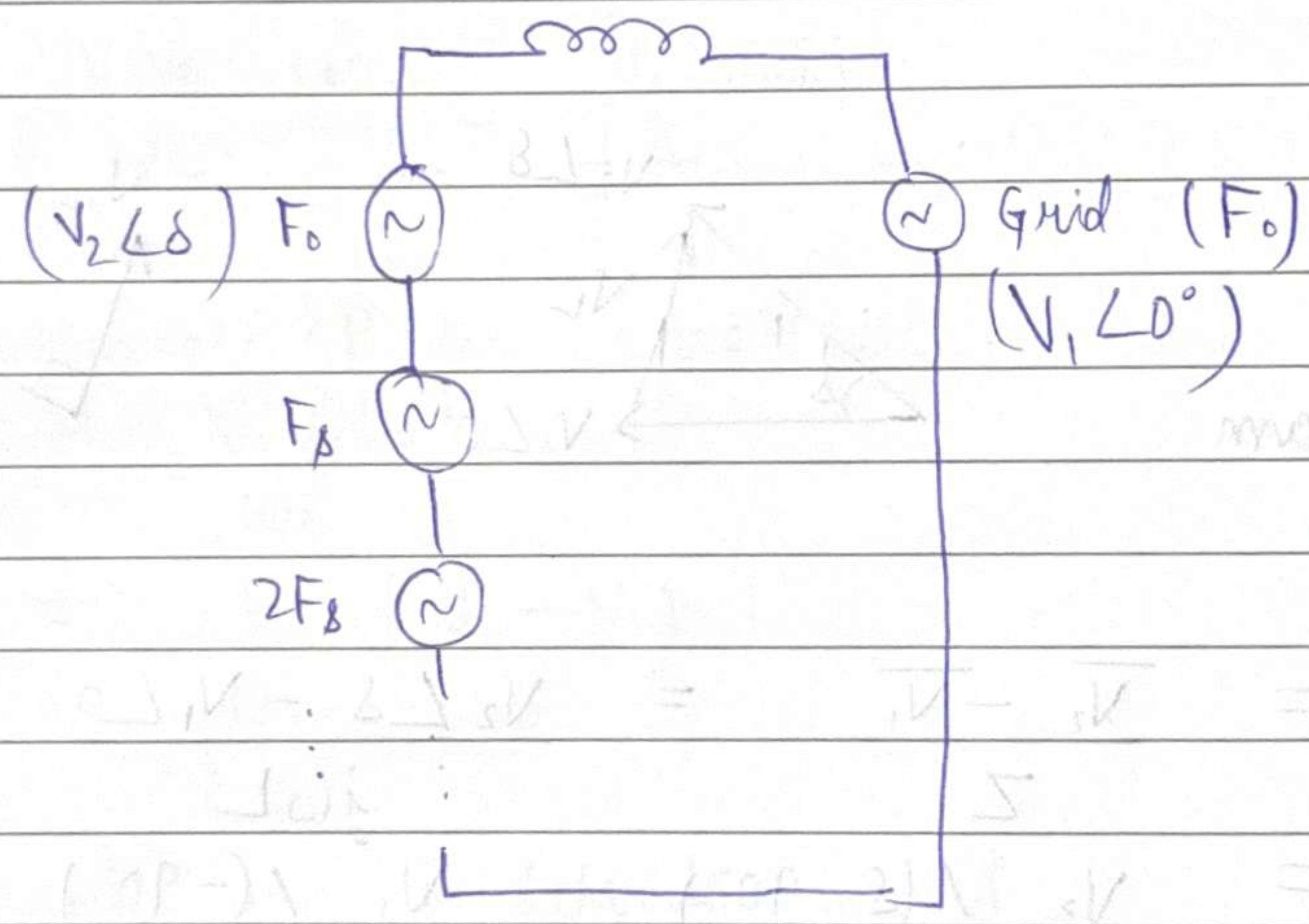
Terminal voltage dependent on current  
 considerable impedance



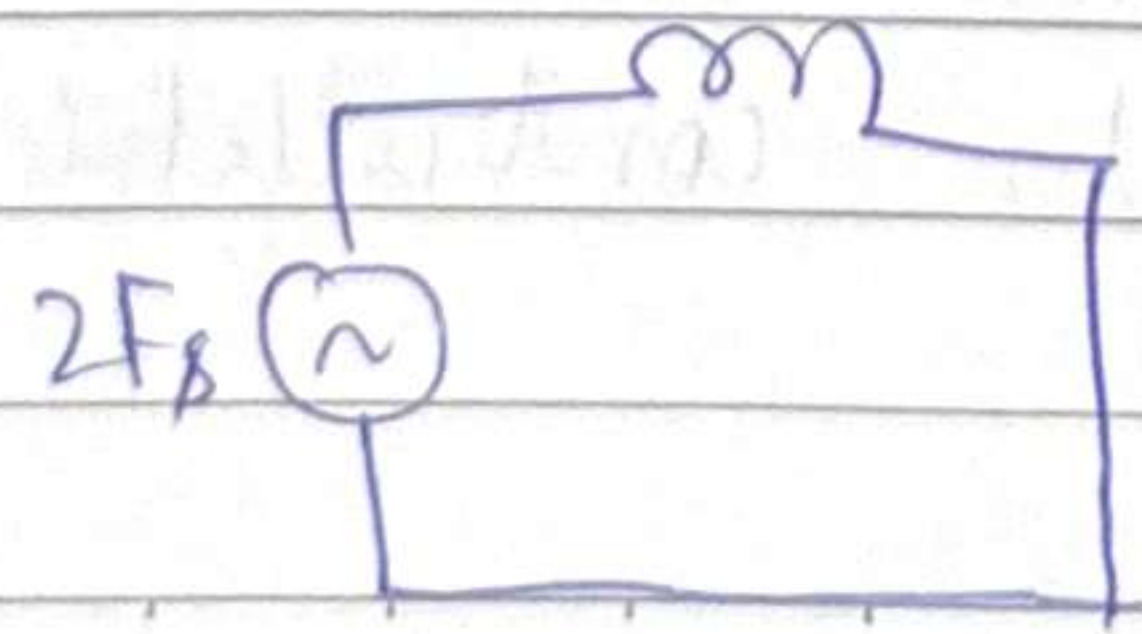
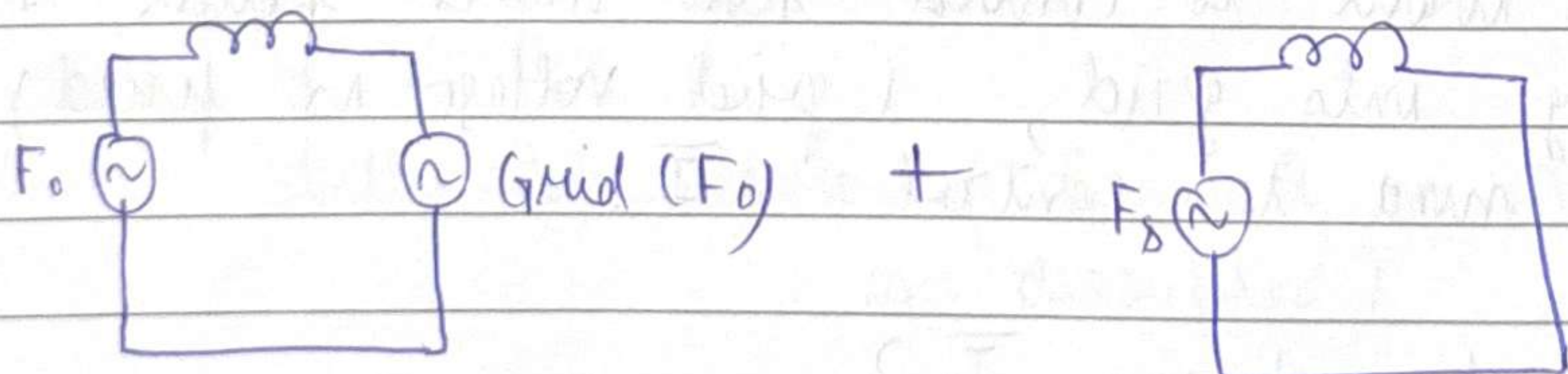
# Feeding power into grid (continued)



Consider the frequency components of  $V_{ab}$



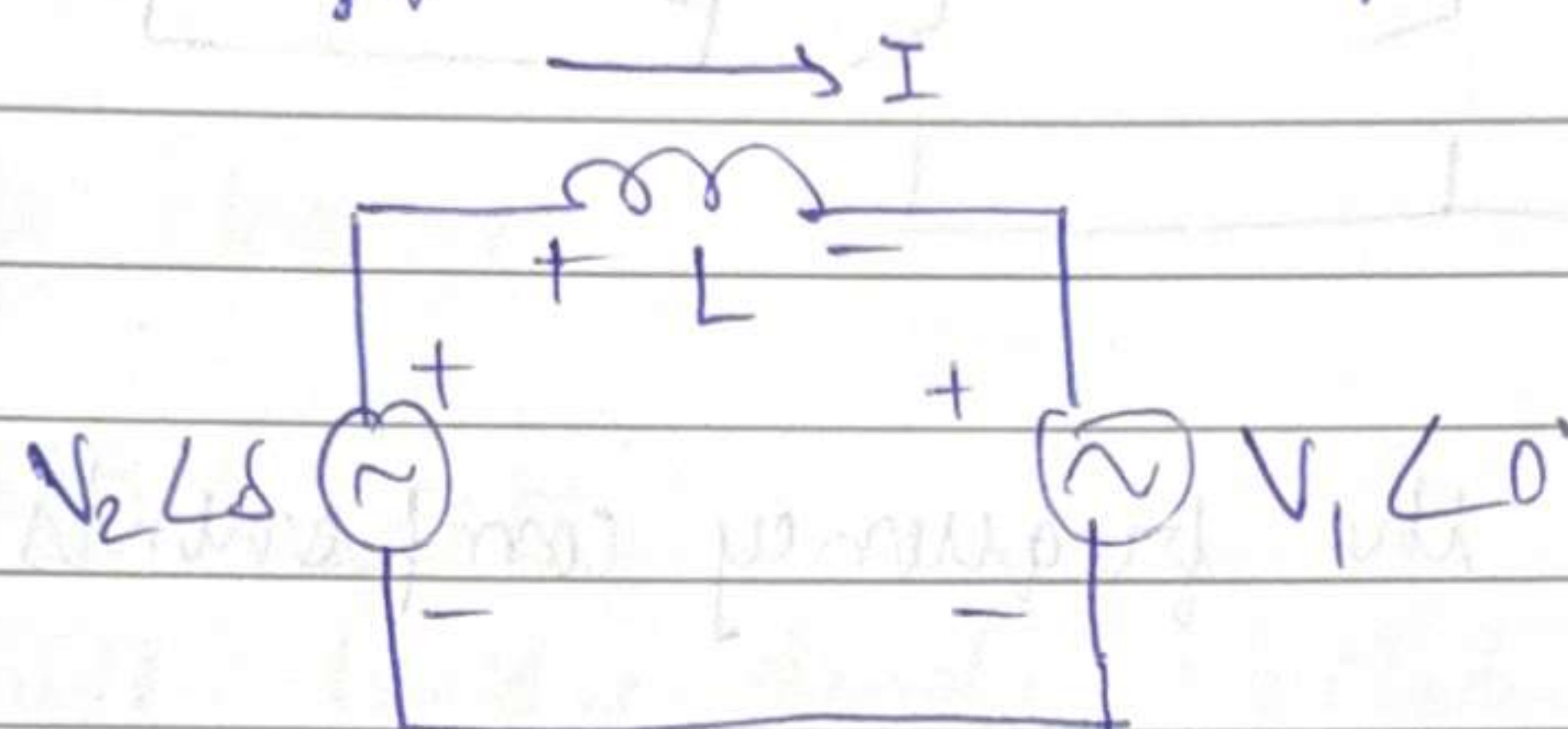
Apply superposition,



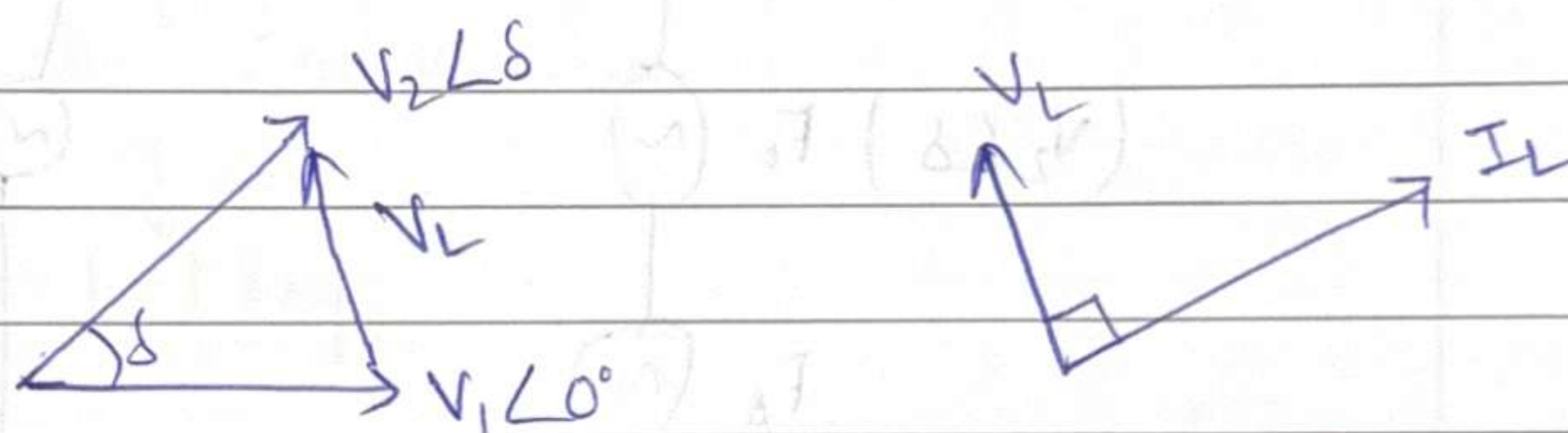


For  $F_x, 2F_x, \dots$ , since frequency is very high,  $L$  offers very large impedance, hence  $F_x$  components contributes only a very small current.

Now let's analyse the  $F_0$  component circuit,



Phasor diagram



$$\overline{I_L} = \frac{\overline{V_2} - \overline{V_1}}{Z} = \frac{V_2 \angle \delta - V_1 \angle 0}{j\omega L}$$

$$\overline{I_L} = \frac{V_2 \angle (\delta - 90^\circ)}{\omega L} - \frac{V_1 \angle (-90^\circ)}{\omega L}$$

If I want to control how much power I am feeding into grid, (grid voltage is fixed), I need to control  $\overline{I_L}$ .

How to change  $\overline{I_L}$ ?

$\omega$  &  $L$  fixed

$\therefore V_2$  and  $\delta$  are the controllable parameters



$$S = (V_1 \angle 0) (\bar{I})^* = P + jQ$$

$$S = V_1 \left( \frac{V_2}{\omega L} \angle 90^\circ - \delta - \frac{V_1}{\omega L} \angle 90^\circ \right)$$

$$P = \frac{V_1 V_2}{\omega L} \cos(90^\circ - \delta)$$

$$\therefore P = \frac{V_1 V_2}{\omega L} \sin \delta$$

$$Q = \frac{V_1 V_2}{\omega L} \sin(90^\circ - \delta) - \frac{V_1^2}{\omega L}$$

$$Q = \frac{V_1 V_2}{\omega L} \cos \delta - \frac{V_1^2}{\omega L}$$

Approximate P for small  $\delta$ ,

$$P \approx \frac{V_1 V_2 \delta}{\omega L}$$

$$Q \approx \frac{V_1}{\omega L} (V_2 - V_1)$$

For small  $\delta$ , P depends on  $\delta$   
 Q doesn't depend on  $\delta$

Hence,  $V_2$   $\xrightarrow{\text{controls}}$  Q  
 then  $\delta$   $\xrightarrow{\text{controls}}$  P

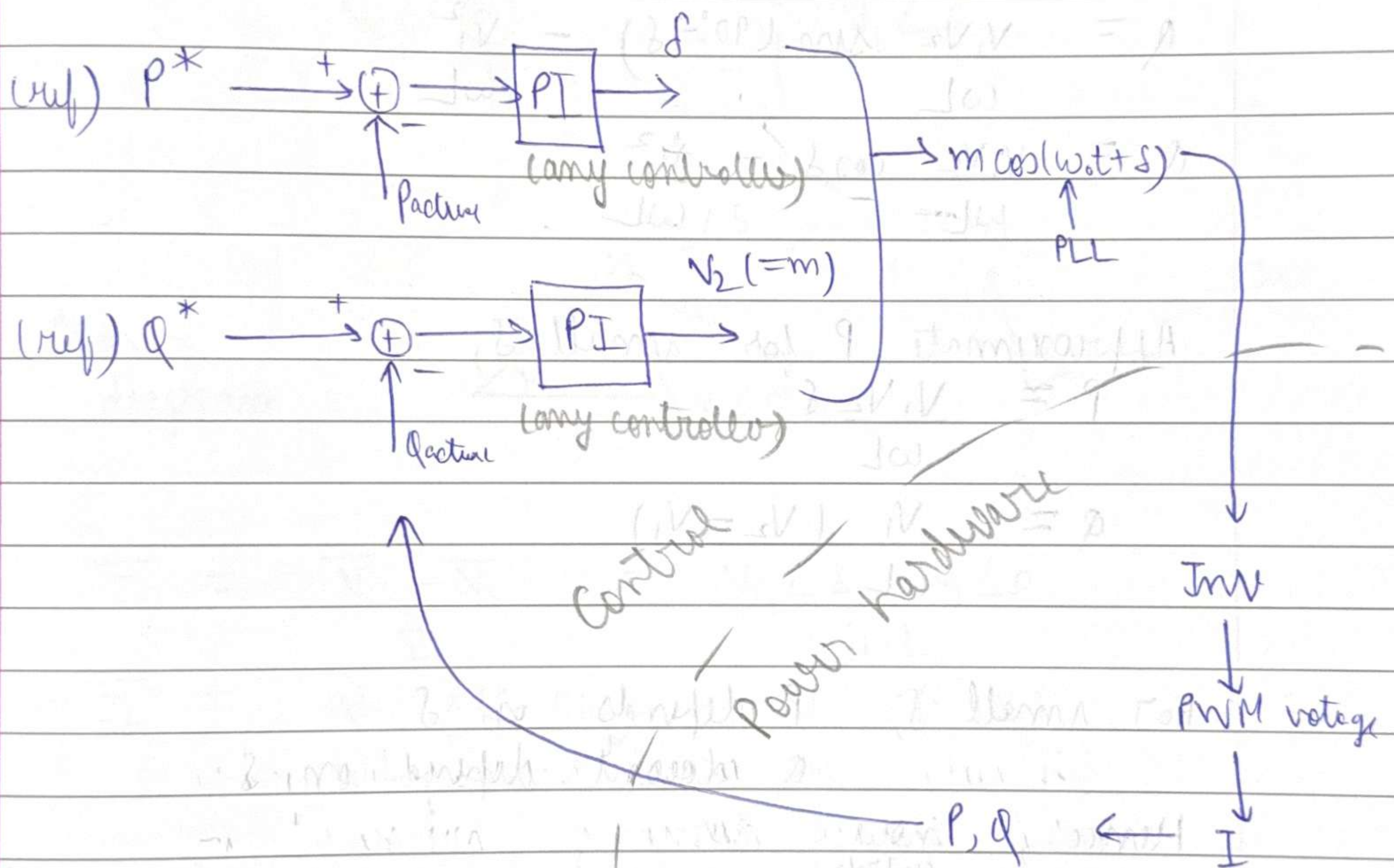
(the dependence of P and Q on  $V_2$  and  $\delta$  is decoupled)

Typically we keep  $\delta < 30^\circ$ , not near  $90^\circ$   
 Why?



- (i) If  $\delta$  is near  $90^\circ$ , there is risk of  $\delta$  going into  $>90^\circ$  region  $\Rightarrow$  risk of instability
- (ii)  $L$  is very small  $\Rightarrow P$  is already reasonably large, even for small  $\delta$
- (iii)  $\delta$  near  $90^\circ$  affects system stability

Control loop to set  $V, \delta$  to achieve desired  $P$



If we set  $P_{ref}$  to  $+10\text{ kW}$ , the inverter will feed  $10\text{ kW}$  into grid.

If we set  $P_{ref}$  to  $-10\text{ kW}$ , the grid will charge the battery of the inverter with  $10\text{ kW}$

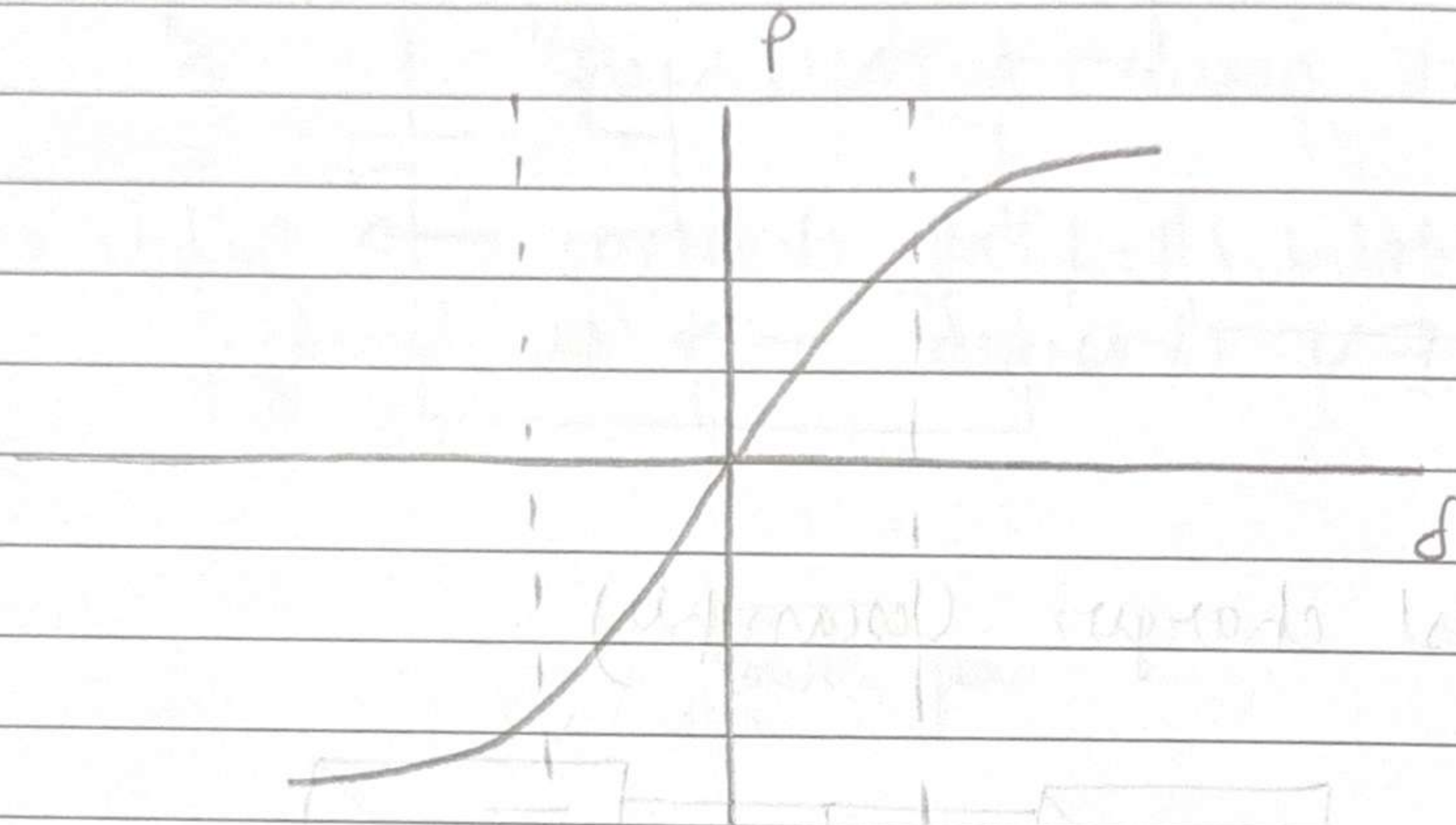
(Battery Energy flows from grid to battery)



Inverter can also be used as a rectifier!

∴ The inverter circuit also works in 'reverse'!

∴ Inverter is just a bridge between DC and AC.



∴ To reverse power flow, I set  $\alpha$  to be negative.

H-bridge inverter also called as a Voltage-source inverter (VSI) or <sup>voltage source</sup> converter (VSC).  
∴ It can allow bidirectional ~~power~~ power flow

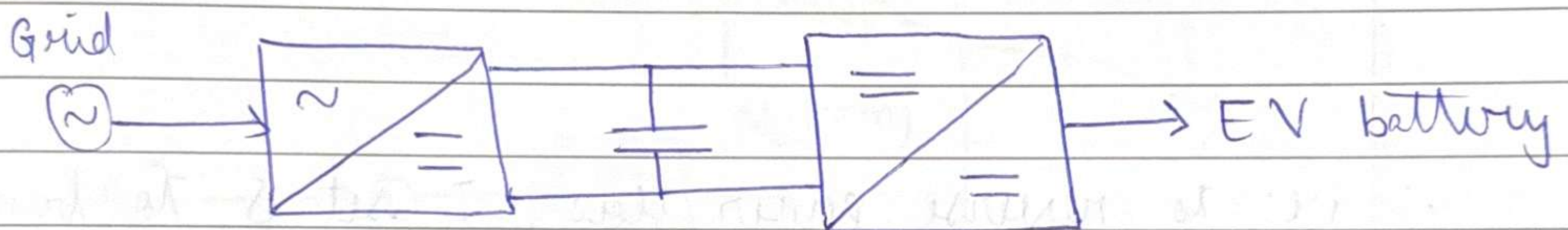
Applications of power transfer from AC domain to DC domain through H-bridge inverter

- Regenerative braking in EVs  
Mechanical energy → Generator → Electrical energy (AC) → H-bridge → Battery is charged
- Charge battery using grid
- EV chargers (use H-bridge to convert AC power to DC power to charge car battery)



- In fact, for most high power applications ( $> 4$  to  $5$  kW), diode rectifiers are not used. H-bridge is used for AC power  $\rightarrow$  DC power conversion
- Mobile/Laptop charger  $\rightarrow$  Diode rectifier  
 EV charger  $\rightarrow$  H-bridge

(d) EV Fast charger (example)



- i) Diode rectifier  $\downarrow$  options  $\rightarrow$  Need isolation to protect EV battery
- ii) VSI

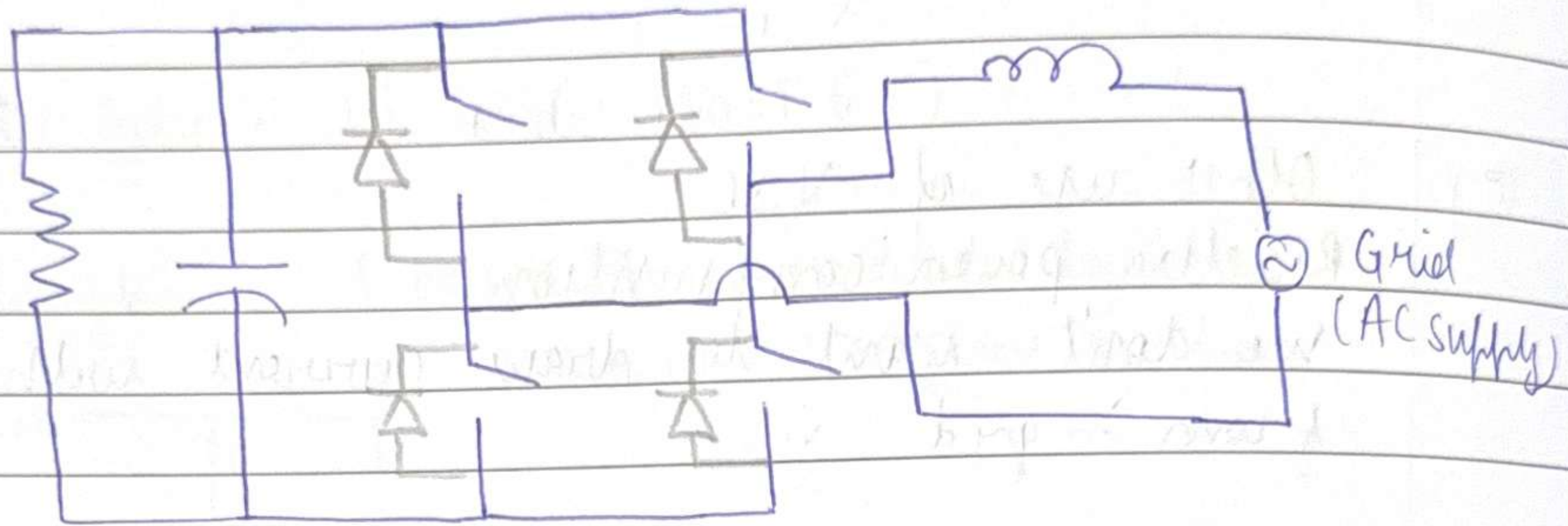
Problem with diode rectifier —

Peaky current waveform  
 High harmonic content  
 Large THD of  $i$   
 $\Rightarrow$  Bad!

$\therefore$  For large power conversion, prefer VSI  $\Rightarrow$   
 hence also called Voltage source converter (VSC)



## VSC based rectifier



This can be interpreted as a generator form of diode rectifier

- (Case 1)
  - No PWM
  - $V_{dc}(\text{at } \cos C) = V_{ac}^{\text{peak}} = 325 \text{ V}$
  - High THD

- (Case 2)
  - PWM
  - at  $m \leq 1$
  - $\Rightarrow V_{dc} \geq 325 \text{ V}$
  - $\Rightarrow$  sinusoidal current

$\Rightarrow$  Very very useful circuit, many applications

By changing control signal, we can choose direction of power flow



AC, fans etc. typically use diode rectifier to keep costs low.

(e) Other use of VSC  
Reactive power compensation

We don't want to draw current with harmonics from grid.

Similarly, we don't want to draw reactive current from grid

→ because reactive  $i$  doesn't contribute to power

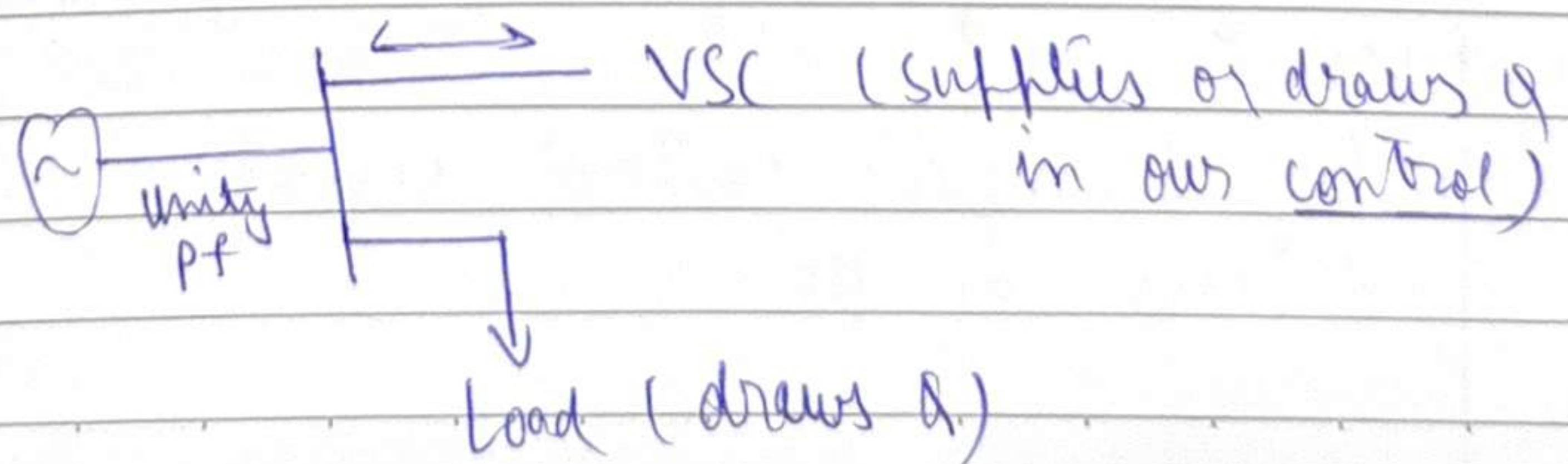
→ but increases losses

Solution?

option 1) Add a C (for induction loads like IM)  
But C value not adjustable. (Crude control)

option 2) Add a VSC, and inject controlled reactive power to counter the load's reactive power (Fine control)

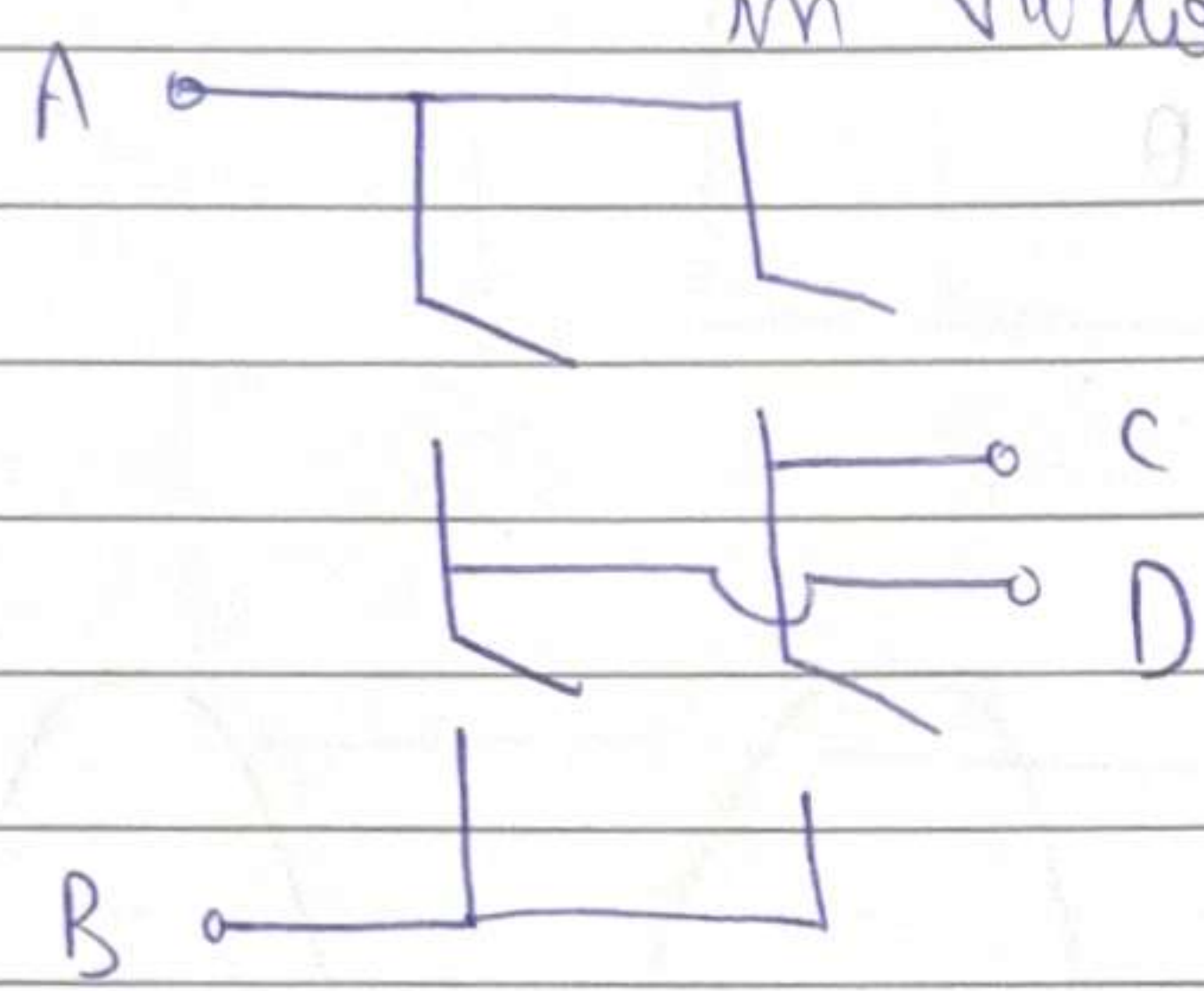
Reactive power compensation  
Stat Com — Static compensation  
Name of Technology





In markets, Power MOSFETs are available in -

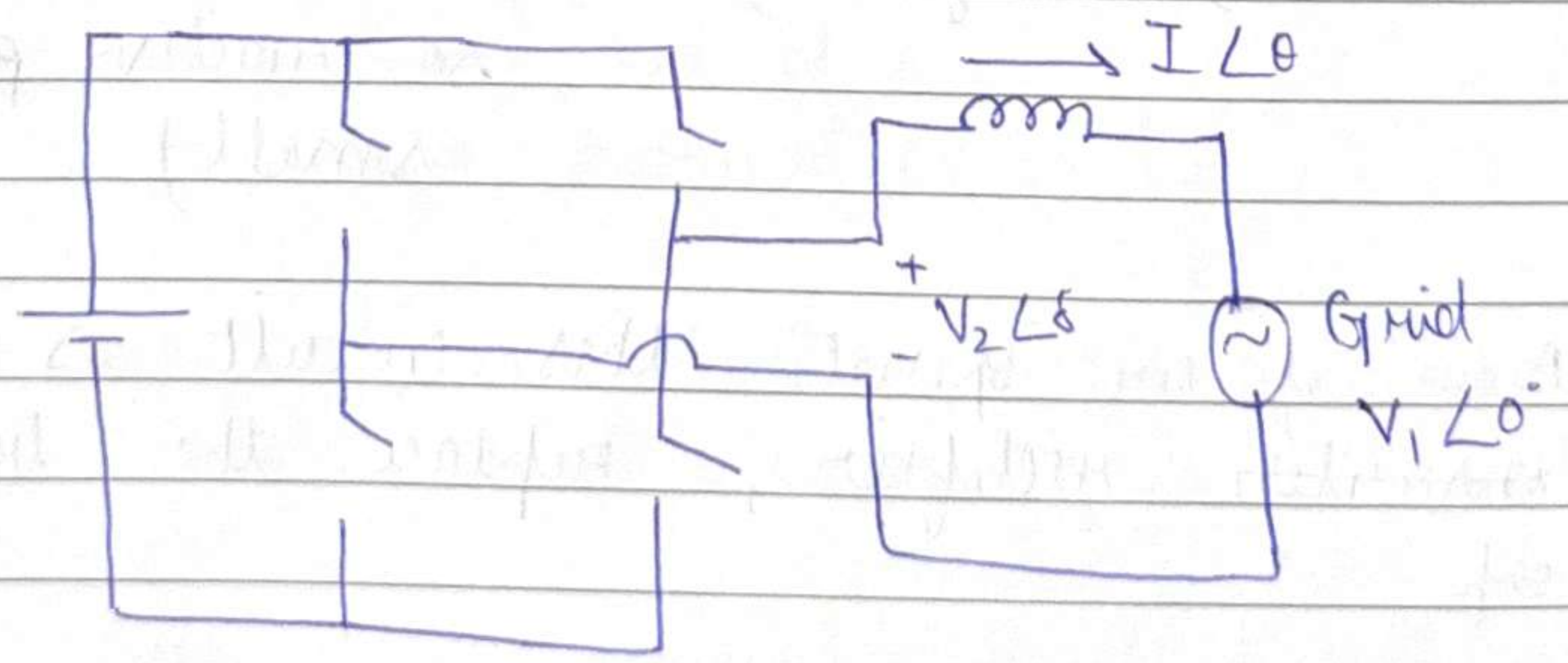
- Discrete type (single MOSFET) =
- Module type (MOSFETs + bundled together in various arrangements like



## FACTS - Flexible AC Transmission Systems

↓  
 Using Power electronics in Power systems  
 (to solve issues that arise in Power Systems)

Consider the inverter feeding the grid



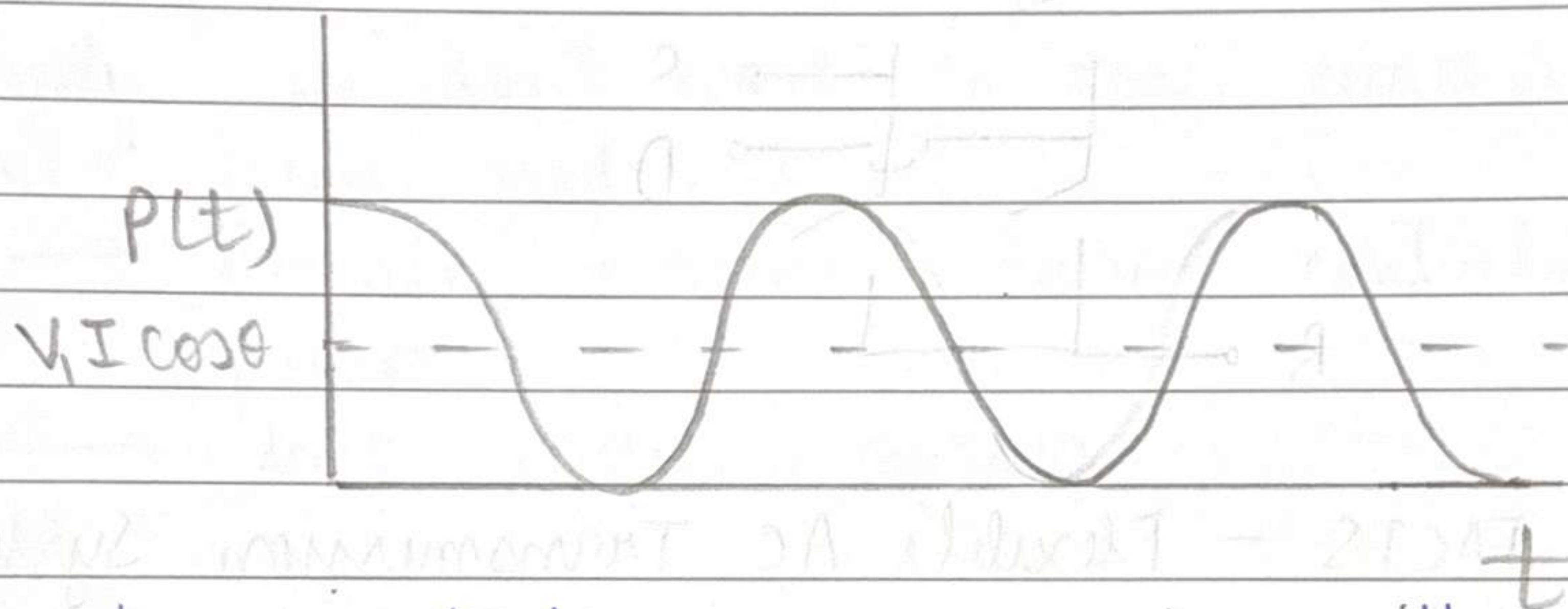


Instantaneous power

$$\begin{aligned}
 p(t) &= v_g(t) i(t) \\
 &= V_1 \sqrt{2} \cos \omega t \times I \sqrt{2} \cos(\omega t + \theta) \\
 &= V_1 I (\cos(\theta) + \cos(2\omega t + \theta))
 \end{aligned}$$

$$\therefore p(t) = V_1 I \cos \theta + V_1 I \cos(2\omega t + \theta)$$

$$\therefore \langle p(t) \rangle_{T_0} = V_1 I \cos \theta$$



$\therefore$  Single phase instantaneous power is oscillating  
(In 3- $\phi$  system, oscillating components cancel each other out to give constant  $p(t)$ )

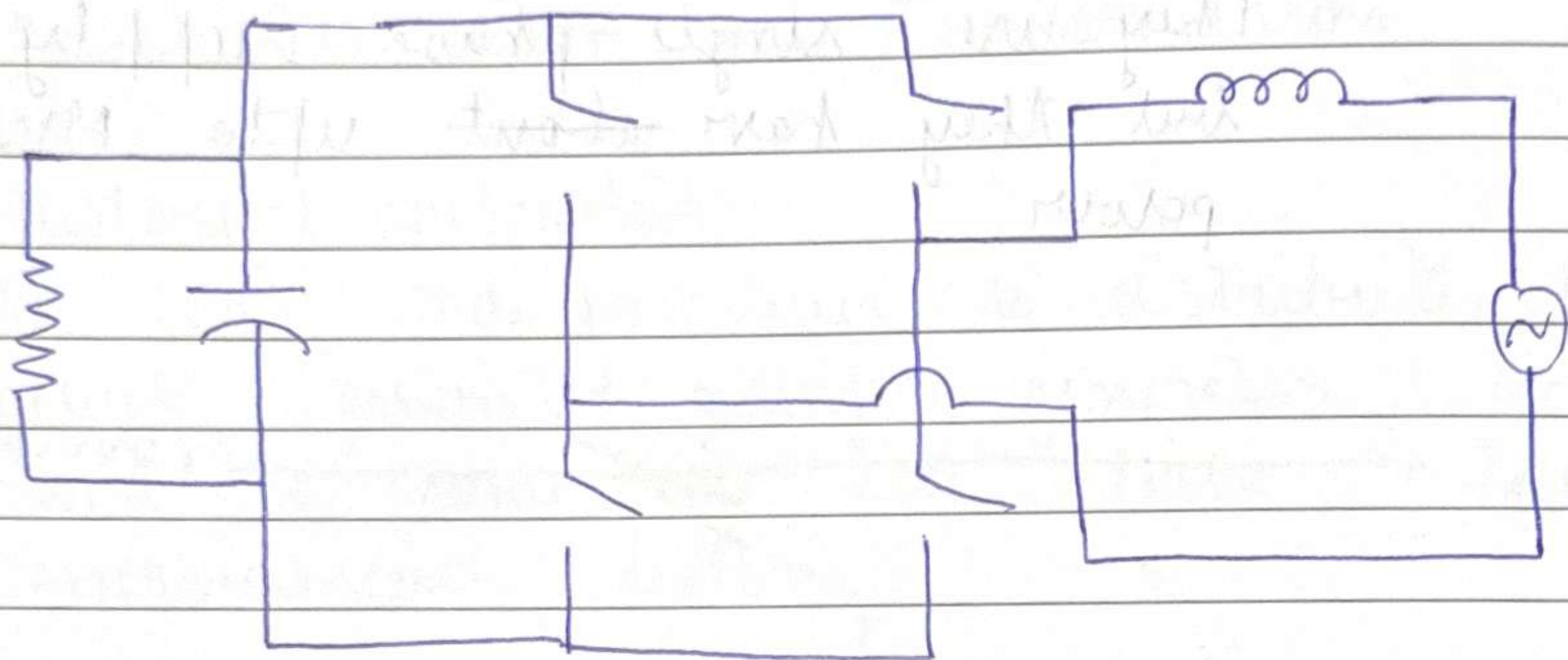
Assuming that power loss in switches is negligible,

$$P_{\text{source}}(t) \approx P_{\text{grid}}(t) \quad (\text{Assume that } L \text{ is small so reactive power of } L \text{ is small})$$

Now if we operate this circuit as an ~~inverter~~ rectifier, replace the battery with a cap



Battery / cap has to "support" oscillating power of  $2f_0$  frequency



Purpose of C is to absorb  $2f_0$  and other high frequency components, and still ~~output~~ have a (nearly) constant terminal voltage with very small ripple.

→ In 1- $\phi$  VSC,  
 Cap takes care of  $2F_0$  and  $F_s$  and harmonics of  $F_s$

→ In 3- $\phi$  VSC,  
 Cap only takes care of  $F_s$  and harmonics of  $F_s$

This is why C is smaller required in 3- $\phi$  systems than 1- $\phi$  systems

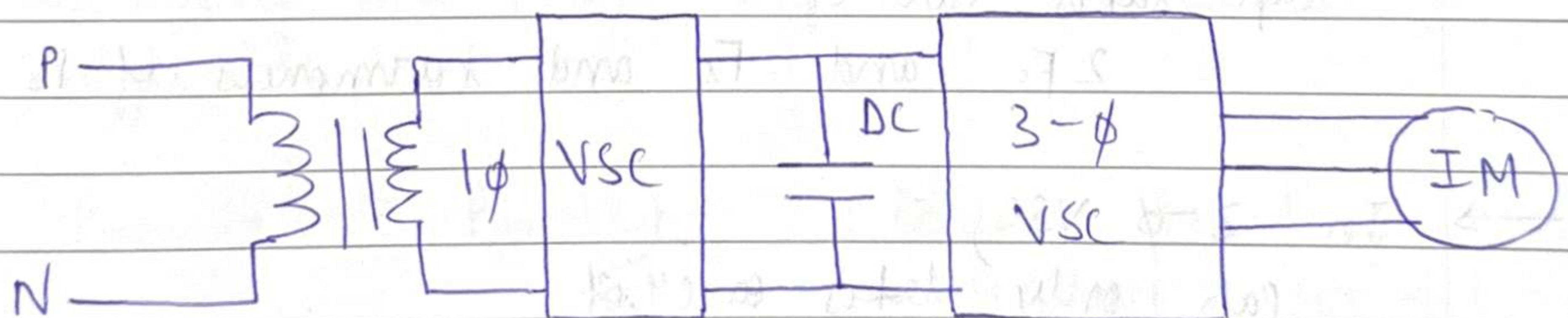
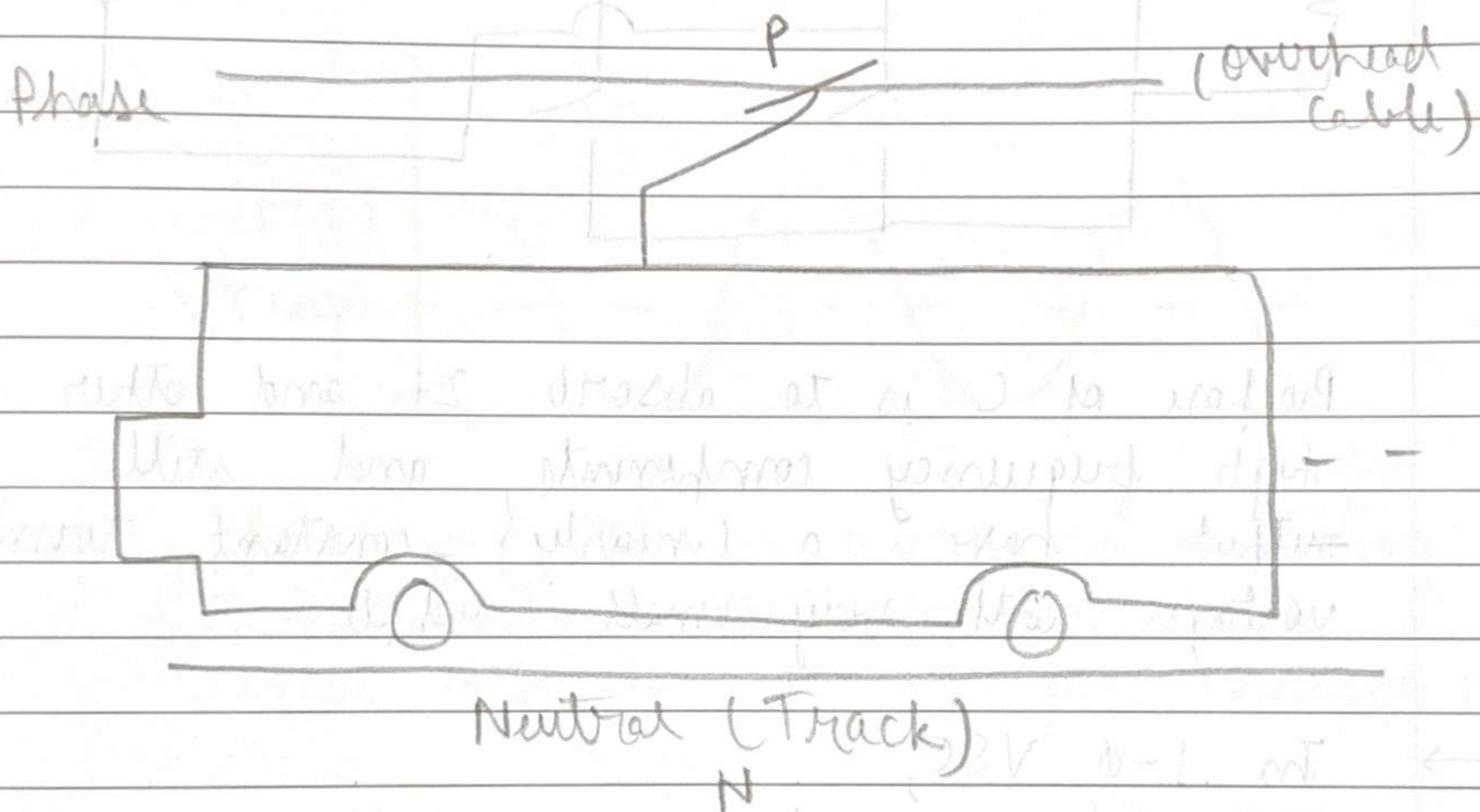
Hence, high power systems usually have 3- $\phi$  systems, so that very huge cap is not required



Typically,  $1-\phi$  systems are used for systems upto  $\sim 5$  kW

Exception — Trains ! (Locomotives)

They use single-phase supply, but they have about upto MW's of power.



(Train's power system)

Some parts of the track gets phase A, some get phase B, some get phase C (similar to how phases are distributed in a building)

→ # Most of the power generated is in 3- $\phi$



Power generation should generally be carried out away from cities

⇒ Need for transmission

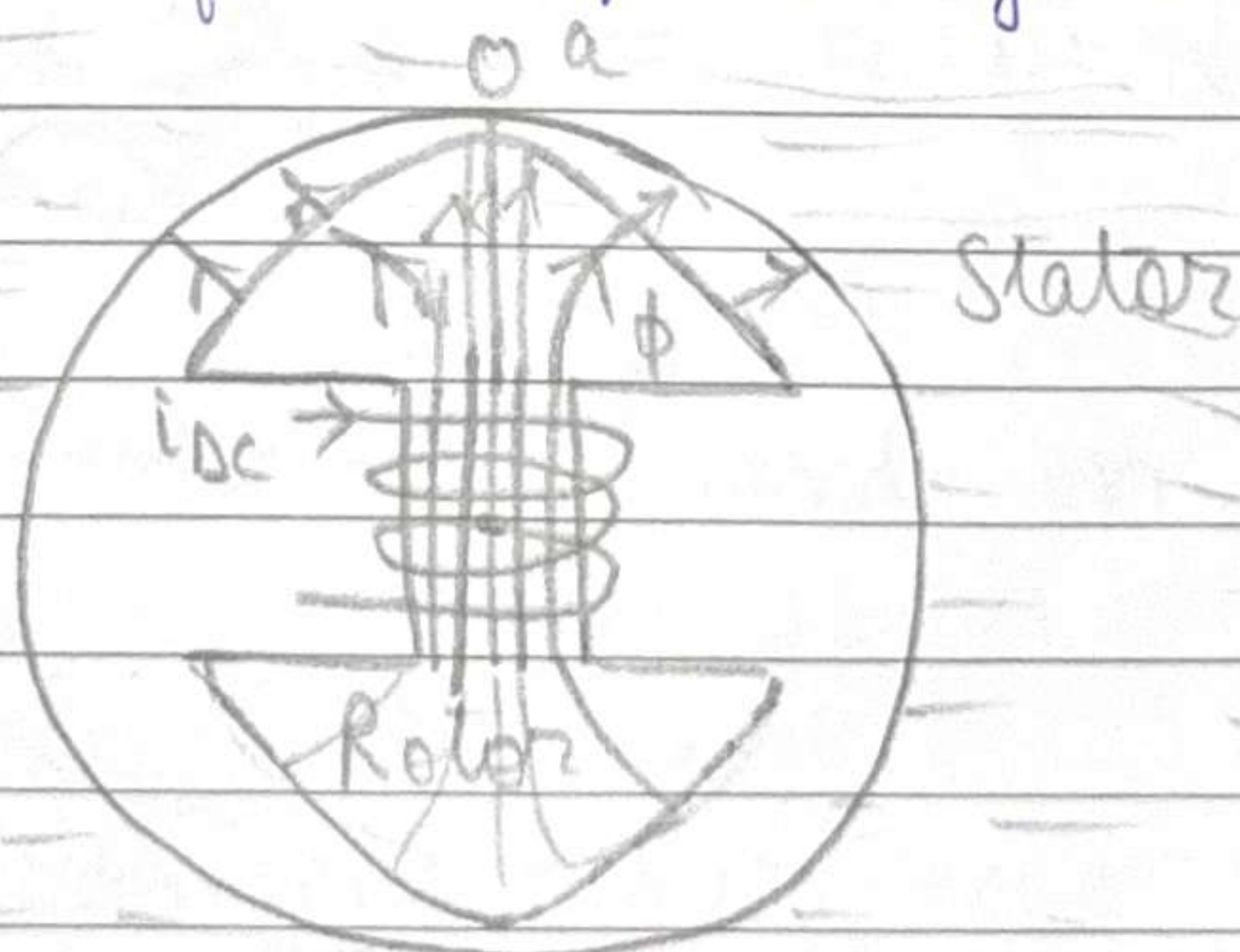
## Power Generation and Transmission

### Historical perspective

- In 1882, Edison tried to distribute DC power, generated using a generator, and send to some place to light  $\sim 200$  incandescent bulbs.
- AC distribution also used to send power to London to power some bulbs.
- 'War of currents' Edison (DC) vs Tesla and Westinghouse (AC) company

### Synchronous Generator

Popularly used for AC power generation



Stator and Rotor made of steel or iron (because they have high  $\mu$ )



By properly adjusting the air-gap length radially, the magnetic flux density  $B_g$  is made to have a sinusoidal spatial distribution in air gap.

$B_g$  (flux density)



sinusoidal distribution in air gap

(Reluctance,  $R = \frac{l}{\mu A} \Rightarrow R \propto l$ )

~~$B_g = B_m \cos(\omega t - \theta)$~~

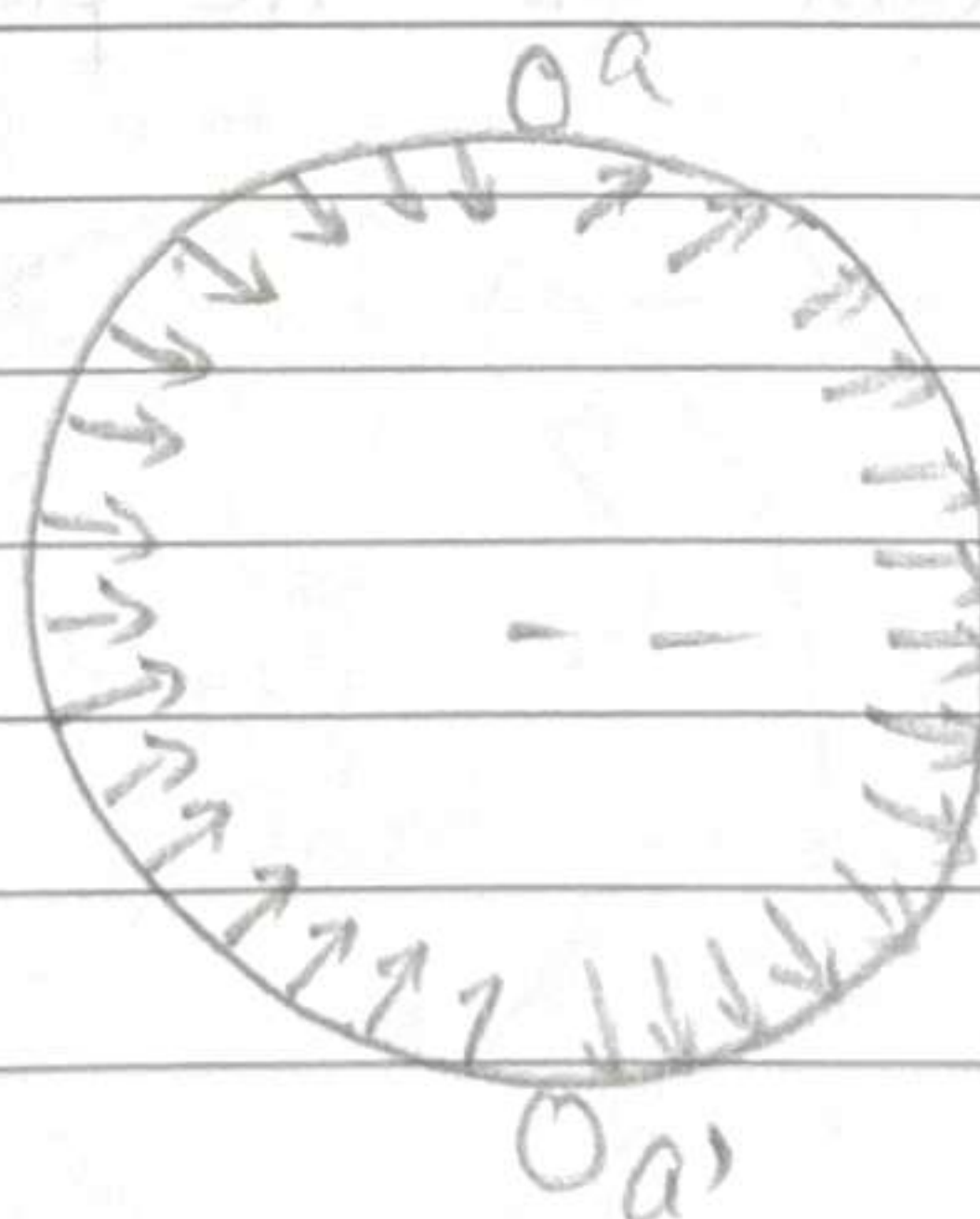
Now, the motor rotates with angular frequency  $\omega$

At any point at radial location  $\theta$ ,

$B_g = B_m \cos(\omega t - \theta)$

Now, we put a (fixed) turn <sup>a-a'</sup> around the (fixed) turn

Let's try to find the voltage induced across a-a'



$B_g = B_m \cos(\omega t - \theta)$

$\theta = 0$

$\phi$  (linking 1 turn) =  $\int_{\pi/2}^{3\pi/2} B_g l r d\theta$



$$\begin{aligned}\phi &= \int_{\pi/2}^{3\pi/2} B_m l r (\cos(\omega t - \theta)) d\theta \\ &= B_m l r \left[ \sin\left(\frac{3\pi}{2} - \omega t\right) - \sin\left(\frac{\pi}{2} - \omega t\right) \right] \\ &= B_m l r \left[ \sin \right] \\ &= B_m l r \left[ -\cos \omega t - \cos \omega t \right] \\ &= -2 B_m l r \cos \omega t = -\phi_m \cos \omega t\end{aligned}$$

$$e = \frac{Nd\phi}{dt} = 2NB_m \omega l r \sin \omega t = mV$$

$\therefore$  Voltage induced in a-a' coil is sinusoidal with frequency same as rotation mechanical rotation. (synchronous motor)

In this case, # Poles =  $P = 2$

$$\therefore \omega_m = \omega_e$$

When  $P \geq 2$

$$\omega_m < \omega_e \quad (\text{Elec angle} > \text{Mech angle})$$

$$\therefore (E_{\text{induced}})_{\text{RMS}} = \frac{N\phi_m 2\pi f}{\sqrt{2}} = 4.44 Nf \phi_m$$

E depends on  $\rightarrow \phi_m \rightarrow$  depends on field current  $I_F$

Now, we can also put more coils b-b' and c-c' at spatial angles of  $\pm 120^\circ$ , which will give us 3- $\phi$  sinusoidal output.

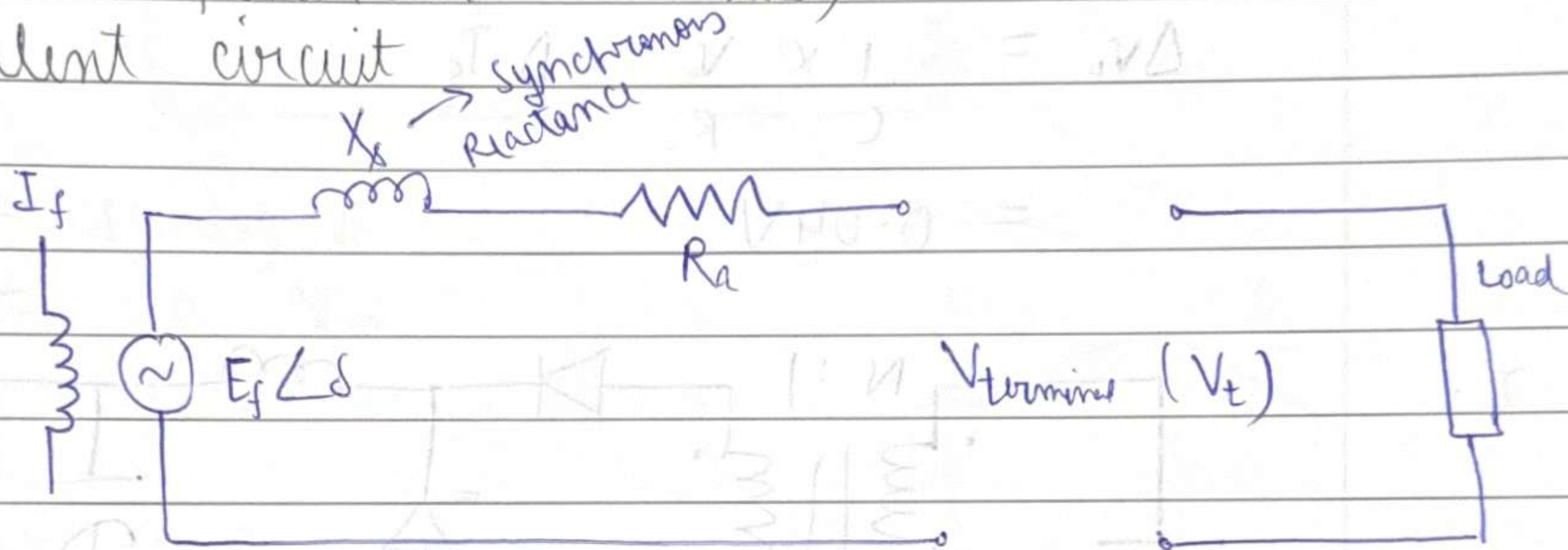
\*  $\omega$  of  $V_{\text{out}}$  depends on rotation speed.



(c)

## Synchronous Generator (continued)

Equivalent circuit



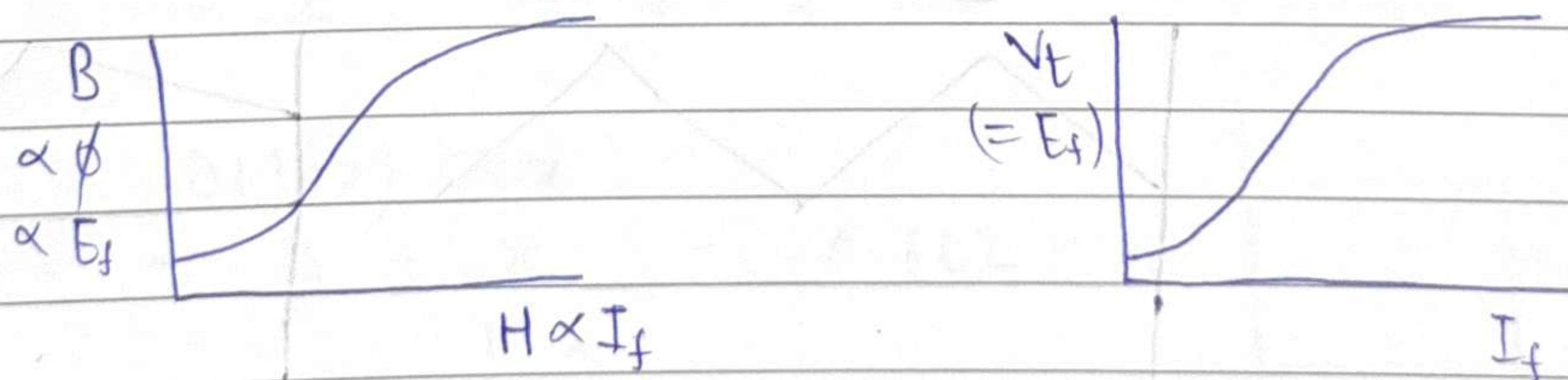
$$E_f = 4.44 N \phi_m f$$

$E_f$  is the induced voltage. It is called the excitation voltage.

## Open circuit characteristics (OCC)

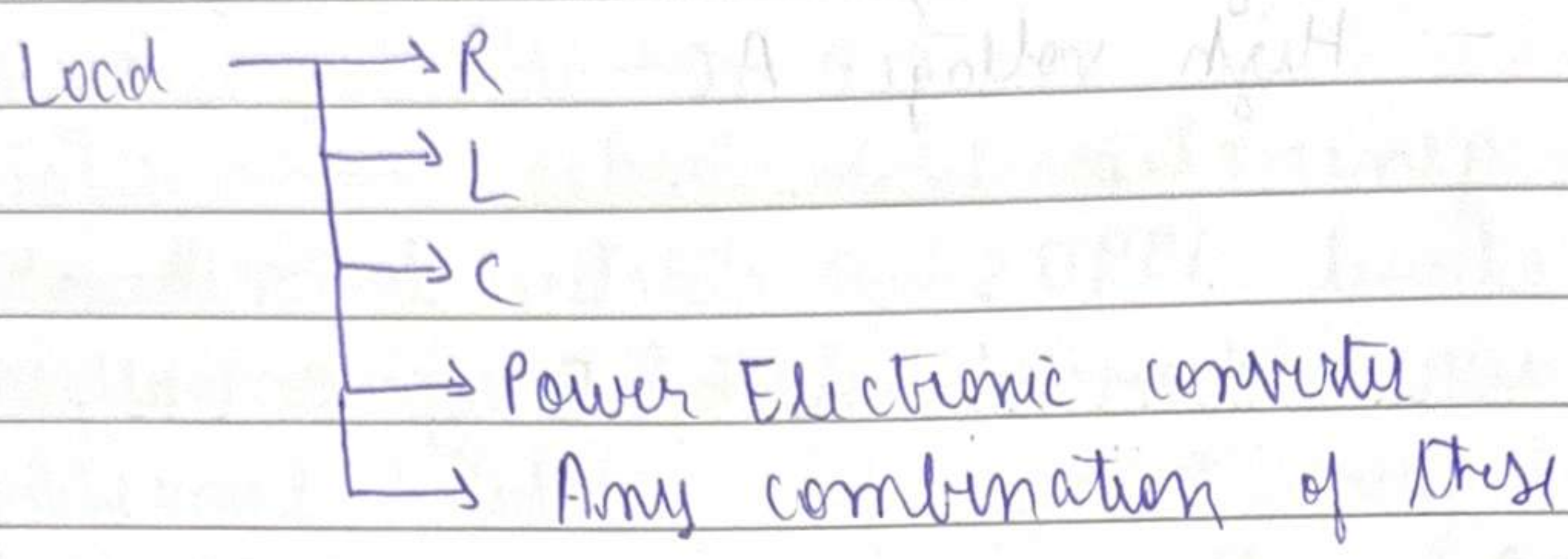
In open-circuit conditions,  $E_f = V_t$

Let us try to understand  $V_t$  vs  $I_f$  behaviour





Now let us connect a load



The generator will work fine for any of these loads.

For some PE loads like rectifiers, current drawn may be peaky = (high THD), so the power drawn may be having THD. But will still work fine as such.

With load,  $V_t \neq E_f$

Need for transmission network

Can we use a single generator to power a city?

Limitations -

- Reliability / Single point failure  
 (What if that one generator fails?)
  - Less utilization factor  
 (If we have dedicated generators for specific areas / localities, that area may not be utilizing generator at all times of the day - when some other area may need extra power - wastage)
- ⇒ Need of "Interconnected generator supply"  
 — Grid!



## Power Generation in India

2 types of connections between two points -

- High voltage DC (HVDC)

- High voltage AC

## Power generation in India

Until about 1990s, electrical network in India was divided into 5 zones (N, W, E, S, NE) with no interconnectors until

In 2013, all zones were combined into single grid (combined  $\Rightarrow$  connected with AC lines)

$$\text{Installed Capacity} = \sum_{i=1}^n P_i$$

$n = \#$  all power plants

$P_i =$  rated power of plant  $i$  (note that any given time, power actually being produced maybe less)

Presently, \* India's installed capacity  $\sim 524$  GW

India also exports / imports some amount of power to / from neighbouring countries ( $\sim 1$  to  $2$  GW) to each country

### Capacity

India  $\sim 524$  GW  $= G_1$  (say)

China  $\sim 5$  times  $G_1$

US  $\sim 2.5$  times  $G_1$

Russia  $< G_1$

Japan  $< G_1$



## Sources of Power generation (Installed capacities)

Coal  $\rightarrow$  221 GW +  $\sim 6.6$  GW (lignite)

Hydro  $\rightarrow$  46 ~~266~~ GW

Solar  $\rightarrow$  143 GW

Wind  $\rightarrow$  55 GW

Nuclear  $\rightarrow$  8.7 GW

Geothermal, tidal, biofuels  $\rightarrow$  very small

Oil / Diesel  $\rightarrow$  0.6 GW

Gas  $\rightarrow$  20 GW

Total  $\rightarrow$  524 GW

How is energy produced from these

Coal — Heat water — turn turbine  $\rightarrow$  generator

Need to use water because on burning coal, heat is spread out, not concentrated

Oil / Diesel / Gas — Directly burn in an engine  
No need to boil water

Nuclear — Boil water

These statistics are of installed capacity

Important to note that coal sources ~~to~~ operate 24 hours.

Solar can only operate for  $\sim 5-6$  hours

Hence actually solar contribution to power generation may be actually much less than coal



## Hydro / Wind — turbine turbines — Generator

In coal / hydro, etc., we can actively control the ~~speed~~ rotation speed of turbine  $\Rightarrow$  because it needs to be in sync with electrical grid frequency

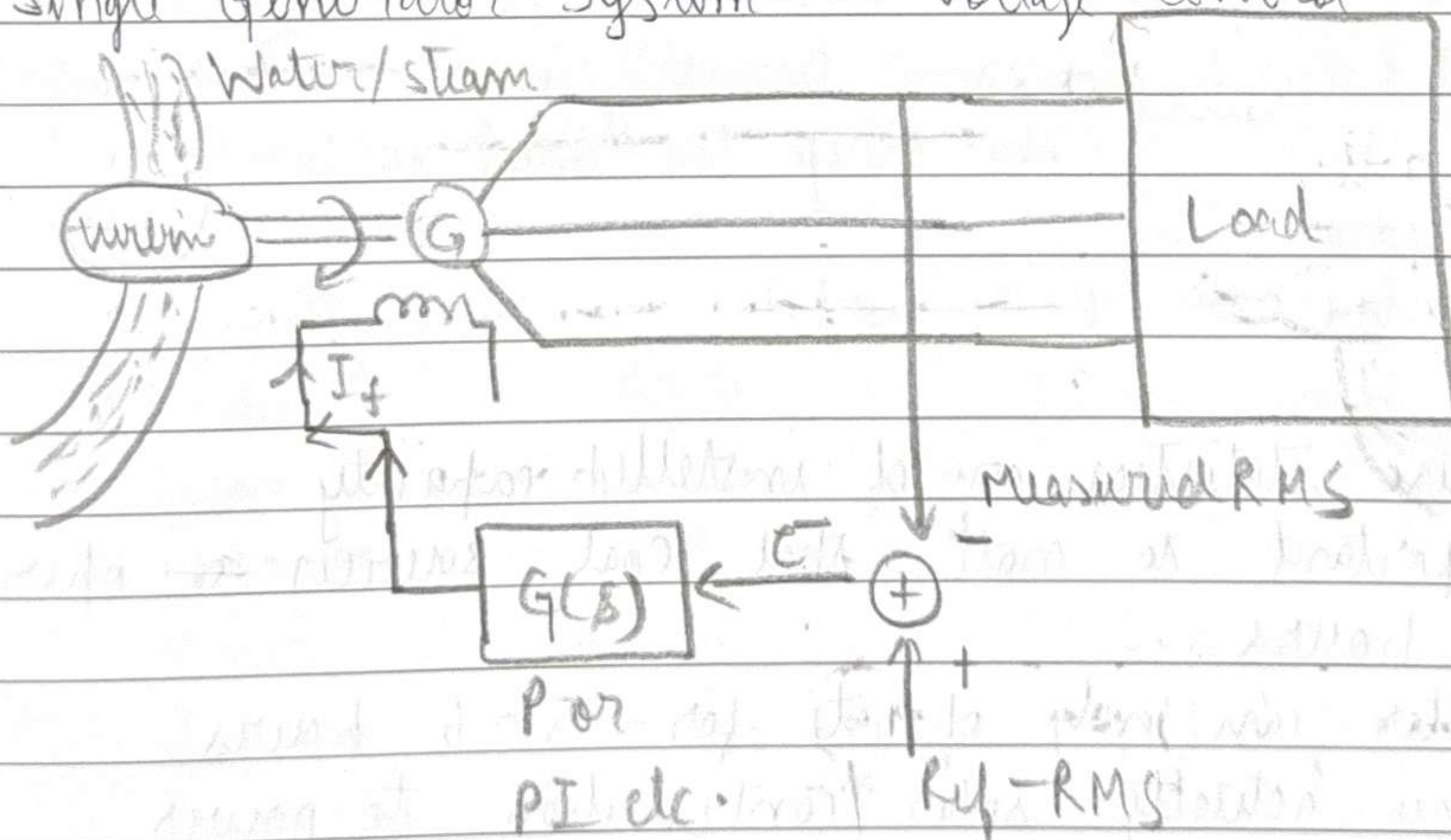
$\rightarrow$  uses synchronous generator

In wind generation, can't control rotation speed (wind is unpredictable)

$\Rightarrow$  use a different generator called DFIG (Doubly Fed induction generator) in which we can set the output voltage's frequency independently of rotation speed.

Solar  $\rightarrow$  Output DC  $\rightarrow$  VSC  $\rightarrow$  AC  $\rightarrow$  Grid

## Single Generator System — Voltage Control



This is called AVR (automatic voltage regulator)



We want constant  $V_t$  even if load changes.  
 We put a controller which adjusts  $I_f$   
 $\Rightarrow$  changes  $E_f \Rightarrow$  controls  $V_t$   
 based on measured  $V_t$

When we put active control on  $V_t$ ,  
 voltage regulation =  $\frac{|V_{nl}| - |V_{fl}|}{|V_{fl}|} \approx 0$

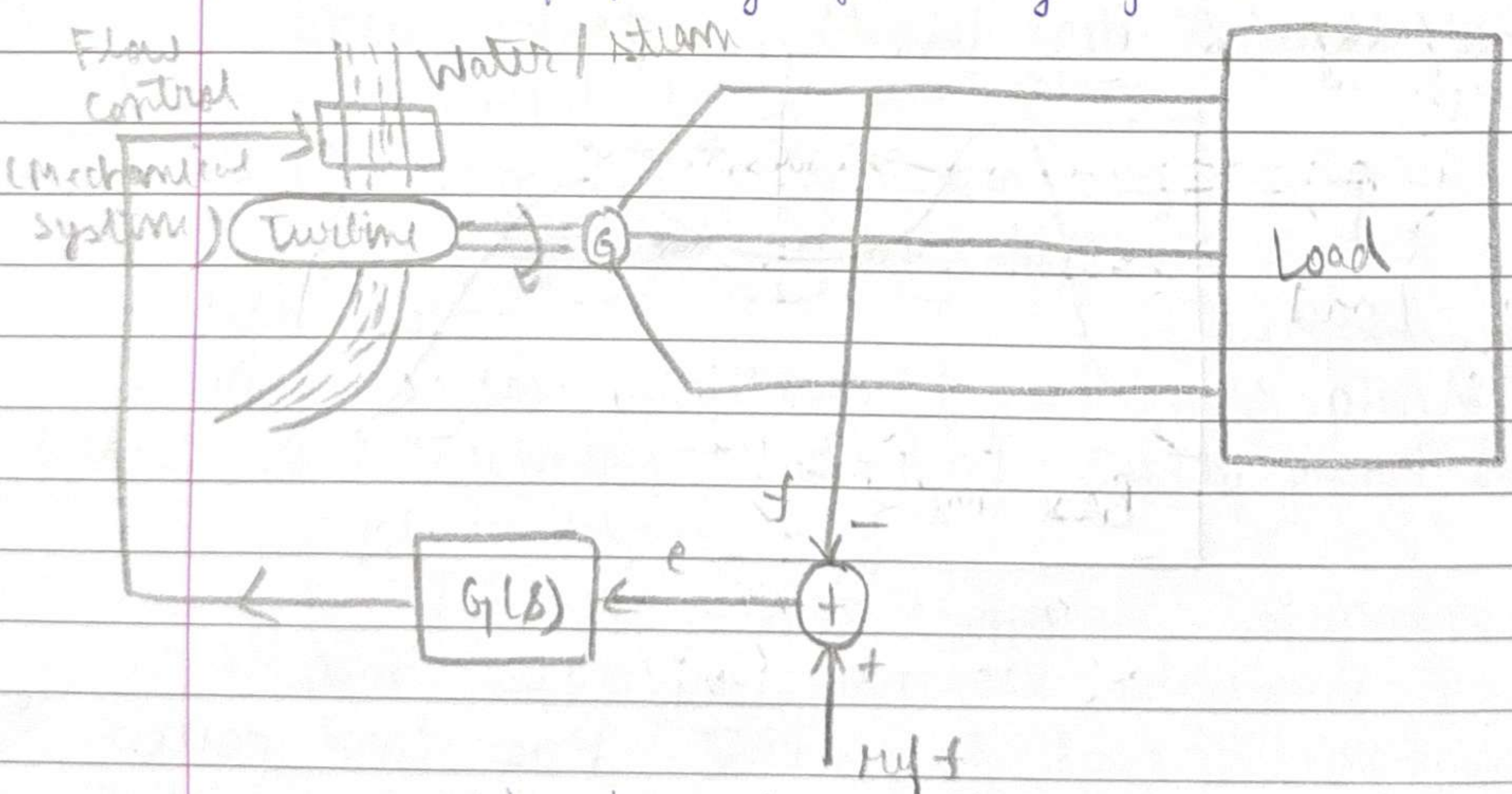
$V_{nl}$  = no-load voltage  
 $V_{fl}$  = full-load voltage

### Frequency control

$$P_{s mech} - P_{elec} = K \frac{dw}{dt}$$

If load  $\uparrow \Rightarrow P_{elec} \uparrow \Rightarrow \omega \downarrow \Rightarrow f \downarrow$

$\therefore$  We need to put active control to maintain constant frequency for varying loads.



This is called speed governor



Example where AVR and speed governor are implemented

DG Sets (Look up)

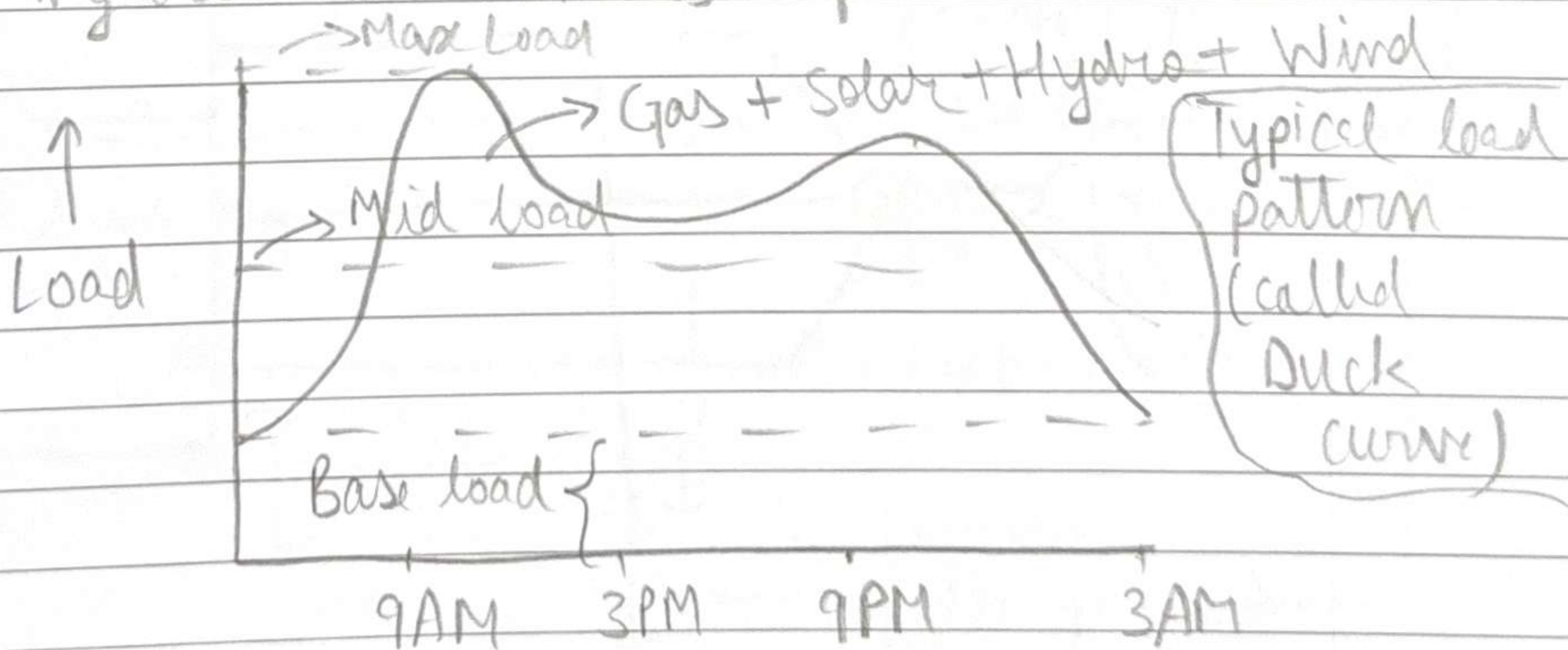
Coal takes a lot of time to ignite  
Water takes time to boil and convert into high pressure steam.

Hence coal power plant cannot be switched on/off quickly. Hence coal plant keeps running even during night. ( $24 \times 7$ )

Gas ignites quickly and does not have to boil water  $\Rightarrow$  can be turned on/off quickly.

$\Rightarrow$  Gas power can tend to sudden spikes in power consumption.

We say,  
Coal handles 'Base' load  
Gas / hydro etc. handles 'peak' loads



If we use a coal to generate Max load power, at most times of the day, it will be underloaded  $\Rightarrow$  losses



Here, coal is only used to supply a lower power level (Mid load)

So that it is under-loaded for smaller durations of the day.

The variation about that mid load level is handled by gas + hydro + solar + wind

gas & hydro - controllable

Hydro - consumes lot of resources

solar & wind - not in our control

Consider a scenario -

At 12 noon

solar was producing 110 GW power

Total load 300 GW

Suppose solar suddenly goes to 0!

What to do?

→ Increase gas, hydro.

Still load requirement not met?

Solution -

→ Have some additional  $\phi$  coal power plants that keep running, but don't supply power in usual conditions. (called spinning reserves)

In case of such a scenario, spinning reserves start supplying required additional power.



But spinning reserve  $\Rightarrow$  high no load losses

Another solution?

Introduce "storage" in grid

$\rightarrow$  BESS (Battery Energy Storage System)

Coal not preferable -

- Environmental factor
- Need to import

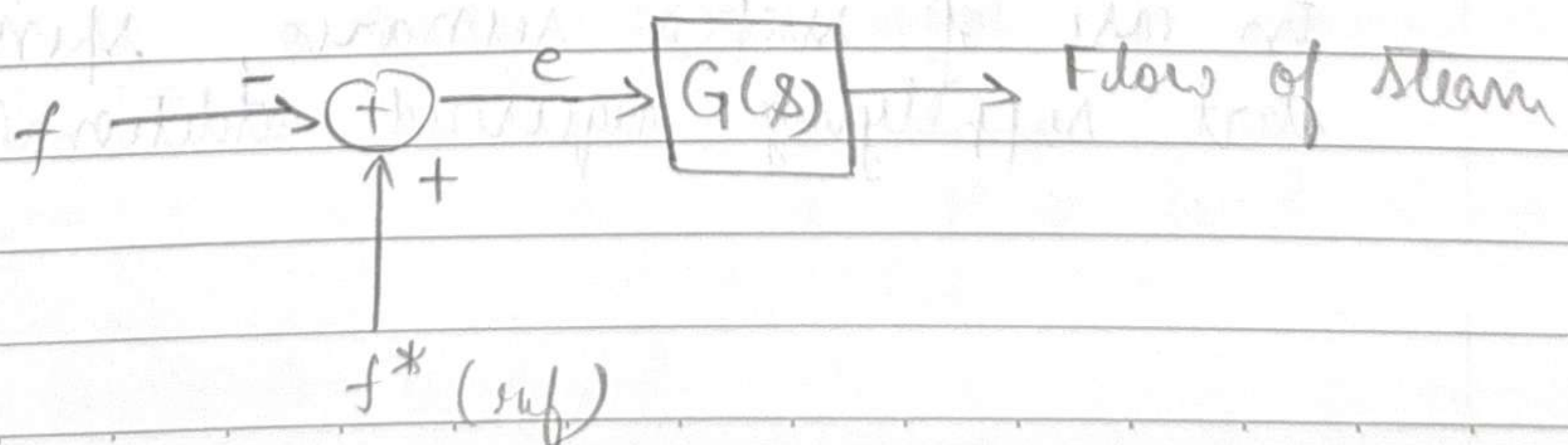
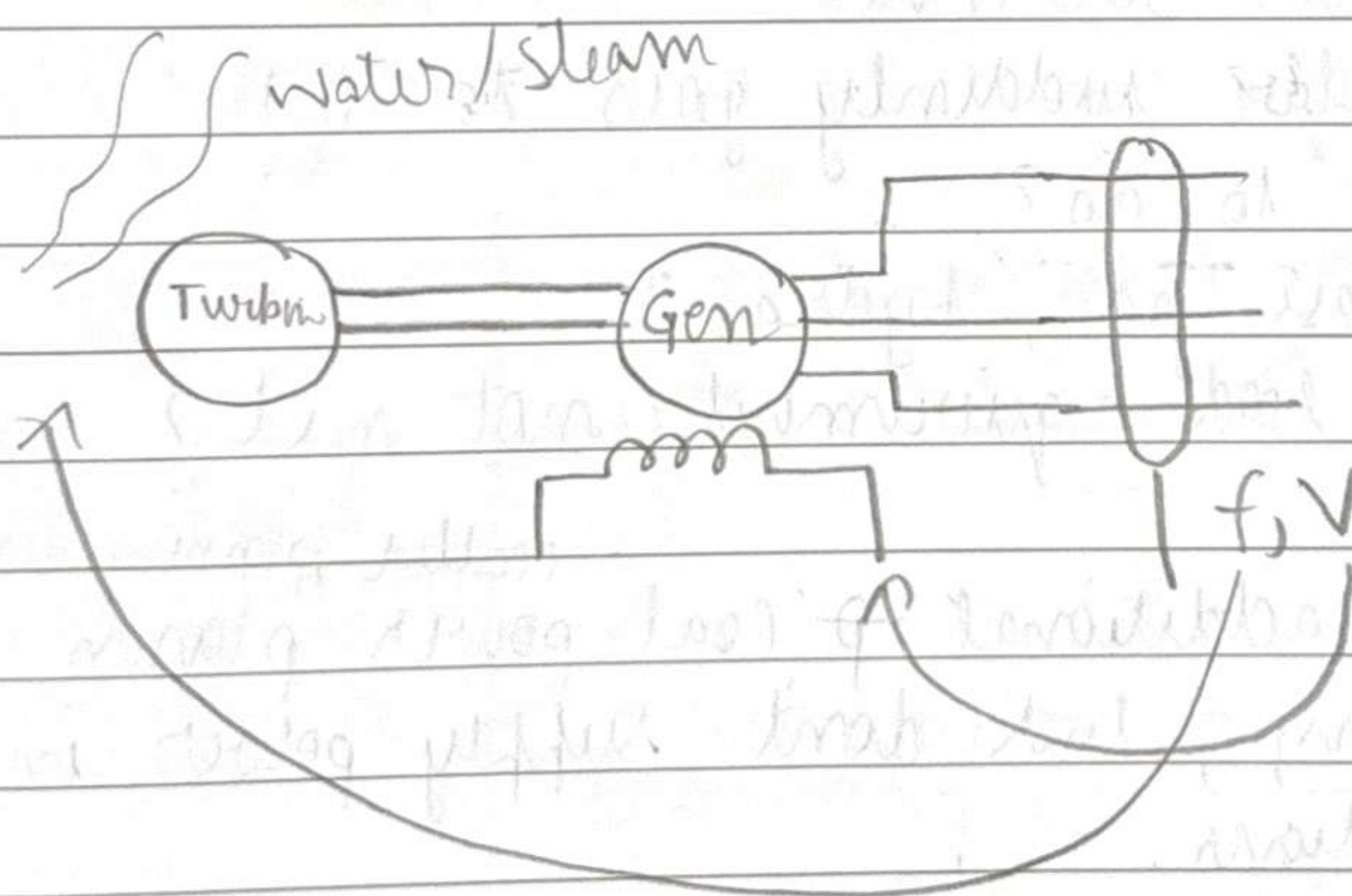
These days, - solar is a preferred option

$\rightarrow$  We want to switch to solar

- Cheap
- No pollution

However, as solar component of produced power increases, we need more storage in grid

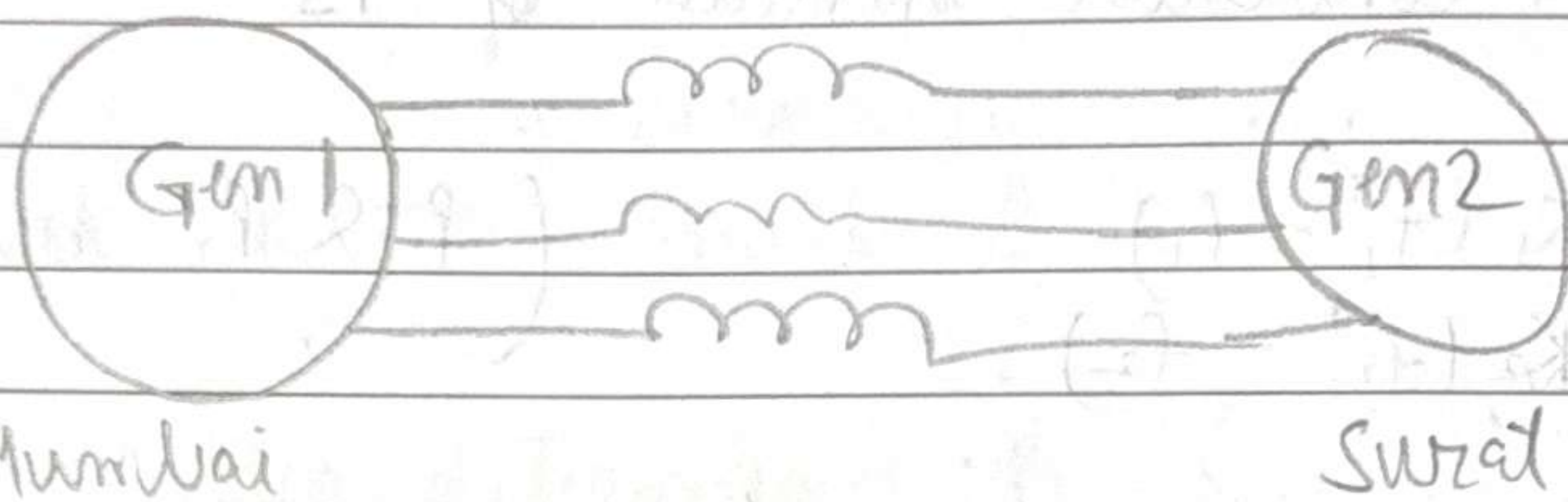
Storage - expensive!!





If  $G(s) = PI$  controller  
 steady-state error = 0  
 $f = f^*$

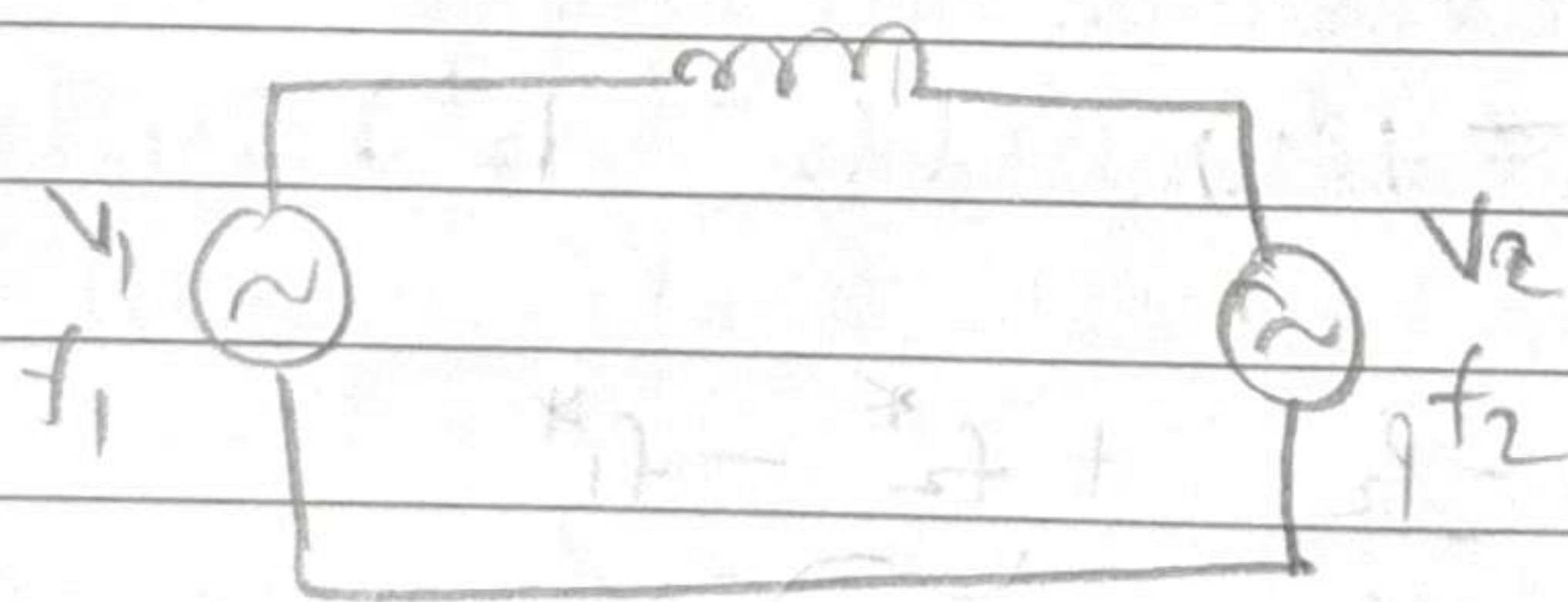
Suppose there were two generators, and we connected them so



Practically  $f_1^* \neq f_2^*$  (it ~~is~~ is not possible to get precisely matching ref ~~frequencies~~ frequencies)  
 $\Rightarrow$  Lot of circulating power of  $\Delta f (= f_2^* - f_1^*)$   
 (beat frequency)

$\therefore$  Cannot connect multiple generators with tightly regulated frequency control.

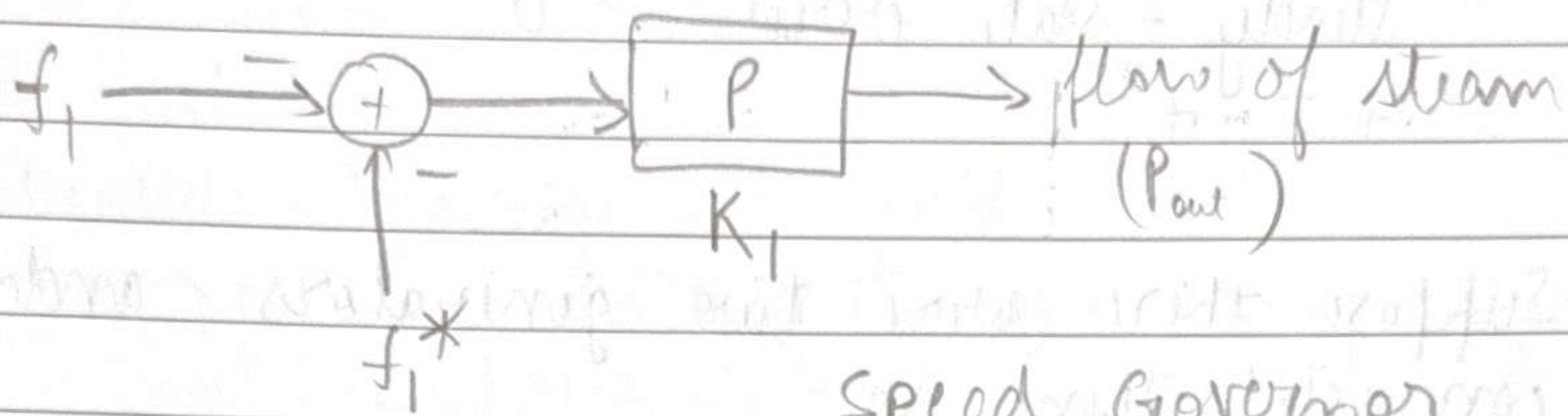
### ★ ★ Homework ★ ★



Solve this circuit and find power (t)  
 observe the circulating power



What to do?



Speed Governor

Use a P controller instead of PI

$$\begin{aligned} P_1 &= K_1 (f_1^* - f_1) \\ P_2 &= K_2 (f_2^* - f_2) \end{aligned} \quad \left( P_1 \& P_2 \text{ are } \cancel{\text{output}} \text{ output power} \right)$$

Assume steady state, i.e.  $f_1 = f_2 = f$  (say)

$$-P_1 + f_1^* = -P_2 + f_2^*$$

$$K_1 (f_1^* - f) = K_2 (f_2^* - f)$$

Because of integrator, steady state error was 0, so  $f_1 = f_1^* = f_2 = f_2^*$

With only P controller, there is non zero steady-state error (less tight regulation — loosely regulated)

$$f_1 = f_2 \text{ even if } f_1^* \neq f_2^* \\ (f_1 \neq f_1^*) \quad (f_2 \neq f_2^*)$$

$$\frac{-P_1}{K_1} = \frac{-P_2}{K_2} + \underbrace{f_2^* - f_1^*}_{\text{very small}}$$

$$\therefore \frac{P_1}{K_1} = \frac{P_2}{K_2}$$

If  $P_1 + P_2 = 5 \text{ MW}$  suppose (total load in Mumbai and Surat)

$$\text{If } K_1 = K_2$$

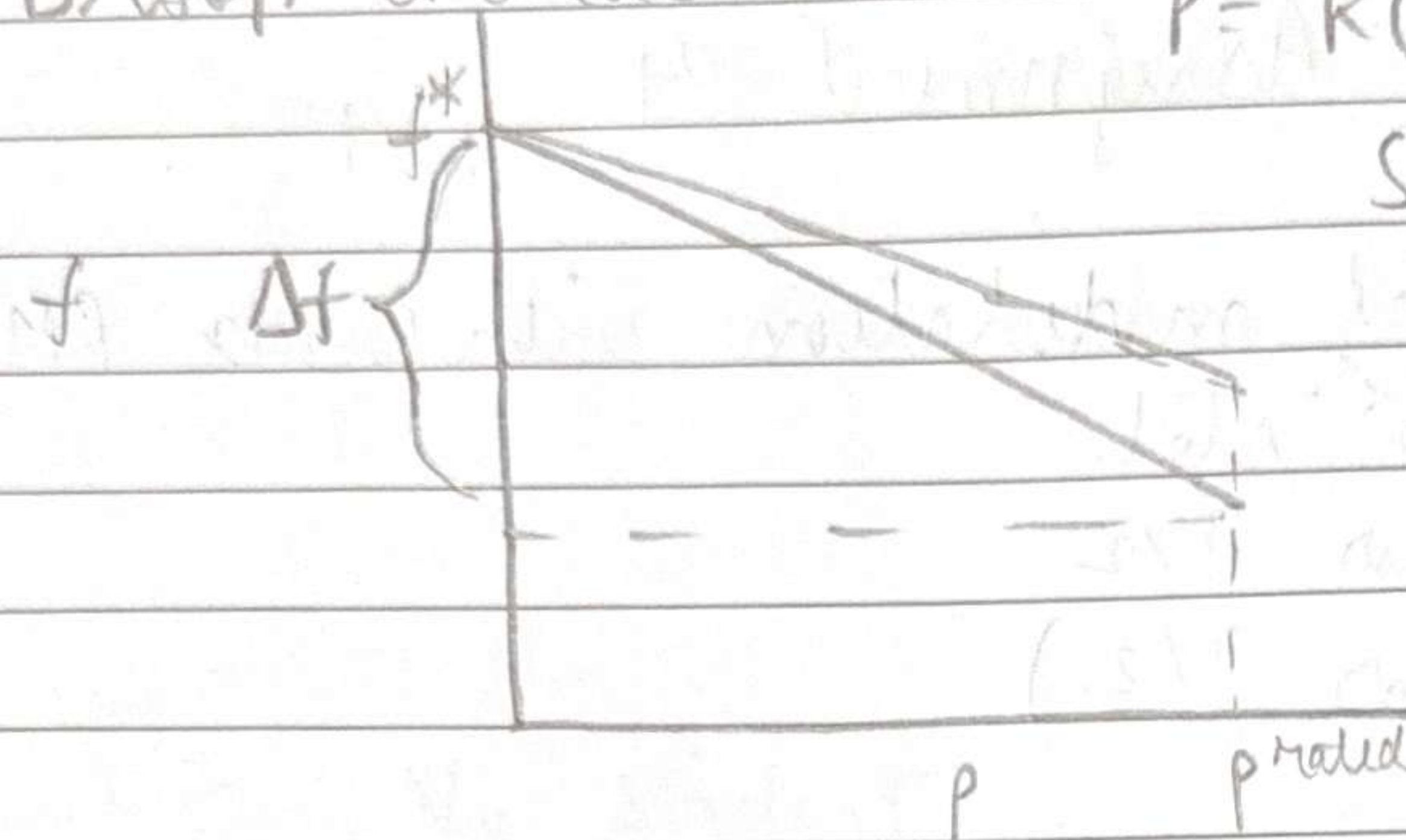
$$P_1 = P_2 = 2.5 \text{ MW}$$



## "Drop characteristics"

$$P = K(f^* - f)$$

$$\text{Slope} = \frac{1}{K}$$



High  $K \Rightarrow$  small  $\Delta f$

|       |                  |       |  |
|-------|------------------|-------|--|
| $K_1$ | ratio determines | $P_1$ |  |
| $K_2$ |                  | $P_2$ |  |

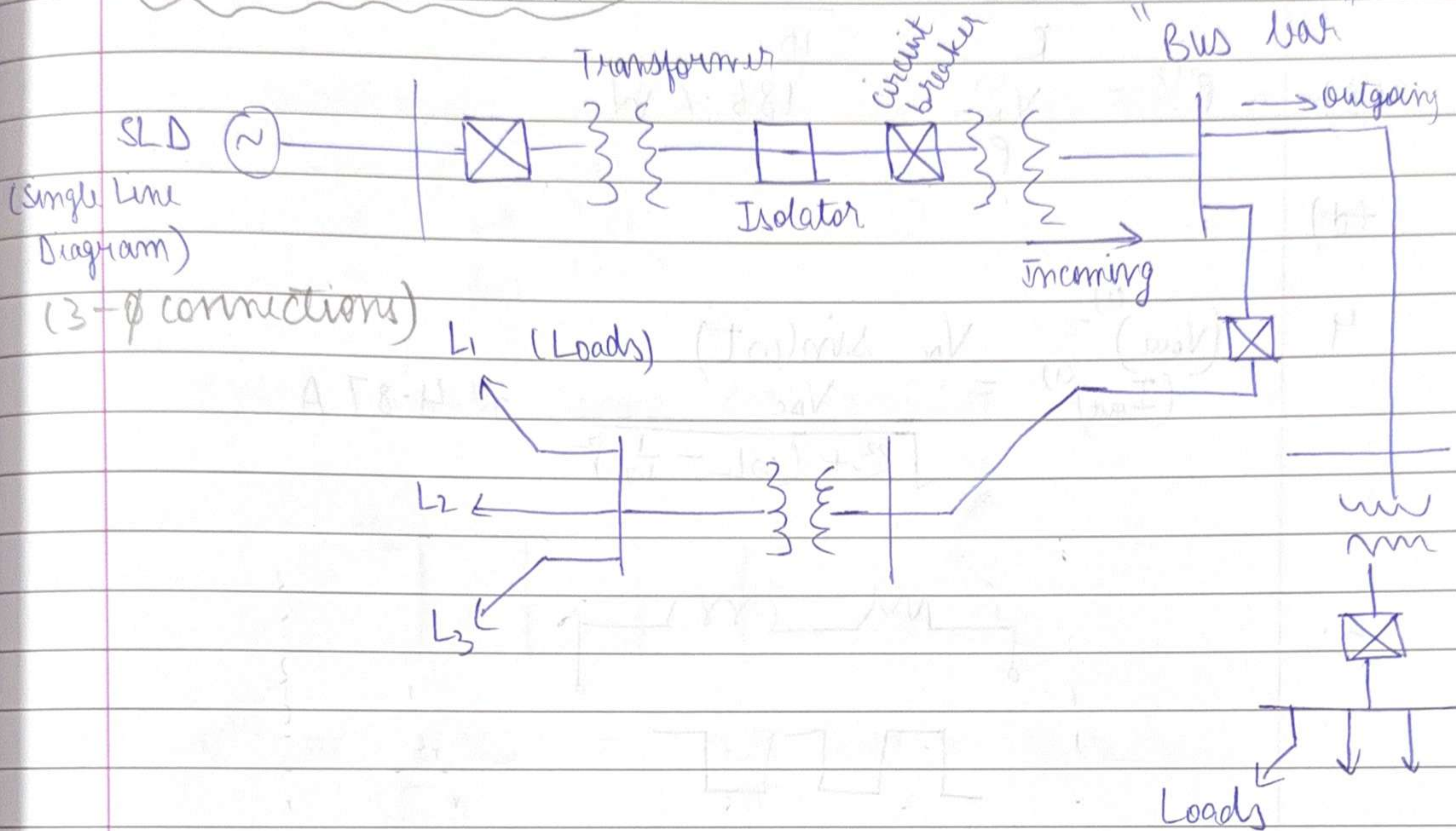
For large  $K$ , as  $P$  goes from 0 to  $P_{rated}$ , variation in  $f$  ( $\Delta f$ ) is small

In case 2 segments of grid get disconnected — they function independently as 'islands'

In order to reconnect them, just ensure their voltages and frequencies match — using 'dark lamp method'.



# Electrical Network



## Circuit breakers

Protects from over-current, fault etc  
"Smart"  $\Rightarrow$  breaks circuit automatically  
Relatively expensive

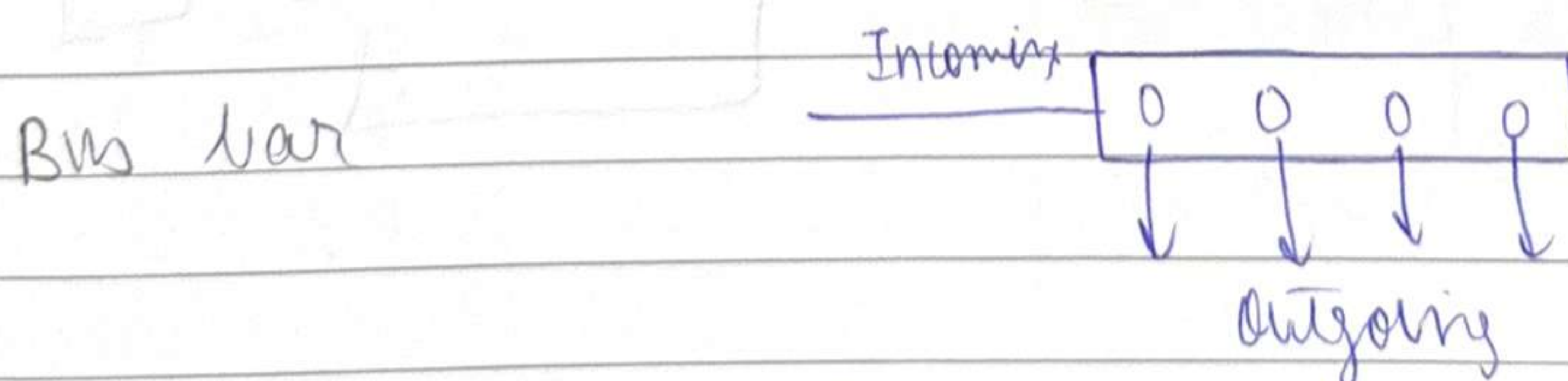
Isolator

Mechanical switch, can manually turn on/off circuit

Not "smart"

## Chapter

Used to de-energise ~~at~~ wires for maintenance etc.



Have multiple outgoing lines connected to an incoming line

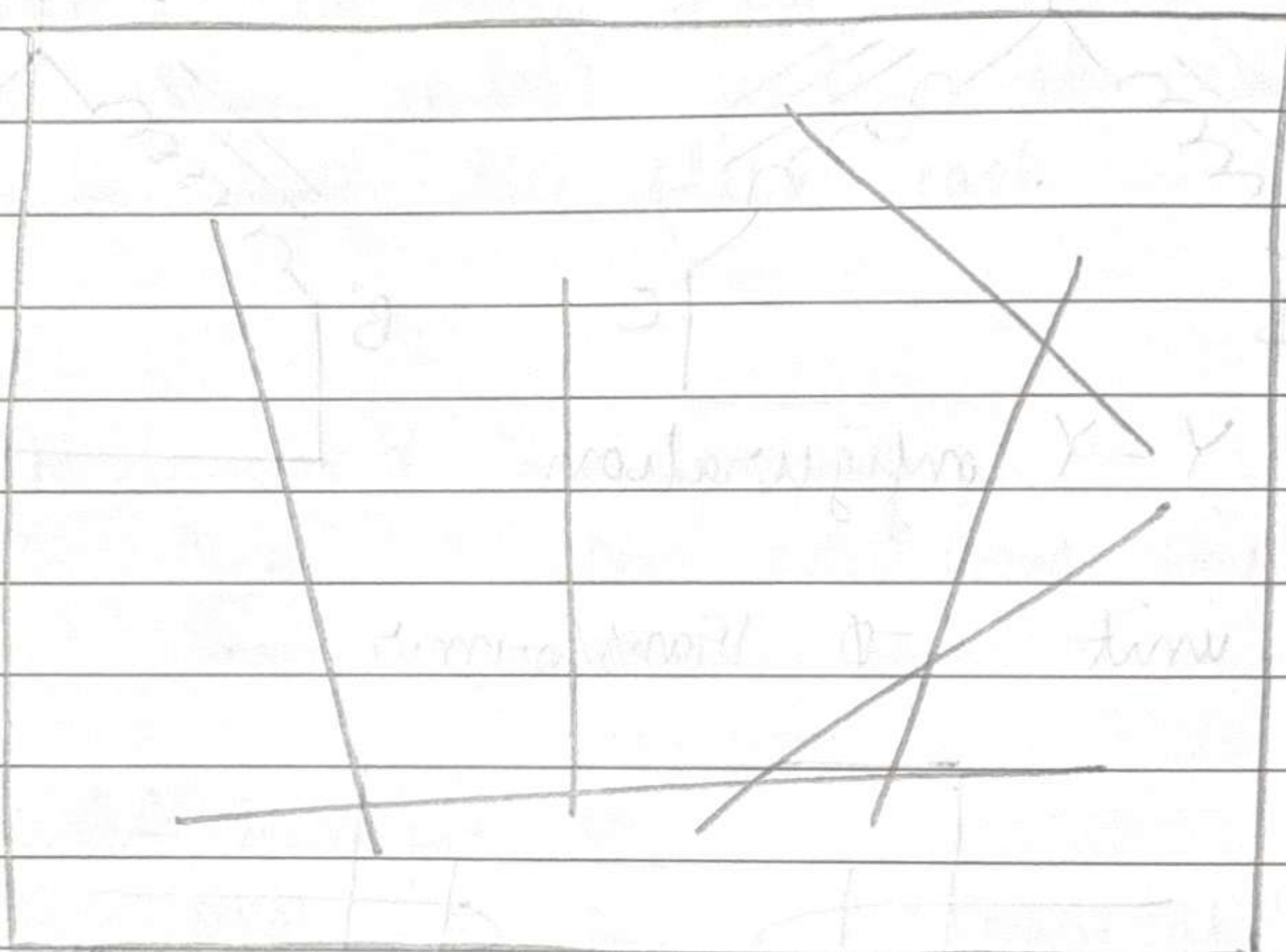


## Single Line Diagram

Doesn't depict distances along and path shapes over which wires are to be laid out.

## AC Transmission

Transmission lines - connecting generation hubs and load hubs.



Map of country  
 (Example sketch)

AC Transmission - 765 kV, 400 kV, 220 ~~kV~~ kV

AC Distribution - 132, 60, 33, 11 ~~kV~~ kV,  
 400 V

Generation typically around 11 kV

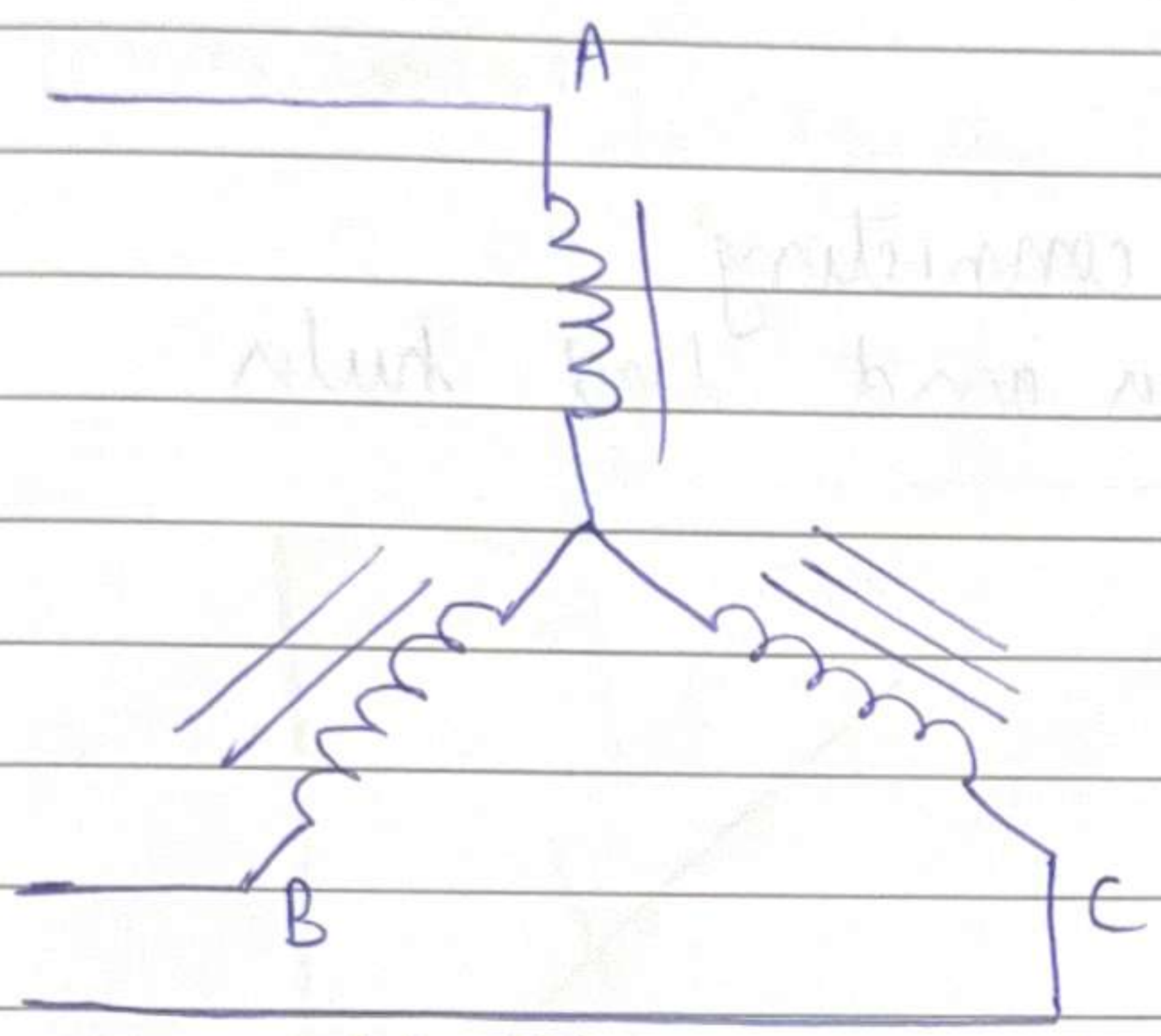
Then stepup → Transmission → step-down  
 → Distribution



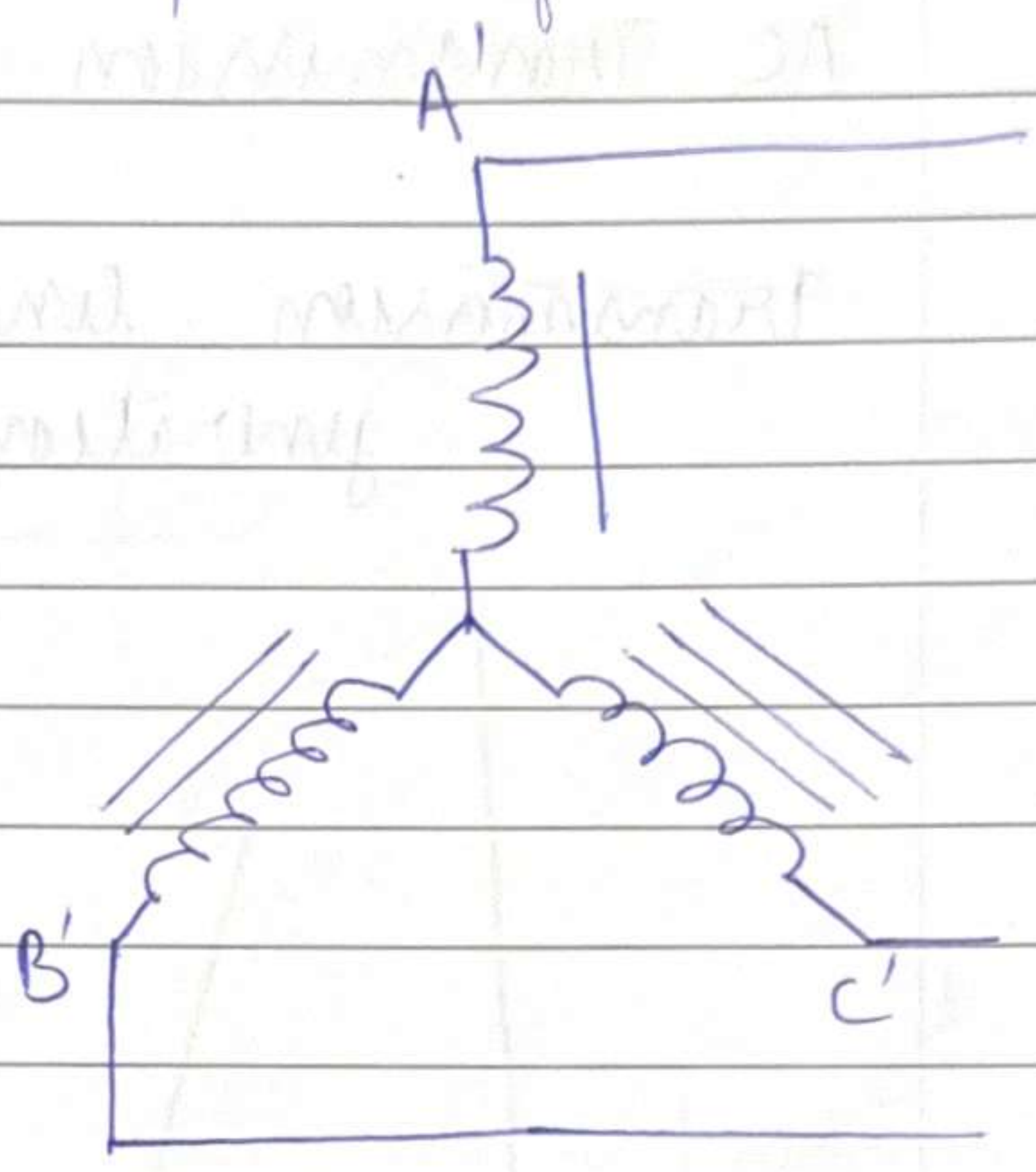
# Three phase Transformer

It could be 3 units of 1- $\phi$  transformer  
 OR

Integrated 3- $\phi$  Transformer

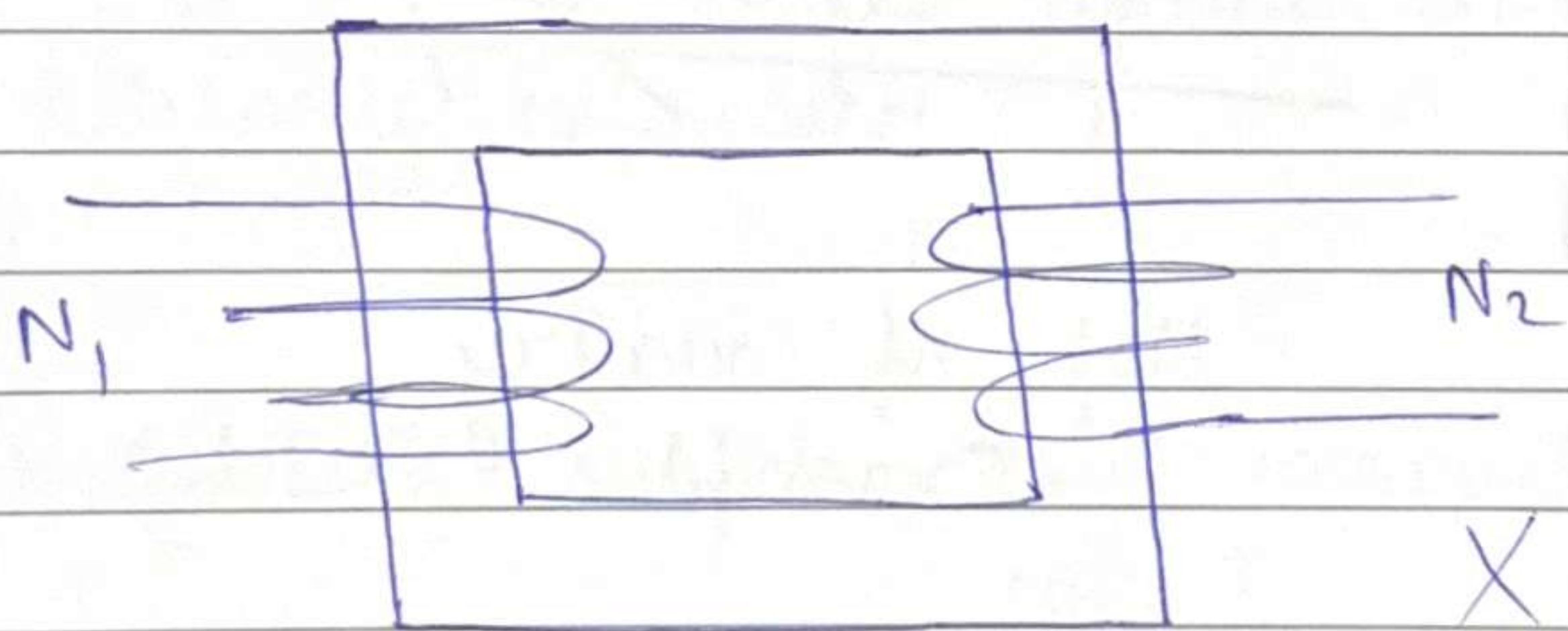


Y-Y configuration



Each unit 1- $\phi$  transformer

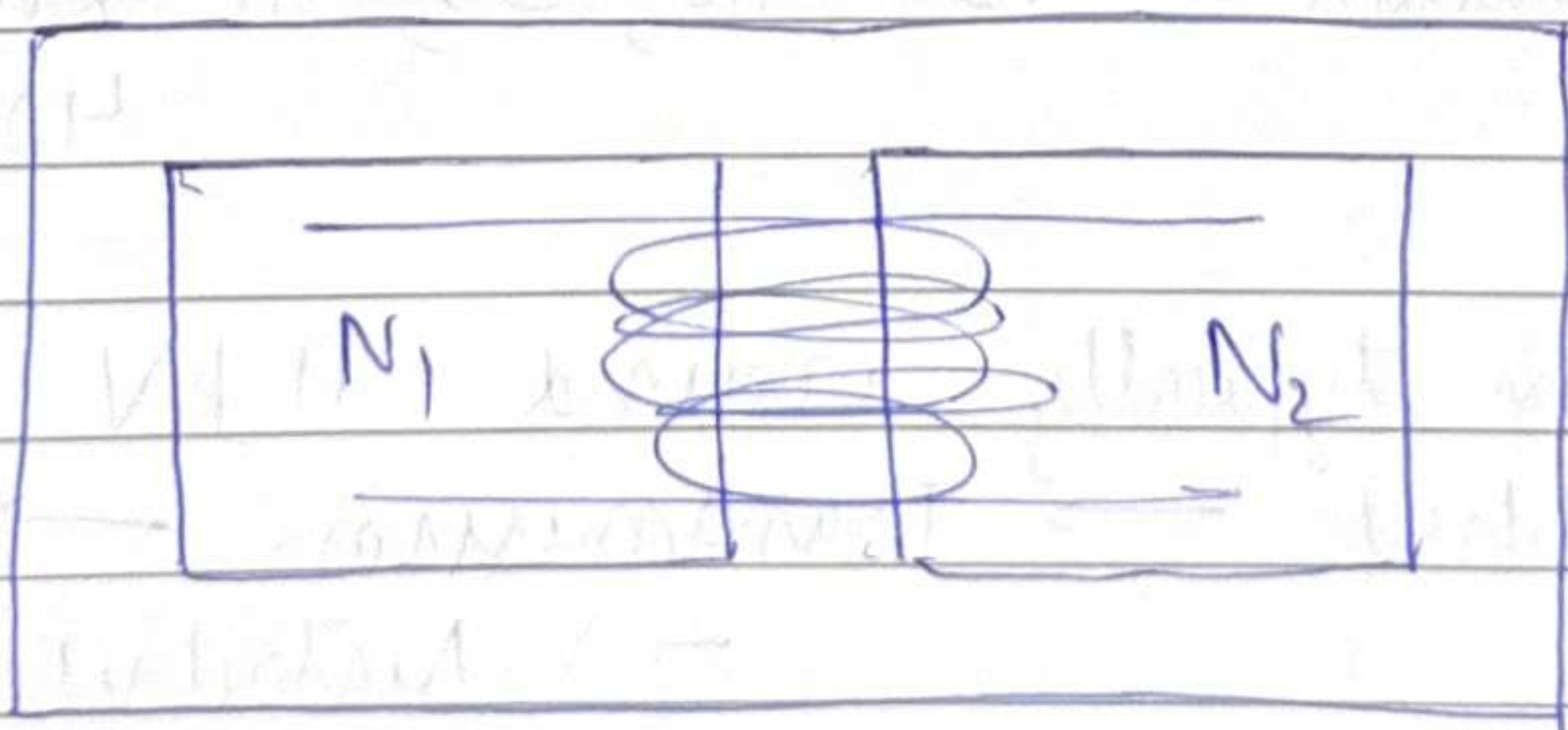
option 1)



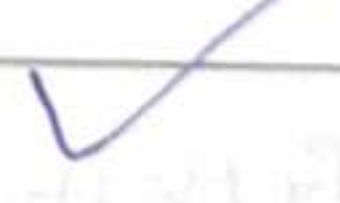
(More leakage)

(not done in actual life)

option 2)



(less leakage)



What is done in real life



Advantage of option 2)  
 less leakage

Disadvantage of option 2)  
 Requires stronger insulation between windings, because of windings are placed very close to each other.

In option 2, we may keep the side limbs thinner than central limb, ~~th~~ since they link only half the flux, each.

For such a Y-Y transformer,  

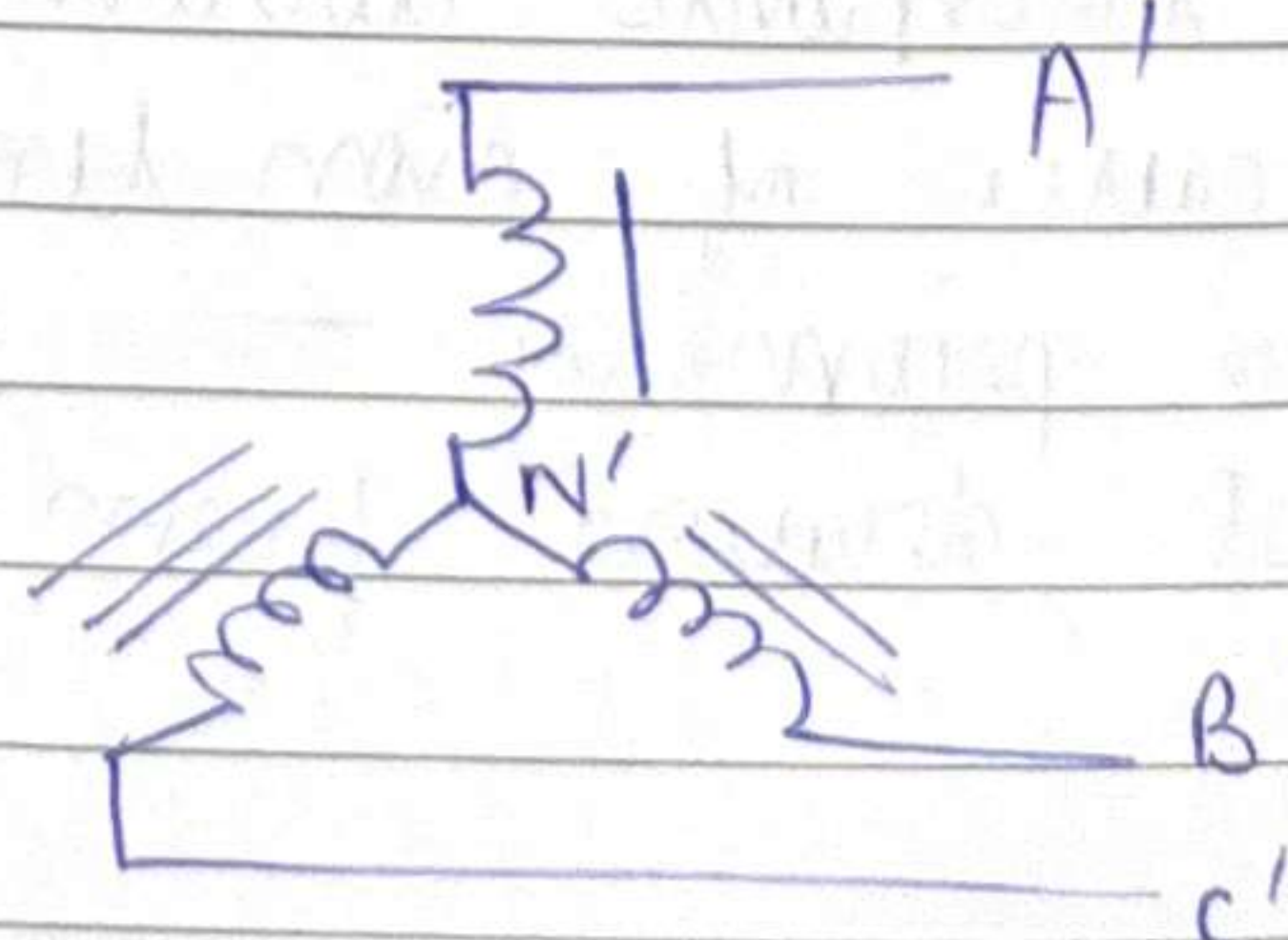
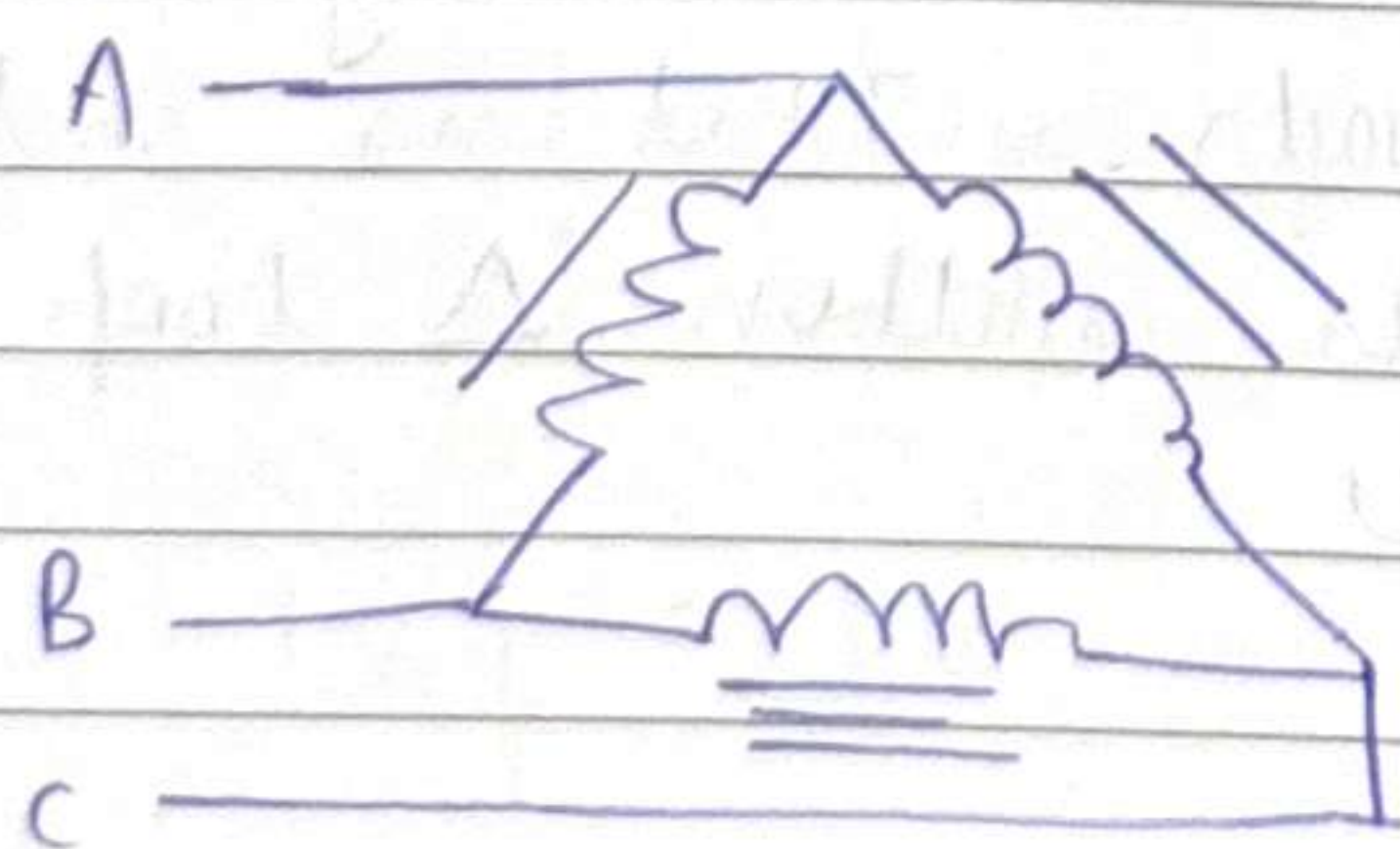
$$\text{Gain} = \frac{V_{A'B'}}{V_{AB}} = \frac{N_2}{N_1} \quad (\text{Phase shift} = 0^\circ)$$

For a  $\Delta\Delta$  transformer,  

$$\text{Gain} = \frac{V_{A'B'}}{V_{AB}} = \frac{N_2}{N_1} \quad (\text{Phase shift} = 0^\circ)$$

For a  $\Delta$ -Y transformer, (most common configuration)  

$$\text{Gain} = \left| \frac{V_{A'B'}}{V_{AB}} \right| = \frac{N_2}{N_1} \sqrt{3} \quad (\text{Phase shift} = 30^\circ)$$



For a Y- $\Delta$  transformer  $\Rightarrow$  HW



Why is  $\Delta$ -Y most popular in grid?

| Primary                                 | Secondary (distribution)            |
|-----------------------------------------|-------------------------------------|
| <p>X</p> <p>Y</p> <p>✓</p> <p>11 kV</p> | <p>Y</p> <p>✓</p> <p>400V</p>       |
| <p>✓</p> <p><math>\Delta</math></p>     | <p><math>\Delta</math></p> <p>X</p> |

Due to presence of non-linear elements like diode rectifiers etc., current drawn contains  $3^{rd}$  harmonics.  $3^{rd}$  harmonic,  $9^{th}$  and so on, don't cancel each other out, so in a Y connection, they would flow back through Neutral.

In  $\Delta$  connection,  $3^{rd}$  harmonic flows within  $\Delta$  loop, and doesn't flow through source.

We want neutral, because

- connect 1- $\phi$  loads across phase
- Earth the system

Y-Y-neutral

$\therefore$  We need Y on this side

—  $3^{rd}$  harmonic current arises on secondary side because of non linear loads. That is reflected onto primary — circulates within  $\Delta$  loop, not drawn from source.

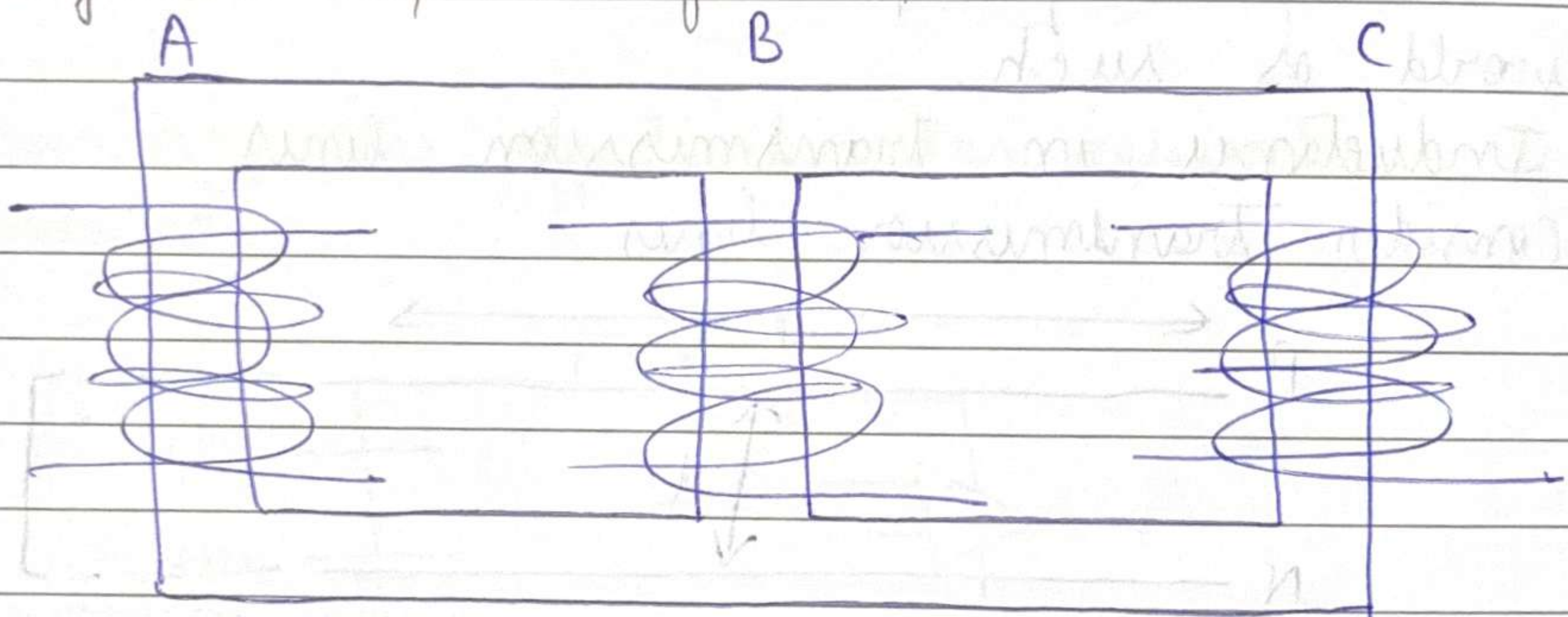


Multiple transformers are cascaded

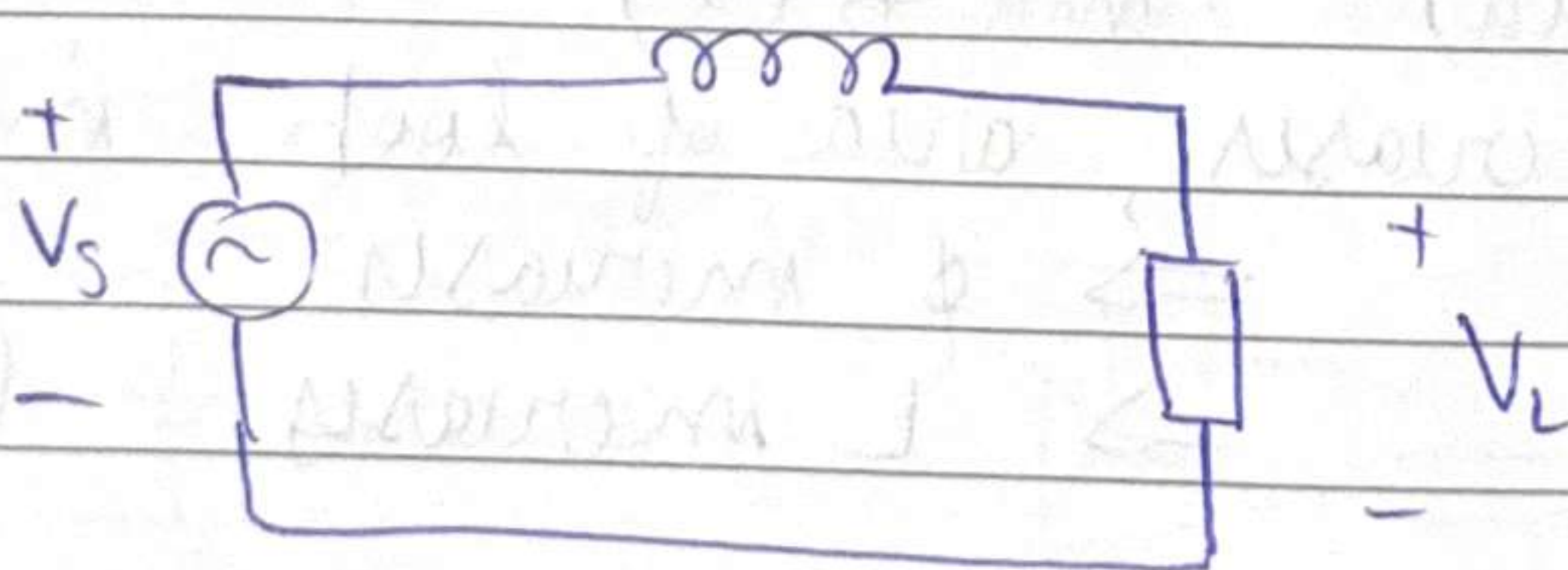
Each one in this sequence is  $\Delta$ -Y transformer

Y at secondary side because of grounding  
 $\Delta$  because of it  $\rightarrow$  leads to zero "sequence current" at primary

Integrated 3- $\phi$  transformer



side limbs have same cross-section area as central limb



If load is purely inductive

$$|V_s| \geq |V_2|?$$

$$|V_2| < |V_s|$$



If load is purely capacitive  
 $|V_L| > |V_S|$

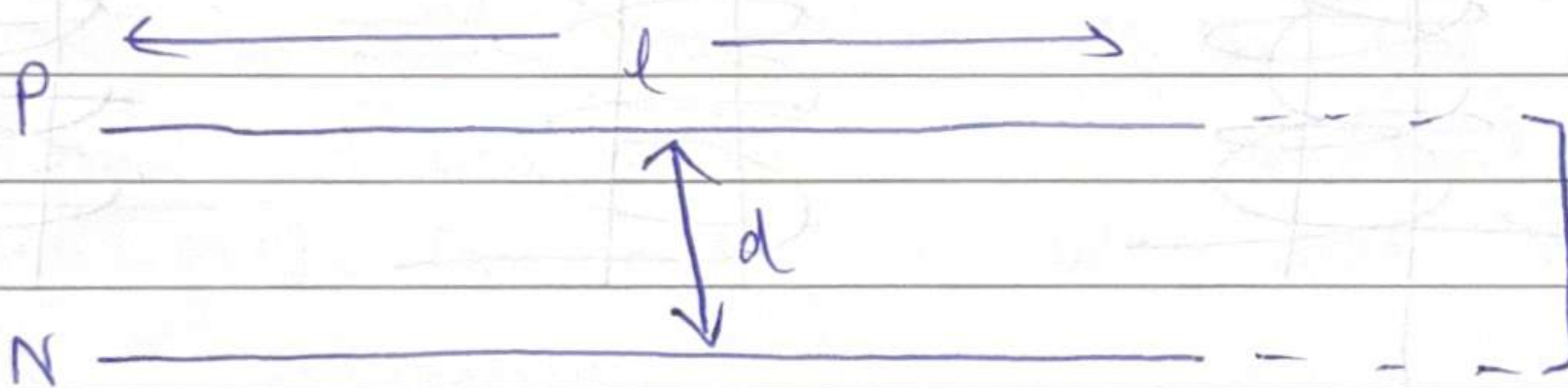
We don't want either situation

- inductive load will lead to cause smaller (e.g. motors) load voltage
- capacitive load will cause larger load voltage  
 $\Rightarrow$  can damage components

There aren't any capacitive "loads" in the real world as such.

Inductance in transmission lines

Consider Transmission lines



Suppose the physical distance between phase and neutral lines is  $d$ .

The loop has an inductance  $L$ .

$$L = f_1(d) \quad \text{and} \quad f_2(l)$$

As  $d$  increases, area of loop increases

$\Rightarrow \phi$  increases

$\Rightarrow L$  increases  $(L = \frac{N\phi}{I})$

These transmission lines carry very high voltages.

To avoid breakdown of air in between

we keep  $P$  and  $N$  at large distance  $d$ .

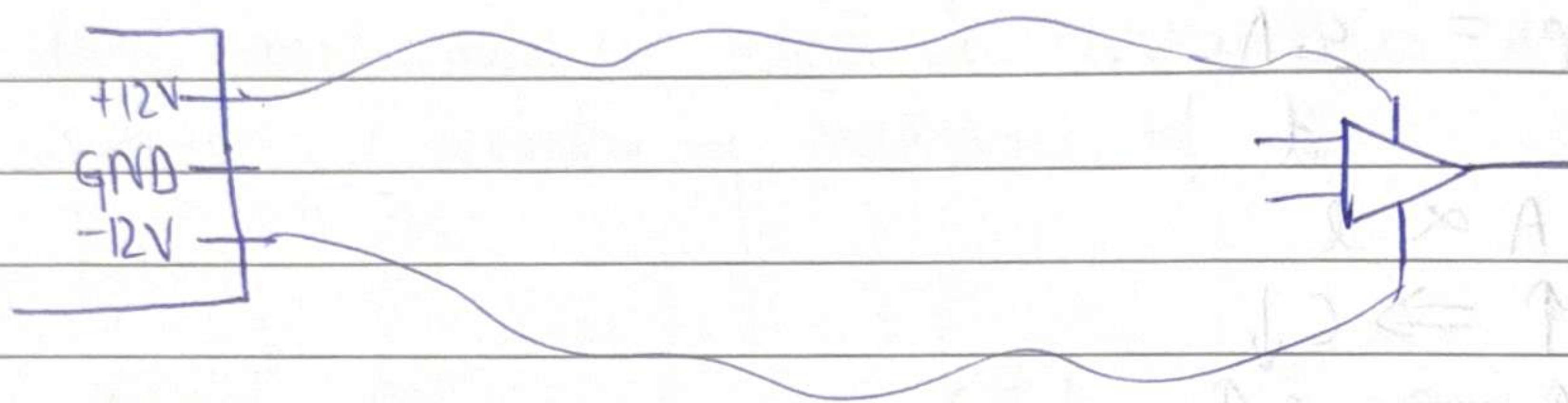
$\Rightarrow L$  is very large.

Also  $l \propto L$

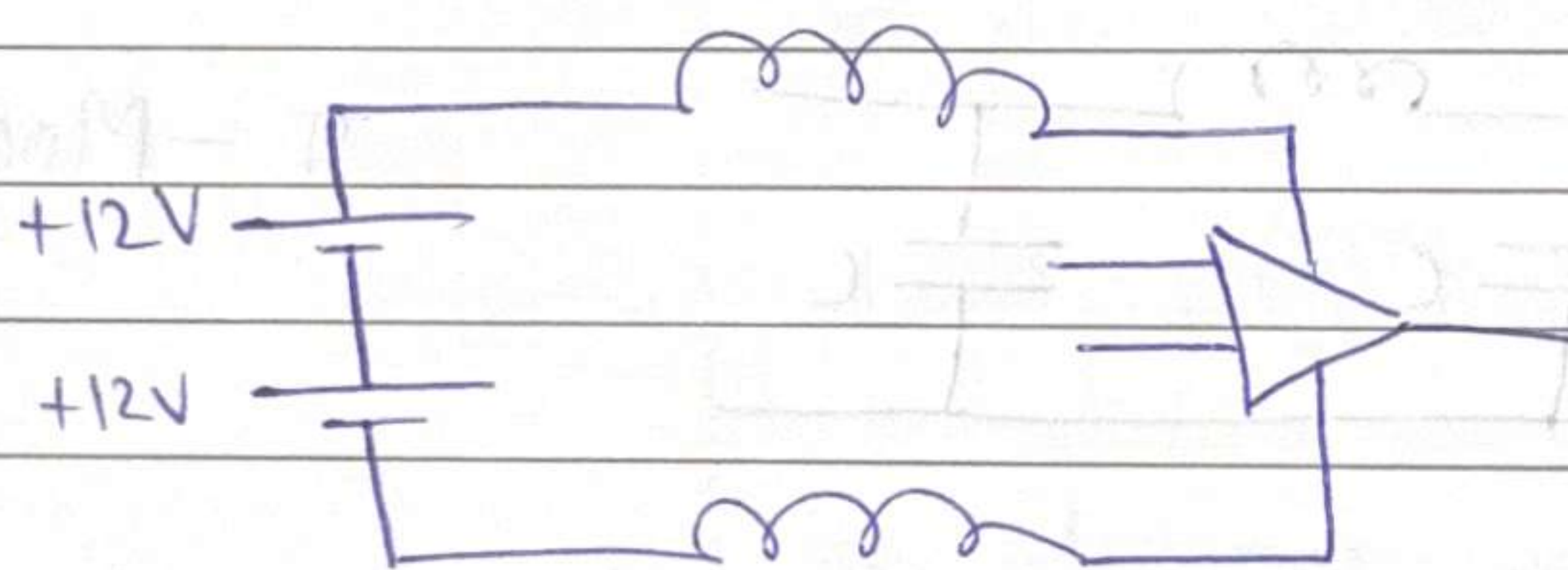


Very high voltage + very large distance  
 $\Rightarrow$  Large  $L$

For example, in ~~at~~ analog lab circuits, to supply power to opamps, we do

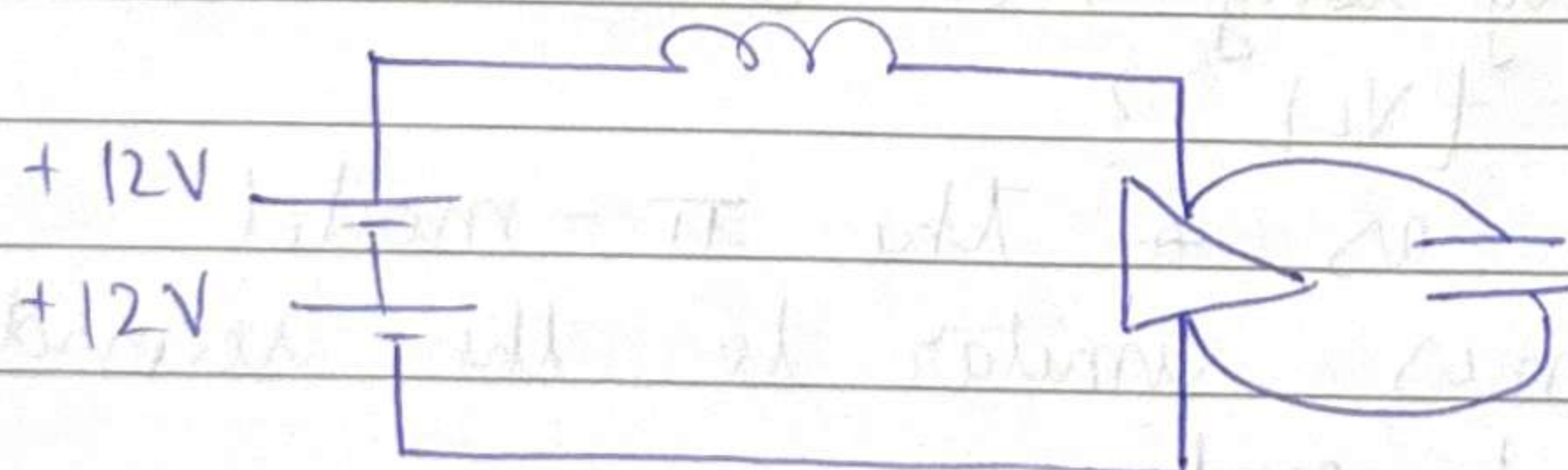


However, this effectively has equivalent series inductance



Solution?

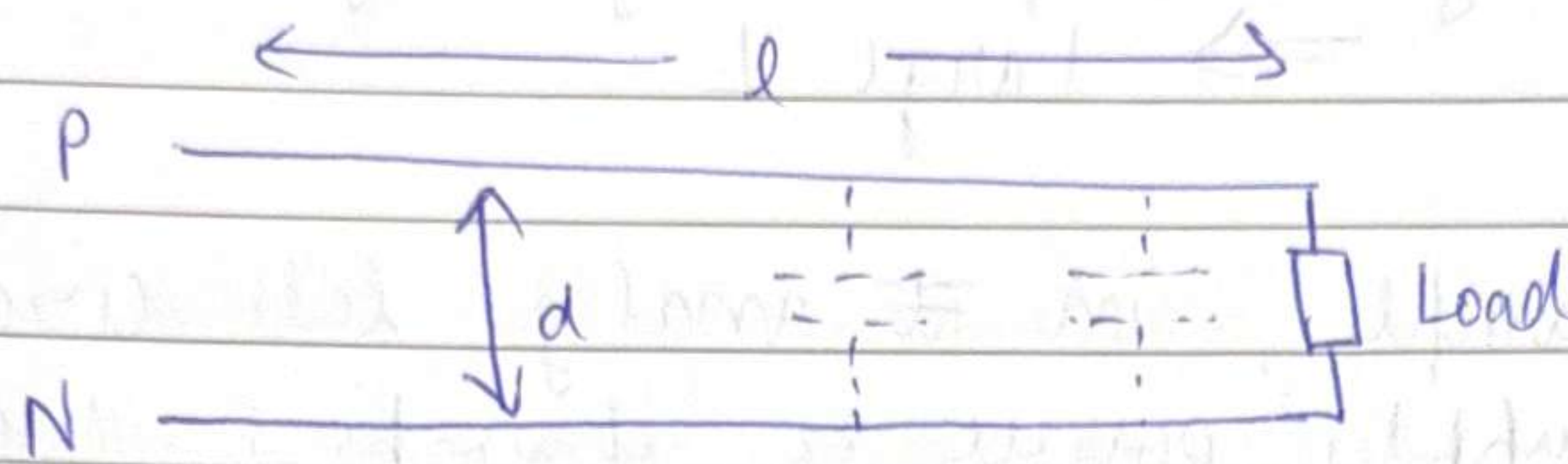
- Put a 'decoupling' capacitor at opamp supply pins



- Use Twisted supply wires
- Use short wires to minimize loop area



## Capacitance in transmission lines



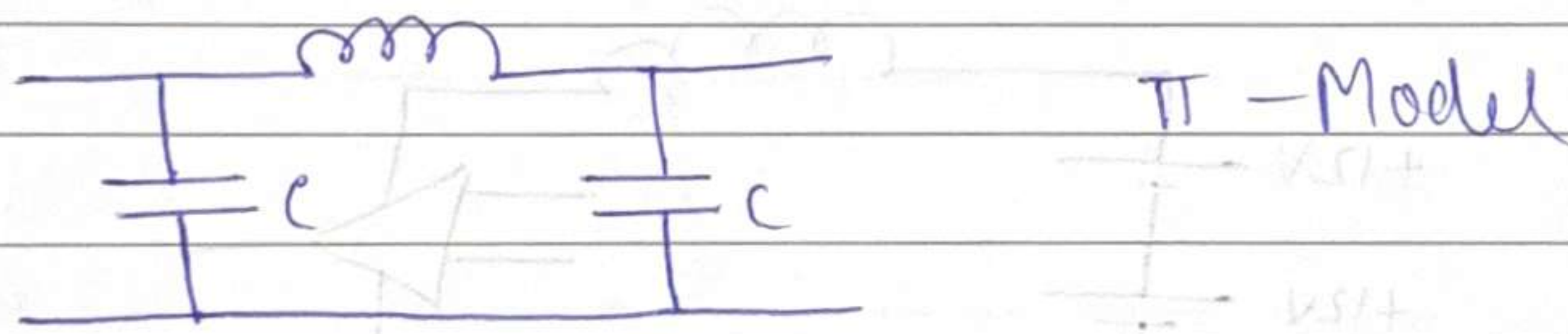
$$C = \frac{\epsilon_0 A}{d}$$

$$A \propto l$$

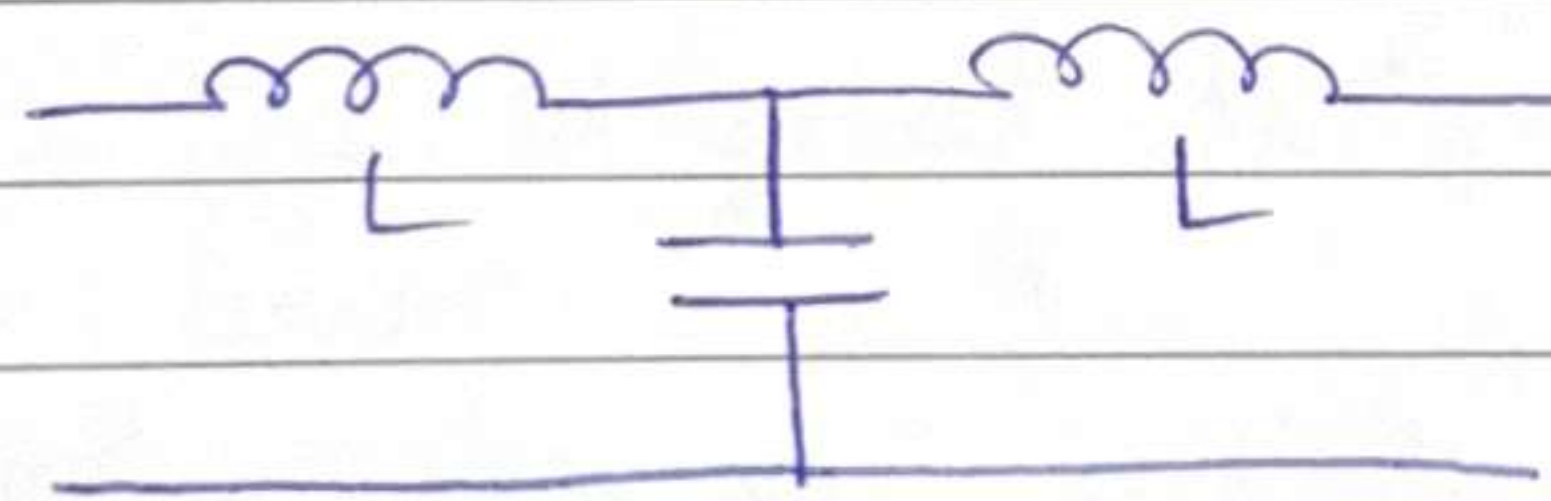
$$d \uparrow \Rightarrow C \downarrow$$

$$l \uparrow \Rightarrow C \uparrow$$

We can model it in various ways



$\pi$ -Model



T-Model

Q. A very very long AC line without any load  
 $|V_s|$  vs  $|V_L|$  ?

Visualize as a  $\pi$ -model

It becomes similar to the second case we looked at earlier

$$|V_s| < |V_L|$$

If we keep increasing  $l$ ,  $L \uparrow$  and  $C \uparrow$   
 $|V_L| - |V_s|$  increases (prove using phasors)



( $\uparrow\uparrow$ )

$\therefore$  Over longer distances, AC Transmission becomes challenging!!

$\therefore$  Problems with AC Transmission over long distances

- (i)  $|V_L| - |V_S|$  becomes large
- (ii) More and more reactive current is drawn from source. (because of presence of  $C$ )

| Lines       | VS | Cable                                                                                 |
|-------------|----|---------------------------------------------------------------------------------------|
| (overhead)  |    |                                                                                       |
| single wire |    | Multiple lines bundled into single cable                                              |
| •           |    |                                                                                       |
| •           |    |  |
| •           |    | Can be put underground                                                                |

Which one should we use for AC transmission?

In case of cable,  $L$  is small

$C$  is very very large!

There will be flow of huge reactive current!

$\Rightarrow$  Bad!

Can use cable only for short distances

( $L$  is small  $\Rightarrow C$  is limited)

$\therefore$  For short distances, can use either cables preferable because can be put underground  
 $\Rightarrow$  safe from lightning, tree fall etc.  
 e.g., distribution within a city

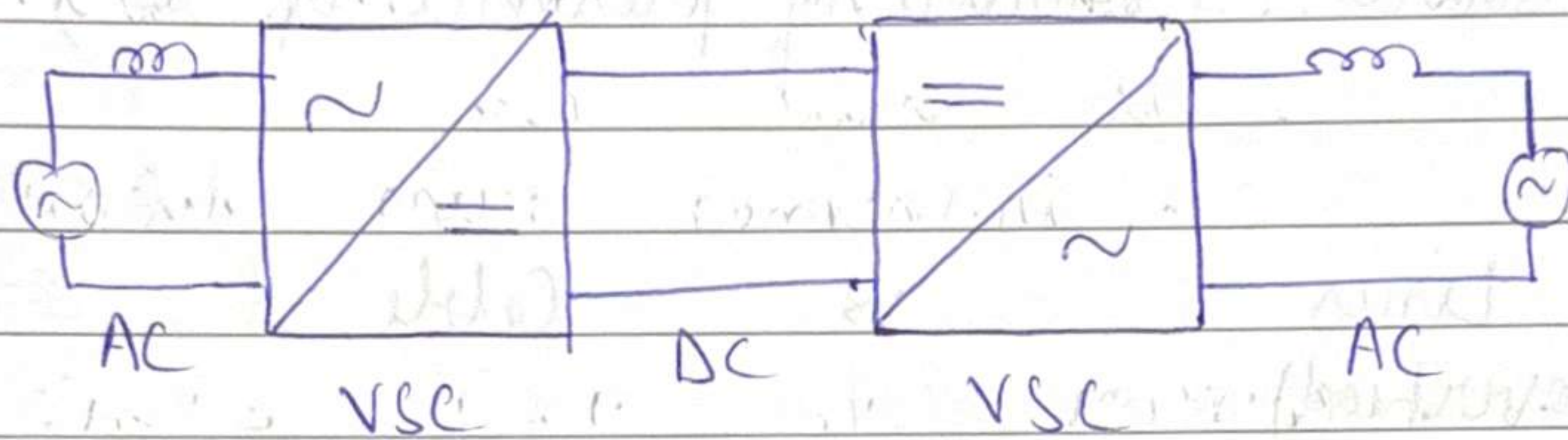


In general, cable is much cheaper and easier to layout than overhead lines.

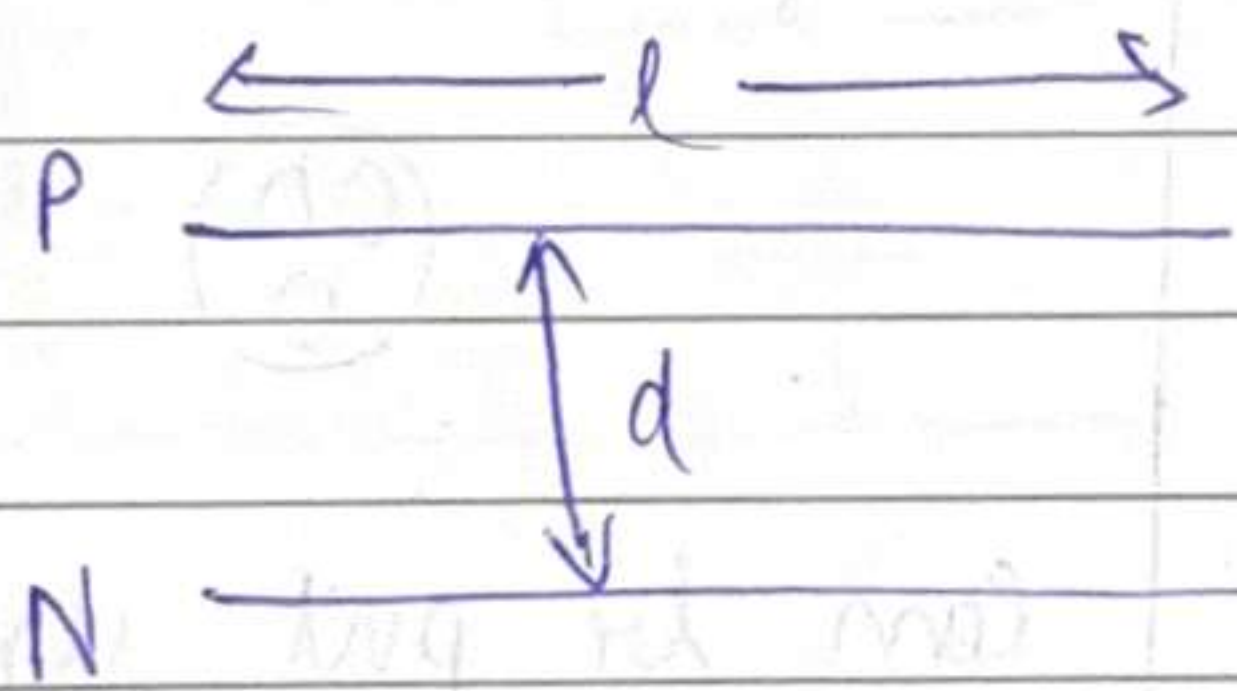
- For long distances, need overhead lines (in cables,  $C \uparrow \Rightarrow$  loading  $\uparrow\uparrow$ )

## HVDC (High Voltage DC Transmission)

We can convert  $AC \leftrightarrow DC$  using VSCs



HVDC  $\Rightarrow$  DC transmission



With DC transmission, presence of stray  $C$  and  $L$  doesn't affect  $|V_L|$  vs  $|V_S|$

$\Rightarrow$  can use either line or cable

$C, L \rightarrow$  does not have same impact on HVDC as on HVAC

HVDC transmission started in Canada  
Till now, globally only 4-5 companies have been able to achieve HVDC

HVDC - very good for long distance transmission rather than HVAC.



— Kudus - Aaruy HVDC project  
 ⇒ Company in Mumbai

developed by Adani

Core Technology supplied by Hitachi  
 (Japanese company)

(India)  
 Still we don't have the core technology  
 (VSC) to build HVDC  
 ⇒ Importing from Japan

This project links HVDC about 70-80 km  
 long

— could have been easily achieved using HVAC  
 Why used HVDC?

HVDC uses underground cables!

Typically HVAC can go to 200 to 300 km

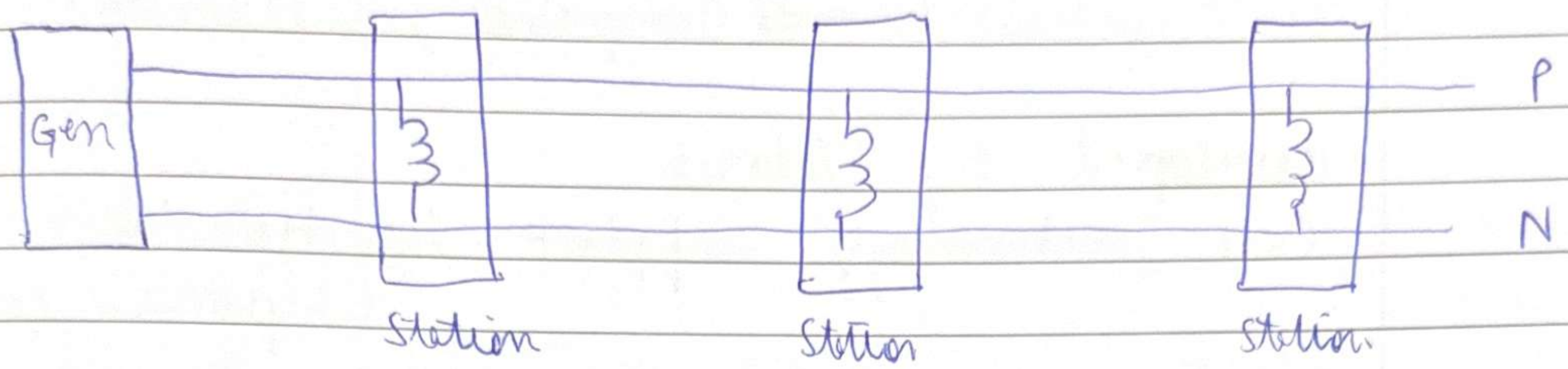
Q If we want HVAC Transmission line for  
 3000 km, how?

Problems —

- Voltage increases along the line
- large reactive current



Solution -



- (i) Have controllable shunt inductances at stations placed over regular distances
- (ii) Have reactive power compensation using VSCs  
 — called Stet Com  
 → FACTS (Flexible AC Transmission System)